

OnlinePermOOB: Detecting the gradual distribution shift in Ever-evolving Data-Streams

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1 Abstract

Understanding and detecting distribution shift is critically important for modern machine learning systems, particularly in post-deployment settings where model performance may degrade over time due to evolving data-generating mechanisms. In such environments, timely identification of distributional changes is essential for reliable monitoring, model maintenance, and informed intervention. Despite its practical importance, the literature on principled online hypothesis testing for distribution shift remains relatively underdeveloped. In this work, we build upon the recently proposed hypothesis testing framework, **PermOOB**, and extend it to an online setting by integrating a multiple testing procedure that enables rigorous false discovery rate (FDR) control over time, denoted as **OnlinePermOOB**. The resulting framework provides a computationally efficient and statistically grounded approach for sequentially detecting performance-relevant distribution shifts as new data arrive in both mini-batch setting and streaming setting. Exponential weighted moving averaged procedure is also leveraged for enhancing the timely detection as well as controlling the False Alarm Rate with the potential extension of the quantile aggregation. Through extensive simulation studies and real-world applications, we demonstrate its highly competitive, superior in more than half of scenario efficiency in detecting distribution shift in comparison to existing methods. Benchmark comparisons highlight its ability to detect point for significant distribution shift through predictive performance degradation while maintaining robust error control. We provide product landing relevance and procedures for the proposed online testing procedure.

2 Methodology

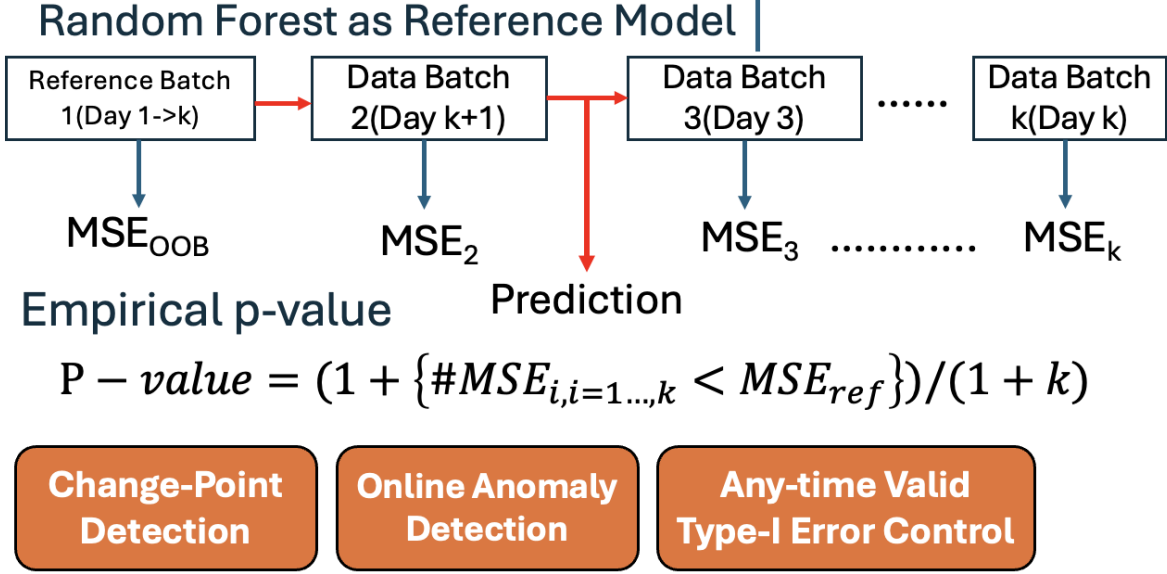
Algorithm overview. In this section we introduce our proposed **OnlinePermOOB** algorithm. The pseudo-code of algorithm is presented in Algorithm 1. Given the reference batch of dataset $\mathcal{D}_{ref} = \{(\mathbf{X}_i, Y_i)\}_{i=1}^{n_{ref}}$ and the burn-in batches $\{\mathcal{D}_{burn,j}\}_{j=1}^b$ with the incoming batches of data $\{\mathcal{D}_k\}_{k \geq 1}$. We want to mention that for each of the batch $\mathcal{D}_k$, both covariates $\mathbf{X}$ and responses Y exist. At first a predictive model f_{ref} is fitted on the reference batch using $\mathcal{D}_{ref}$ and the validation(out-of-bag for random forest) error is recorded on the in-sample validation set and the mean is denoted as $E_{ref}[MSE_{ref}]$. Then for each burn-in and incoming batch, the discrepancy between the batch-wise predictive error and the reference baseline is computed. These discrepancies are accumulated over time and compared against historical values to form empirical p-values, which quantify the extremeness of current predictive performance relative to past behavior. The proposed p-values are constructed via a rank-based comparison between the current prediction degradation statistic and its historical counterparts obtained from the burn-in and monitoring stages. Under the null hypothesis of no distributional shift, the predictive discrepancies from incoming batches are expected to be exchangeable with those from the burn-in period. Consequently, the rank of the current statistic among past values is approximately uniform, yielding the empirical p-value

$$p_k = \frac{1}{b+k} \sum_{i=1}^{b+k-1} \mathbf{1}\{T_{b+k} \leq T_i\},$$

Then the Online FDR procedure(s[34] and [49]) are leveraged to adjust the significance level for deciding significance across different batches to ensure the global False Discovery rate Control. In practice the f_{ref} serves as the deployed model in production where the continuously coming batches can serve as the reference for the downstream online distribution shift detection procedure.

This construction closely parallels conformal p-values, where nonconformity scores are evaluated against a calibration set. In our framework, the statistic T serves as a nonconformity score that quantifies predictive degradation relative to the reference model, while the burn-in batches form the calibration set. The sequential comparison thus produces valid p-values under *exchangeability*. This connection enables the integration of conformal-style validity guarantees with online multiple-testing procedures, facilitating principled sequential detection of distributional change. Moreover, as the batch size shrinks to one, the problem naturally reduces to a change-point detection framework, wherein

When the distribution shift become significant for already-deployed model?



- Very Robust under High-Dimensional Covariates
- Change-Point Detection via Monitoring Predictive Performance

Figure 1: Diagrams for the illustration of *OnlinePermOOB*

the onset of significant distributional shift can be identified through deviations in predictive performance, providing actionable insights for model monitoring and intervention. The batch-size can be either 1 or an arbitrary positive integer. By applying online multiple-testing corrections to these empirical p-values, the algorithm provides a principled mechanism for identifying the onset of significant distribution shift in a streaming environment. We want to emphasize that this algorithm is not limited specifically to **Random Forest** but can be extended to arbitrary machine learning models where in our experimental settings we leveraged XGBoost([5]) and MLP(multi-layer neural network, 2 hidden layers are added, [48]) to build the predictive model. The procedure detects distributional change by monitoring the predictive degradation of a reference model trained on an initial stable dataset. The overview of the procedure is summarized in Figure 1, where the procedure not only serves as an effective detector of distribution shift but also provides concrete, actionable early signals for model refreshment. Specifically, the out-of-bag error of the reference model serves as a baseline performance measure.

We observed that the average p-value procedure carries the exactly equal weight over all historical information, which increases the risk of random rejections and prevents the method from focusing on recent oscillations. Therefore, we propose an exponentially weighted moving average (EWMA) procedure for p-value calculation, where each past observation is weighted by the exponential of its distance from the current time point. -

$$p_{n+1} = \frac{\sum_{i=1}^n e^{-(n-i)} I\{MSE_{new} > MSE_i\}}{\sum_{i=1}^n e^{-(n-i)}}$$

which enables us focusing on the and forgetting the remote information in the previous batches and enforcing the procedure to focus on the most recent change of the data-generating processes.

Another aspect worth noting is that when the batch-size is large, the mean of the MSE along would no longer be sufficient - other modalities(e.g. variance, quantiles or distribution shapes) becomes highly desirable. This further underscores the importance of the weighting procedure for the p-value aggregations. In practice, one may want to select the batch-size appropriately - for large small batch sizes, the procedure is subject to high volatility where for the large batch-size the mean would not be capable of capturing the comprehensive information. Thus the quantile aggregation procedure is introduced to incorporated the information beyond the mean or median of the predictive error across this batch of data - $MSE_{agg} = \frac{1}{n_q} \sum_{i=1}^{n_q} w_i MSE_{q_i}$ where $q = (q_1, ..., q_{n_q})$ is corresponding to the varying

Algorithm 1 Online Detection of Significant Distribution Shift via Random Forest(**OnlinePermOOB**)

Input: Reference dataset $\mathcal{D}_{\text{ref}} = \{(\mathbf{X}_i, Y_i)\}_{i=1}^{n_{\text{ref}}}$; burn-in batches $\{\mathcal{D}_{\text{burn},j}\}_{j=1}^b$; incoming batches $\{\mathcal{D}_k\}_{k \geq 1}$, the batch size can vary from 1 to an integer.

Output: Sequential p-values $\{p_k\}_{k \geq 1}$ for shift detection.

- S1** Fit a predictive model(can be any kinds of model that subject to the types of the response) f_{ref} using $\mathcal{D}_{\text{ref}}$. Compute the validation error(out-of-bag error for random forest) and denote its mean by

$$E_{\text{ref}} := \mathbb{E}[\text{MSE}_{\text{ref}}].$$

- S2** For each burn-in batch $\mathcal{D}_{\text{burn},j}$, $j = 1, \dots, b$, compute the predictive error under f_{ref} and define

$$T_j := \mathbb{E}[\text{MSE}_j] - E_{\text{ref}}.$$

- S3** For each incoming batch $\mathcal{D}_k$, $k \geq 1$, evaluate

$$T_{b+k} := \mathbb{E}[\text{MSE}_{b+k}] - E_{\text{ref}}.$$

Define the empirical p-value

$$p_k = \frac{\#\{i = 1, \dots, b+k-1 : T_{b+k} \leq T_i\}}{b+k}.$$

- S4** Apply an online multiple-testing correction - e.g., SAFFRON([34]), ADDIS([49]), Alpha-Investing([11]) to $\{p_k\}$ to determine significance and detect distribution shift.
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quantile of the $(\text{MSE}_{b,1}, \dots, \text{MSE}_{b,n_b})$ with the flexibly tuned weights $w = (w_1, \dots, w_{n_q})$. In our implementation, 10 quantiles(ranges from 10% to 100% by step 10%) are aggregated for batch-size 20 and 5-quantile(ranges from 20% to 100% by step 20%) are adapted for batch-size 10, each with uniform weights $w = (1/n_q, \dots, 1/n_q)$.

The rest of the writeup is organized as follow. Section 3 discusses the theoretical guarantee of the proposed **OnlinePermOOB** procedure in terms of the proper False Alarm Rate Control. Section 4 discusses the simulation results and the False Alarm Rate control for the proposed procedure under stationary data-generating processes. Section 5 discusses the real-data applications and their applicabilities. Section 6 justifies the rationale of leveraging random forest as the component model in practice. Section 7 provide summaries and present future directions in this streaming testing procedure. Practical considerations in terms of the flexibilities of the procedure, computational complexity and production landing relevances are presented and carefully discussed in Appendix A.

3 Theoretical Guarantee

We hereby establish the statistical theoretical guarantee for this *OnlinePermOOB* procedure under online FDR control. Let $\mathcal{D}_{\text{ref}}$ denote the reference dataset and the RF_{ref} denote the random forest trained once on $\mathcal{D}_{\text{ref}}$. For each of the incoming batch $\mathcal{D}_t$ define the performance deviation statistic $T_t = \hat{\mathbb{E}}[\text{MSE}_t] - \hat{E}[\text{MSE}_{\text{oob}}]$ where $\hat{E}[\cdot]$ denote the empirical mean across this batch. We want to emphasize that in the streaming setting, batch-size $t = 1$. We consider the sequences of hypothesis

$$H_0^{(t)} : \text{No Distribution Shift at batch } t, H_a^{(t)} : \text{Significant Distribution Shift at batch } t$$

where the shift is characterized by a systematic increase in the predictive error. Define the filtration $\mathcal{F}_t = \sigma(\mathcal{D}_{\text{ref}}, \mathcal{D}_1, \dots, \mathcal{D}_t)$ represent all information available up to the batch t . An online multiple testing procedure with adaptive significant level $\{\alpha_t\}$ is then applied to $\{p_t\}$ producing the rejection results $R_t \in \{0, 1\}$. Define the rejection counts $R_T = \sum_{t=1}^T R_t$ and the false rejection counts $V_t = \sum_{t=1}^T R_t \mathbf{1}\{t \in \mathcal{H}_0\}$ where $\mathcal{H}_0$ denotes the set of true null indices. The false discovery rate(FDR) are $E[\frac{V_T}{\max(R_T, 1)}]$ and the modified FDR(mFDR) are $mFDR_T = \frac{E[V_T]}{E[\max(R_T, 1)]}$.

We present the main theorem here where the $mFDR$ is controlled and FDR is controlled up to a scaler with very small order.

- **Theorem 1** Under the following assumptions **A1**: The testing level α_t is $\mathcal{F}_{t-1}$ -measurable for all t , **A2**(conditional super-uniformity), for every null time $t \in \mathcal{H}_0$, $P(p_t \leq u | \mathcal{F}_{t-1}) \leq u$, $u \in [0, 1]$, **A3** Regularity conditions for SAF-

FRON/ADDIS are satisfied. Then with target level $\alpha \in (0, 1)$, for every horizon $T \geq 1$,

$$mFDR_T = \frac{E[V_T]}{E[\max(R_T, 1)]} \leq \alpha$$

First we verify the assumptions, for **A1**, the SAFFRON/ADDIS compute α_t using only information available prior to observing p_t : namely, past rejections $\{R_s\}_{s < t}$, past candidacy indicators (typically of the form $C_s = \mathbf{1}\{p_s \leq \lambda\}$ for a fixed $\lambda \in (0, 1)$ and fixed algorithmic parameters. Since $\{R_s, C_s\}_{s < t}$ are functions of $(p_1, \dots, p_{t-1})$ which are $\mathcal{F}_{t-1}$ -measurable, it follows that α_t is $\mathcal{F}_{t-1}$ -measurable.

For the conditional super-uniformity condition, we fix $t \in \mathcal{H}_0$ and condition on $\mathcal{F}_{t-1}$. Let $N_{t-1} := t - 1$ denote the number of past statistics. Define the (random) rank count $K_t := \sum_{i=1}^{t-1} \mathbf{1}\{T_t \leq T_i\} \in \{0, 1, \dots, N_{t-1}\}$ then the p-value can be written as $p_t = \frac{1+K_t}{t}$. Thus conditional on $\mathcal{F}_{t-1}$, T_t is a fresh draw from the same continuous conditional distribution as the past null statistics. Continuity implies that ties occur with conditional probability 0. Hence, conditional on $\mathcal{F}_{t-1}$, the value T_t is equally likely to fall in any of the t open intervals determined by the order statistics of $\{T_1, \dots, T_{t-1}\}$. Equivalently, the number of past values exceeding T_t is uniform:

$$K_t | \mathcal{F}_{t-1} \sim U[\{0, 1, \dots, t-1\}]$$

Therefore for any $u \in [0, 1]$, $P(p_t \leq u | \mathcal{F}_{t-1}) = P(\frac{1+K_t}{t} \leq u | \mathcal{F}_{t-1}) = P(K_t \leq tu - 1 | \mathcal{F}_{t-1})$ since K_t is discrete-uniform on $\{0, 1, \dots, t-1\}$, $P(K_t \leq tu - 1 | \mathcal{F}_{t-1}) \leq u$ where the inequality is due to the discretization. Thus the conditional super-uniformity is justified. Then the Theorem 3 is the direct application of theorem 1 in [10] which implies that **SAFFRON** and **ADDIS** control mFDR at level α without requiring independence or PRDS.

Then we establish the relationship between mFDR and FDR where we let $\mu_T = E[\max(R_T, 1)]$ then $FDR(T) - mFDR(T) = E[\frac{V_N}{\max(R_N, 1)}] - \frac{E[V_N]}{E[\max(R_N, 1)]}$ thus by the fact that the false discovery proportion is bounded between 0 and 1, $|FDR(T) - mFDR(T)| \leq \frac{1}{\mu} E[|\max(R_T, 1) - \mu|] \leq \frac{SD(\max(R_T, 1))}{\mu}$ so $FDR_T \leq \alpha + \frac{SE(\max(R_T, 1))}{E[\max(R_T, 1)]}$ by Jensen's inequality. So if we assume that $Var(R_T) \leq C E[R_T]$ for some constant $C > 0$ and $E[R_T] \rightarrow \infty$, then the relatively tight bound is established $FDR_T \leq \alpha + O(\frac{1}{\sqrt{E[R_T]}}) \sim \alpha + o_P(1)$

We want to emphasize that, for the EWMA (exponential weighted moving average) procedure, the conditional super-uniformity won't be satisfied here - however, as the downstream experiments show, it would enjoy early alerts generation, faster than the existing methods while maintaining the similar degree of the false alarm rate - thus, this weighted procedure would serve as an efficient improvement on top of the current methodology.

3.1 Asynchronous Testing Procedure

Another practical complication is the overlap between successive batches—e.g., $\mathcal{D}_1$ and $\mathcal{D}_2$ share observations - which induces dependence among p-values and thus invalidates the standard OnlineFDR procedure we discussed in the previous sections. We hereby leverage the asynchronous online multiple testing procedure proposed by [65] for the variants on top of the ADDIS([49]) and SAFFRON([34]) procedures. Specifically they proposed to leverage a running window with the size γ to ensure no overlaps exist between the running windows. Specifically, the SAFFRON*([34]) and ADDIS_async procedures are leveraged. As there are overlaps between the neighboring sliding windows, we adapt the original OnlineFDR procedure to an Asynchronous manner with a lag to ensure no overlap between the batch used for drawing the rejection decision and the current batch. The SAFFRON_async and ADDIS_async procedures are leveraged to conduct the onlineFDR control procedure, along with the fixed- α procedure. We provide a heuristic demonstration of the theoretical support here. The readily available FDR control for this SAFFRON_async is derived in [65] where as the independent p-values are required for each other and of the not nulls - specifically. The theorem 2 and 3 in [65] is leveraged here for the mFDR (marginal FDR) control with the super-uniformity condition is satisfied which has been proved in the previous subsections. The theorem is presented as below:

- **Theorem 2** Suppose the null p-values are super-uniform conditional on $\mathcal{F}^{-x^{E_t}}$, meaning $P(P_t \leq u | \mathcal{F}^{-x^{E_t}}) \leq u \forall u \in [0, 1]$ and $t \in \mathcal{H}^0$. Then both the ADDIS_async and SAFFRON_async with target FDR level α both guarantee that $mFDR(t) \leq \alpha$ for all $t \in N$.

However, similar to the synchronous procedure, because of the fact that the p-values are dependent across batches, the strict FDR control is not viable, a similar procedure to bound the FDR to $\alpha + o(1)$ can be achieved. However, we will showcase the proper FDR control for the algorithm under stationary data-generating processes.

4 Simulation & False Alarm Rate Control

In practice, we want to balance the false alarm rate and the timely detections. Thus we simulate the null-DGP where the data-generating process remains the same across continuously coming batches of data records. Specifically, the following three types of the data-generating processes are leveraged, under each circumstance, 10k observations are simulated. The first 2000 observations are served as the training set with the later observations as the reference evaluation set.

- Random Noise Data is generated where $\{\mathbf{X}_i, Y_i\}_{i=1}^n$ follows the joint normal distribution $(\mathbf{X}, Y) \sim N(0, \Sigma_{p+1})$ with the sample-sizes $n = 10000$, 30 features are included in the data-generating process.
- Linear Model is leveraged where $Y = \mathbf{X}\beta + \epsilon$ for the data-generating process where the coefficients β satisfies with the covariance matrix $\Sigma_{ij} = I\{|i - j| \leq 5\}\rho^{|i-j|}$.
- Nonlinear Data-generating process is leveraged with the Mars Model where $Y = 0.1 * \exp(4X_1) + \frac{4}{1 + \exp(-20 * (X_2 - 0.5))} + 3X_4 + 2X_4 + X_5$ with 25 standard normally distributed nuisance features simulated.

The false alarm rate for each of the setting is demonstrated for different models - with 100 simulations repeated and the false alarm rate is recorded as the proportion of the points that is wrongly identified as the change-point. As demonstrated in tables 2 and 3, the following insights can be extracted, firstly the **OnlineFDR** procedure is much more efficient than the Fixed α procedure in preventing the False-Alarm, secondly the EWMA procedure can slightly reduce the False Alarm Rate in comparison to the unweighted procedure.

We conduct additional sensitivity analyses on the burn-in rounds and noise levels. As shown in Figures 2 and 3, the detection efficiency is largely insensitive to the choice of burn-in length. Nevertheless, to mitigate excessive variability in early-stage p-values, a short burn-in period (e.g., 3-5 rounds) may be beneficial in stabilizing the procedure and preventing oscillatory p-value behavior from obscuring detection efficiency. We further assess sensitivity to the goodness-of-fit of the component model trained on the reference set $\mathcal{D}_{ref}$. As illustrated in Figure 3c, consistent with the static two-batch **PermOOB** setting, a poorly fitted random forest substantially reduces detection efficiency, leading to a lower detection rate. Thus in the real-data applications the burn-in period is set as 1 to enable the parsimonious procedure.

Another important aspect is the sensitivity of the hyperparameters in the onlineFDR procedure for ADDIS, SAFFRON and their asynchronous correspondence. For both of the ADDIS and SAFFRON procedures, the hyperparameters w -wealth(initial wealth), λ (adaptive threshold) and τ (candidate threshold) are tuned w.r.t. the grid in the Baseline procedure to minimize the False-Alarm Rate as well as enhancing the timely detection. In the false alarm rate evaluation sanity check, the linear DGP, Mars DGP and Random Noise DGPs are adapted. Thus the modified procedure here would be essentially helpful to help people achieve. Specifically, to achieve the best performance for these methodologies, we would primarily need this procedure to be robust across a variety of to minimize the False Alarm rate under stationary data generating processes. As demonstrated in Tables 44 and 43 the False Alarm Rate for both ADDIS and SAFFRON are fairly robust in terms of the hyperparameter settings, much more efficient than the Fixed α procedure. Thus, in practice, the default hyperparameter setting is sufficient to ensure the validity of the **OnlinePermOOB** procedure.

The false-alarm rate evaluation is conducted for the stationary data streams(with no change in the data-generating processes) under the following three component models: MLP, Random Forest, XGBoost; the following three data generating processes: random noise, linear and nonlinear; the following three p-value aggregation procedures: unweighted, EWMA(exponentially weighted moving average), QA(quantile aggregated for the mini-batch settings, the mean for the 10-90 quantile within each batch is adapted) and the following two batch-sizes including 1(streaming) and 10(mini-batch), with each of the setting repeated for 5 times to calculate the average False Alarm Rate in comparison with all of the benchmark methodologies. As demonstrated in Tables 32 and 34, the **ADDIS_async** and **SAFFRON_async** maintains very efficient False Alarm Rate control, especially in the unweighted regime without the exponentially weighted moving average procedure for the overlapped sliding windows.

To summarize the simulation results demonstrated highly robust False Alarm Rate control under a variety of types of data-generating processes with no distribution shift and the hyperparameter setting of the OnlineFDR procedure. Thus in practice, we would recommend practitioners to stick on the default hyperparameter settings whereas the procedure incorporates the very efficient & robust false alarm rate control and reduced detection delay in comparison to almost all the other methodologies.

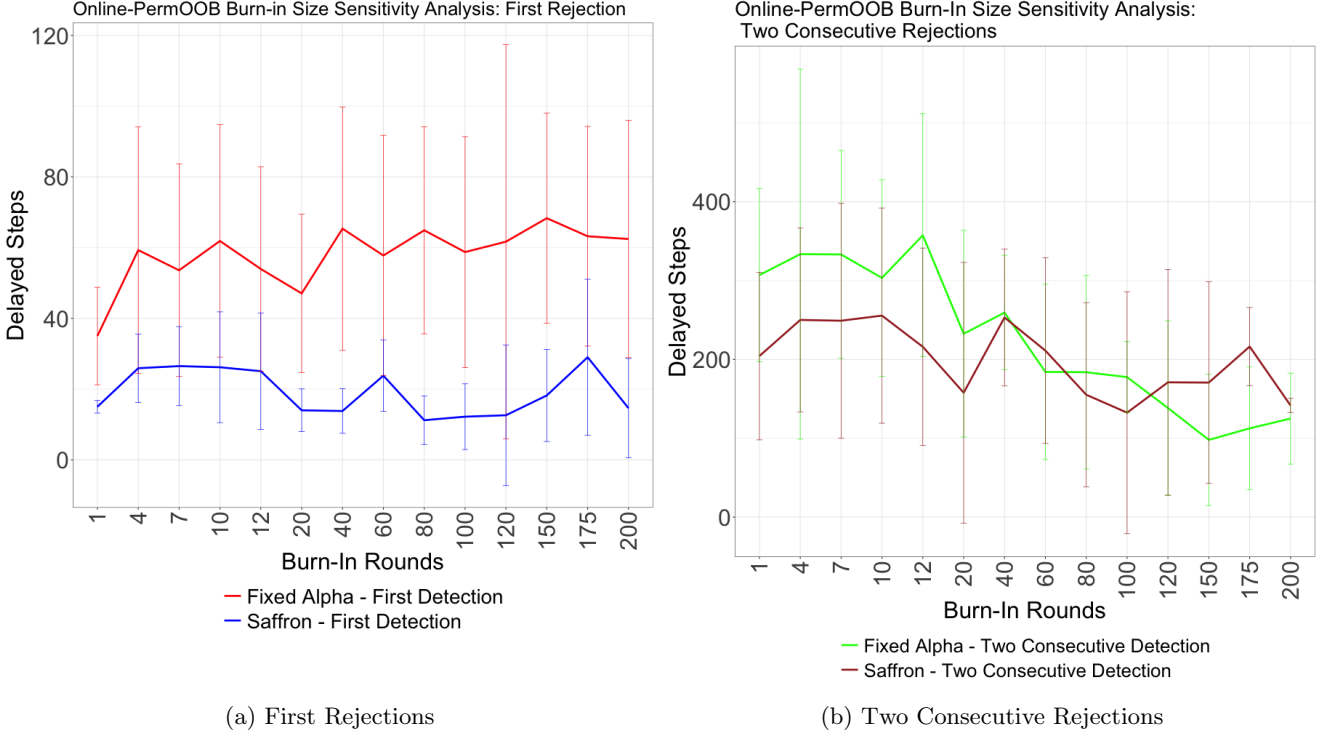


Figure 2: Sensitivity Analysis for the Burn-in Rounds w.r.t. the detection delayed steps, significance level $\alpha = 0.05$

5 Real-Data Applications

We list a variety of the real-world datasets to apply our testing procedures. We first briefly state the settings of the datasets and give a brief introduction for the data pre-processing procedures. The datasets we leveraged and their corresponding feature extraction methods are presented in Table 1. We first introduce the benchmark methods we leverage to evaluate against our proposed algorithm.

5.1 Benchmark Methods

We compare the proposed **OnlinePermOOB** method with a wide variety of existing statistical methodologies ranging from online anomaly detection methods, online change point detection methods and martingale based procedures. In this subsection we present the list of the algorithms we leverage to compare against our proposed OnlinePermOOB algorithm. Several online bayesian change-point detection (BOCPD) methods ([17, 9]) are benchmarked via the *Bayesian Online Change Point Detection* Python package. The benchmark classical sequential monitoring procedures such as the Page-Hinkley test ([26]) and the adaptive windowing algorithm *ADWIN* ([4]) – with readily available **Python** packages. Additionally also incorporate martingale-based e-value detectors as recent alternatives for sequential hypothesis testing using two calibrations of the martingales $e_t = \frac{2}{3}p^{-1/3}$ (Martingale 1) and $e_t = \frac{4}{5}p^{-0.2}$ (Martingale 2) ([58, 33]). In addition, we benchmark widely used concept-drift detection algorithms including STPED (Online Kolmogorov–Smirnov test, [25]), ECDDWT (Exponential Weighted Moving Average drift detection, [36]), and HDDMA (Hoeffding drift detection with A-test, [12]), as well as representative online anomaly detection methods including Half-Space Trees ([45]), Online Maximum Mean Discrepancy ([60]) and FITNESS(Fine-Tune on New and Similar Samples, [37]). We record both the first significant time point and the first occurrence of one, two and three consecutive significant time points. We evaluate the *average detection delay*(in number of observations after the reference period) of the proposed method against a set of benchmarks from traditional mean shift detection, streaming drift detection, to online anomaly detection methods.

- **[1] NYC Yellow Taxi Trip Data (2023).** We analyze the publicly available 2023 Yellow Taxi Trip Data released by the New York City Taxi and Limousine Commission (TLC; [41]). This large-scale operational dataset consists of trip-level records submitted by licensed yellow taxi Technology Service Providers, where each observation corresponds to a completed taxi ride in New York City. The dataset contains detailed temporal, spatial, and financial attributes, including pickup and drop-off timestamps, trip distances, passenger counts, payment types, and itemized fare components.

The data reflect complex urban mobility dynamics shaped by commuting cycles, traffic congestion, seasonal variation, and demand fluctuations, thereby providing a natural testbed for studying distribution shift in high-volume, real-world systems. Predictive targets are defined based on fare or revenue-related quantities, while

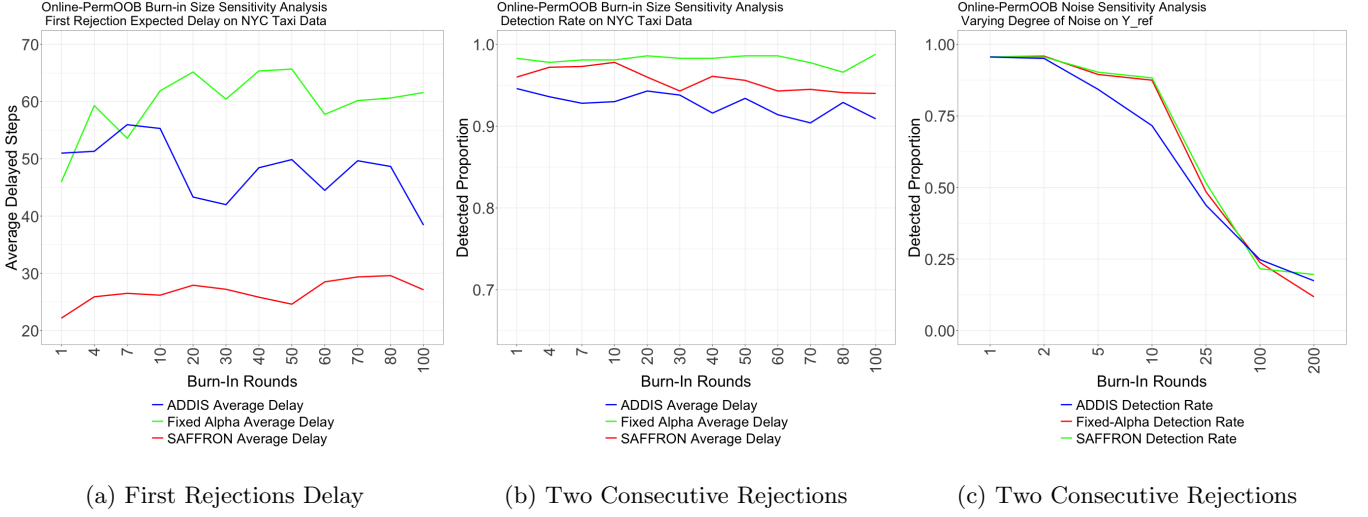


Figure 3: Sensitivity Analysis for the Burn-in Rounds w.r.t. the detection delayed steps, significance level $\alpha = 0.05$, for the NYC taxi data

temporal and contextual variables serve as covariates. The large scale and inherent nonstationarity of this dataset make it particularly suitable for evaluating both batch and streaming detection procedures. The results are presented in Tables 27, 26, 24, 25 and 4 & 5. As demonstrated, the ADDIS/SAFFRON is highly efficient in terms of both timely detection in both nonweighted procedure and the EWMA scenarios.

- **[2] UCI Air Quality Dataset.** We employ the Air Quality dataset from the UCI Machine Learning Repository ([52]) to evaluate the online change-point detection procedure in an environmental monitoring context. The dataset contains hourly measurements from gas sensor devices alongside reference gas concentration levels and meteorological variables.

In our setup, the response variable is $NO_2(GT)$, representing the ground-truth concentration of nitrogen dioxide. The covariates include sensor readings and environmental measurements:

$$[PT08.S1(CO), PT08.S2(NMHC), PT08.S3(NO_x), PT08.S4(NO_2), PT08.S5(O_3), T, RH, AH].$$

Each day is treated as a batch, and the first 60 days are designated as the reference set. This configuration reflects a realistic monitoring scenario in which predictive models must adapt to evolving atmospheric conditions and sensor drift. As demonstrated in Tables 20 and 21, the OnlineFDR control procedure is highly efficient in terms of timely detection in comparison to the fixed α procedure. More importantly, the EWMA procedure is highly competitive in this dataset in comparison to the unweighted procedures.

- **[3] Stock Emotion Dataset.** We utilize the Stock Emotion dataset ([18]) to examine distribution shift in financial forecasting driven by textual sentiment. The dataset contains approximately 15,000 daily observations linking sentiment signals extracted from textual sources to subsequent stock returns.

Let X_t denote sentiment-derived features at time t and Y_{t+1} denote the next-period return, defined as

$$\text{return} = \frac{\text{Closing}_{t+1} - \text{Closing}_t}{\text{Closing}_t}.$$

Textual information is transformed into structured covariates using TF-IDF vectorization. In the batch setting, we adopt a batch size of 50 observations. For streaming analysis, we examine robustness across different initial-ization points by allowing monitoring to begin at multiple calendar anchors (e.g., the start of each month). As in the taxi application, early detection of predictive degradation is critical for timely model updates in financial decision systems. As demonstrated in Tables 18 and 19, the onlineFDR procedure would provide very timely detection - especially for ADDIS and SAFFRON. Moreover, as demonstrated in 19, the EWMA procedure would further enhance the detection efficiency.

- **[4] Interstate Traffic Volume Dataset.** We analyze the Traffic Volume dataset ([14]), which records hourly interstate vehicle counts from October 2, 2012 to September 30, 2018, with the goal of predicting traffic flow one hour ahead. To incorporate contextual and temporal information, we construct sentence-level embeddings using Sentence Transformer models ([7]) and reduce dimensionality via the first ten principal components. The covariate set further includes meteorological variables—*temperature*, *snow_1h*, *rain_1h*, and *clouds_all*. To evaluate distributional changes, we initialize candidate monitoring points at indices 0 and 10000-th observation and

apply the OnlinePermOOB procedure sequentially, using the first 2,000 observations as the reference set. This dataset provides a dynamic transportation forecasting environment characterized by seasonal effects, infrastructure changes, and evolving traffic patterns. As demonstrated in Tables 22 and 23 the OnlineFDR Control method is highly efficient. However, the EWMA procedure didn't work out for this dataset.

- **[5] Walmart Benchmark Dataset** [38] We leverage the Walmart M5 demand forecasting dataset for conducting the gradual shift testing on the subset across each of the item-id. The dataset contains the sales/demand statistic in 300 stores across the 10 states with daily-level observations recorded between 2011 to 2016. Thus the sales statistics in the past period of time would be highly predictive on the current price. Bearing this in mind, we build the features to capture the temporal dependence(how many orders in the past 7/14/28 days) manually. The online extension of the permutation test is adapted to understand the time-point where the gradual distribution shift starts to become significant. As demonstrated in Tables 7, 6 and 8, the ADDIS and SAFFRON based procedure are among one of the most timely detected methods to detect the significant change-point of the data-generating processes. Additionally, the exponentially weighted procedure would be fairly helpful in enhancing the timely detection of the significant distribution shift in comparison to the unweighted procedure.
- **[6] Image Data - Online Extension** Building upon these batch-level evaluations, we adapted the mds-A dataset ([24]) to assess performance under gradual distribution shifts. We utilized high-dimensional embeddings extracted via advanced architectures, specifically Vision Transformer ([31]) and ConvXNet ([20]). As demonstrated in Table 16, the onlineFDR procedure is highly efficient at detecting significant changes in the data-generating process in a timely manner. Gradual shifts are notoriously difficult to isolate due to their insidious nature, yet the online procedure successfully maintains strict control over the false discovery rate while remaining highly sensitive to temporal drift dynamics.
- **[7] Motion Planning Data** To demonstrate the practical utility of the proposed streaming testing procedure in real-world applications, we adapt the Waymo Open Dataset([42]) via the Autonomous Vehicle Driving behaviors modeling. The dataset provides detailed time-indexed records with a 0.1 second interval, including positions, heading directions, acceleration statistics for the car - providing granular characterization of the driving behavior for the AV(Autonomous Vehicles). A rich set of the features are developed to predict the speed in the next time-point including the **jerk** (derivative of acceleration rate), average speed over rolling windows (1-minute, 3-minute, and 10-minute horizons), and directional kinematics such as **yaw-rate** and **aspect ratio** (turning angle characterization at each position). Additional features include the kinetic energy and frontal areas(width times the height for the scan). By construction, both gradual concept drift and covariate shift exist except for abrupt change of the driving patterns(bump or sudden changes forced hard brakes).

The streaming testing procedures are adapted for evaluating how timely the detection procedure can perform in the gradual shift of the driving patterns. As demonstrated in Table 35, 37 and 38 the ADDIS procedure equipped by the Online FDR control is among the most efficient procedure for detecting the time point where gradual shift become significant in comparison to almost all of the benchmark methods both in terms of the total detection rate and the time-to-first-detection. As shown in Tables 35 and 36, the online FDR control procedures—particularly ADDIS and SAFFRON—remain among the most effective approaches for balancing timely detection with false alarm rate control in the Waymo point-cloud scenario. Notably, when paired with the EWMA-based detection procedure, ADDIS further enhances timeliness, underscoring its practical advantage in streaming autonomous driving applications.

- **[8] Point Cloud Data** Point cloud data, characterized by the high density, millimeter-level geometric precision and rich three-dimensional spatial information, has become a modality in a lot of real-world applications including wireless driving, robotic navigation and digital twins. However, unlike static images or structured data, point clouds captured by the streaming sensors or LiDARs with generating multi-dimensional and unstructured spatial-temporal data streams. Ever-evolving distribution shift exists along the time-points with environmental changes as well as the gradual shifts in terms of the sensor shifts. For the empirical evaluation, we employ the Toronto Point Cloud dataset [46]. To effectively characterize the streaming data, we extract a comprehensive set of features, including historical positional trajectories over the time windows, inter-frame angular deviations capturing relative shifts and the local spatial point densities reflecting the geometric distribution of the captured scenarios. The streaming testing procedures are adapted for evaluating the timely detection of the gradual shifts. As demonstrated in Tables 39, 40 and 42, the ADDIS procedure is much more efficient in comparison to existing methods both in terms of the detection delay and the occurrence of detection which further demonstrated the marvelous degree of practicability for the streaming testing procedure.
- **[9] Moving Trajectory Data** To enhance the capability of the Statistical Methodology, the experiments are conducted on the Third Generation Simulation Data(TGSIM, [44]) I-395 trajectories collected from the trajectories on the freeways of non-automated passenger cars, trucks, buses, and automated vehicles on an expressway within an urban environment. Predictive modeling is conducted leveraging the speed, (x,y) positions as well as

the smoothed width,heights as covariate with the acceleration variable as the response. The dataset is split into 285 batches based on the time-order with each batch containing 15000 observations. Random Forest, XGBoost and MLP models are adapted for the streaming change-point detection procedure. As demonstrated in Tables 47, 45 and 46, the **OnlinePermOOB** algorithm equipped with **ADDIS** procedure consistently outperform the in terms of both detection delay and number of detections in comparison to most of existing benchmark methods.

- [10] **Audio Data** The audio data is another scenario to showcase the power of the developed statistical methodology - with both gradual shift and abrupt shift exists. The data-generating processes for the audio data is ever-evolving, unfolding across multiple temporal scales and semantic layers simultaneously. At the lowest physical level, the raw acoustic waveforms evolve continuously at the microsecond scale, capturing the transient fluctuation in terms of both amplitude and the frequency. Thus the timely detection procedure for the change-point is highly desired especially under the application scenario of audio segmentation([39]). Specifically, we leverage the [32] dataset with the CLAP hstat([56]) and SPEAR semi-supervised learning embeddings([59]) for the representation extraction in this audio data. The sliding window is set as 10 seconds with the step size set as 2 seconds(with 5 steps lag in the asynchronous online multiple testing procedures) and the sampling rate here is set as 0.25 with the interval specified targetedly. The asynchronous testing procedure via both ADDIS and SAFFRON are leveraged to benchmark with existing methods. As demonstrated in Tables 13 and 15, the **OnlinePermOOB** algorithm equipped with **ADDIS** procedure consistently outperform the in terms of both detection delay and number of detections in comparison to most of existing benchmark methods.
- [11] **Video Data** We employ the video datasets including MSR-VTT([57], relatively shorter videos) and LV-bench datasets([54], extremely long videos). In the recent years, numerous methods have been developed to extract the representation of video data ranging from foundational model, image transferring models and the image-video joint embedding extraction methods([62], [55], [61]). Among all of the information extraction algorithms, we leverage the most recently popular frameworks for video representation extraction - Video Vision Transformer(ViViT, [1]) equipped with the spatial-temporal attention mechanisms on the time scale as well as the frame level for representing both temporal evolution as well as the spatial information in the video datasets. The 768-dimensional vector for each of the sliding window is extracted with some overlaps between the neighboring ones. We want to emphasize that the sizes of the overlaps between the neighboring sliding are no larger than the one third of the size of the individual sliding window, making them proper for the asynchronous online multiple testing procedure. Taking one of the illustrative example as demonstrated in Figure 4, the nearly 2 hours video of a soccer game encompass continuous shifts across different frames with the time interval for every 0.1 second. Thus the timely detection of the significant point of distribution shift is highly desired in the streaming video information through the online distribution shift detection algorithms. The hyperparameter lag is set as 2 for the asynchronous online multiple testing procedures. The asynchronous testing procedure via both ADDIS and SAFFRON are leveraged to benchmark with existing methods. As demonstrated in Tables 31, the **OnlinePermOOB** algorithm equipped with **ADDIS** procedure consistently outperform the in terms of both detection delay and number of detections in comparison to most of existing benchmark methods.

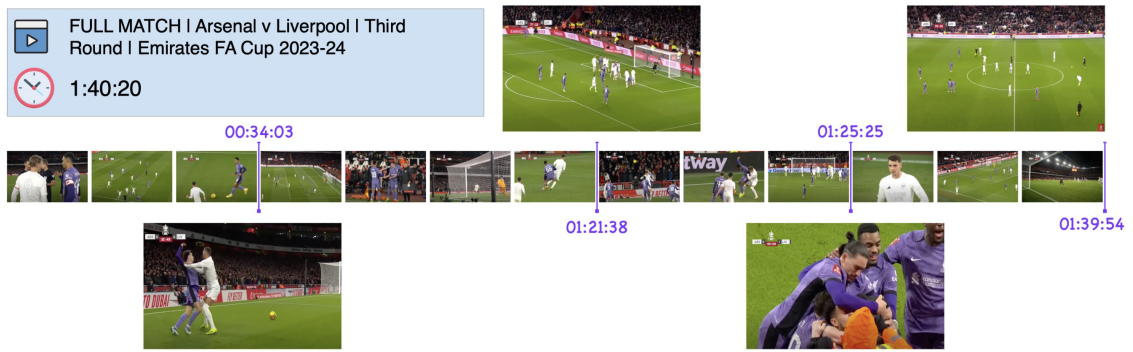


Figure 4: Illustration for one of the examples of Video Benchmark Datasets with selected frames presented

5.2 Summarization of Results

Our proposed **OnlinePermOOB** procedure is highly competitive and ranks among one of the most efficient one in more than half the evaluation scenarios. To summarize the main findings, (1) The EWMA based procedure outperforms the unweighted procedure slightly in most of the datasets. (2) Among all of the benchmark comparison methodologies, the BOCPD(bayesian online change-point detection) and ECDDWT are among one of the most efficient procedure in balancing the timely detection and proper false alarm rate control. (3) The **OnlinePermOOB** procedure equipped

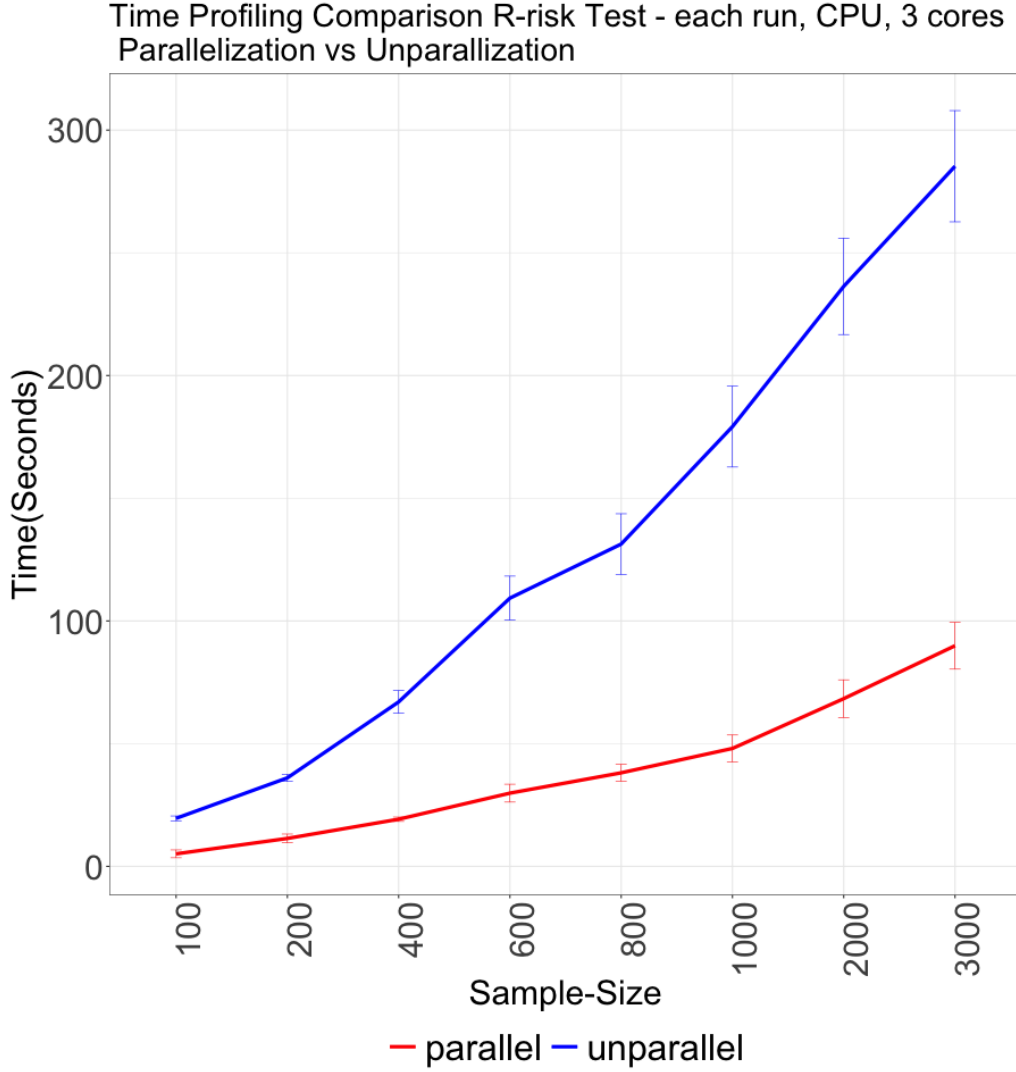


Figure 5: Time Complexitiy Comparison between the Parallel implementation and the Original Procedure

with ADDIS & SAFFRON is highly competitive in comparison with the fixed α procedure in the detection delay. Considering the much higher false alarm rate introduced by the fixed α procedure, the online FDR control is preferred in practice. (4) In the asynchronous testing procedure, the **OnlinePermOOB** procedure equipped with the **ADDIS** procedure is among the most efficient in comparison to almost all of the existing methods, thus highly preferred in practice (5) In most of the circumstances, although the QA(quantile aggregation) procedure won't give significant advantage under moderately large batch-size(30 or 50), we want to emphasize that it would be suitable for ultra-large individual batches with more than 10k. Combining the results above, we demonstrate the practical efficiency of our proposed **OnlinePermOOB** procedure relative to existing methods, rendering it a concrete Statistical Methodology for detecting significant distribution shift in the deployed model $\mathcal{D}_{ref}$.

6 Out-of-Distribution and the Model Choice

Although enormous flexibilities lies in the streaming testing procedure, we do want to emphasize that the random forest model is among one of the most robust(as demonstrated across many challenging scenarios and data-generating processes in many previous studies and existing literature including [21](robustness under low signal-to-noise rate, SNR), [64](robust under shallow tree_depth), [19](a proxy as bias reduction), [29](robustness under high-dimensional data), [47](robustness under imputed random missing data) and [3](consistency for classification rate). Thus as long as the component random forest model fits well, the robustness can be achieved under a lot of circumstance. Our empirical results further indicate that alternative methods including XGBoost and MLP, won't yield superior detection efficiency. Moreover, the MLP model imposes a substantially higher computational cost, making it much less practical in this framework. We further emphasize that, as demonstrated in [27] and [28], random forest exhibits significantly lower tunability in comparison to the parallel machine learning algorithms. It implies that the default hyperparameter

setting can be directly adopted in the streaming setting (i.e. our **onlinePermOOB** procedure). Therefore, when implementing our proposed framework in real-data, the most parsimonious and light-weighted setting - employing only one random forest model with its default hyperparameter setting - is recommended. Such a design yields substantial gains both in terms of practicability and computational efficiency for the **onlinePermOOB** algorithm.

From another perspective, compared with incrementally training procedures based on the deep neural networks - whose online optimization procedure in the post-reference stage often introduce substantial bias in the existing model. Thus in practice, masking the effect of out-of-distribution itself is essential where the robustness of random forest can help - when employed as the component model in the reference batch, can isolate the effect of distribution shift. This advantage stems from its exceptional robustness across a broad range of the challenging scenarios, as extensively documented in existing empirical studies and real-world applications.

7 Discussions

In this work we develop a highly flexible, computationally efficient and generalizable procedure for detecting the distribution shift in the ever-evolving data batches or streams. This procedure is well suited for gradual distributional shifts and tell people "When and why the distribution shift becomes significant that the model update is required". High degree of robustness of the procedure under hyperparameter settings is demonstrated and the False Alarm Rate Control has been justified via simulations. Real-Data Applications in a wide variety of types of datasets demonstrated its efficiency for the timely detection of the significant distribution shift. In this section we provide careful discussions about the algorithm.

Most of the existing procedures/approaches for detecting the distribution shift in the continuously coming datasets primarily focus on identifying changes in the covariate distribution without capable of characterizing the change of the conditional dependence of the response Y given covariate $\mathbf{X}$. *In contrast, the proposed OnlinePermOOB framework detects distributional change through monitoring predictive performance degradation.* Compared to traditional change-point detection methods, OnlinePermOOB can offer improved efficiency in high-dimensional settings, as the random forest reference model provides stable performance estimates even in the presence of noisy or weakly informative covariates. Moreover, the proposed method exhibits robust efficiency across batch sizes varying from 1 to 20 and 30, consistently outperforms nearly all existing online change-point detection benchmarks which are only fitted on streaming data settings.

The online multiple testing procedure is essential to the success of this algorithm. In comparison to the fixed α procedure, the onlineFDR control (particularly **ADDIS** procedure) version demonstrates substantially improved efficiency in preventing false alarm rate. More importantly the EWMA (exponentially weighted moving average) procedure would be highly competitive in comparison to the unweighted procedure while still be capable of preserving the false alarm rate. While the *OnlinePermOOB* framework is particularly adept at identifying gradual distribution shifts, its application in real-world environments necessitates a nuanced understanding of sequential testing trade-offs. The fixed α procedure may fail to provide reliable false alarm control in practice for sequential monitoring scenarios. Although a minor theoretical gap between the control of mFDR and FDR exists, empirical results demonstrates the effective false alarm rate control under stationary data-generating processes. Empirical results also justified the hyperparameter tuning agnostic nature of the **OnlinePermOOB** procedure in terms of the False Alarm Rate control, although in practice one may want to fine-tune the SAFFRON/ADDIS procedure to achieve the optimal detection performance on top of the status quo. Another aspect worth mentioning is that, in contrast to conventional methodologies that prioritize the detection of shifts in covariate distributions—often a computationally fraught task in high-dimensional spaces—the *OnlinePermOOB* framework shifts the focus toward monitoring predictive performance degradation. This paradigm offers superior efficiency in high-dimensional settings, as the Random Forest reference model provides stable performance estimates even when the feature space is populated by noisy or weakly informative covariates.

Moreover, our **OnlinePermOOB** algorithm is model-agnostic and evaluation-metrics agnostic and can be applied to a wide variety of types of data. The only thing you will need is the concrete version of the evaluation metrics over the sliding windows with the predictive model if our primary goal is to identify the time-point where the distribution shift starts to become significant in the ever-evolving data-generating processes - enabling this procedure a highly efficient, generalizable and flexible Statistical Methodology.

In the practical deployment settings, the *OnlinePermOOB* framework demonstrates notable robustness across a wide range of noise levels and drift regimes. This robustness arises primarily from its performance-centric formulation, which evaluates distributional change through predictive degradation rather than direct comparison of covariate distributions. The rank/quantile based construction of the empirical p-values further enhances stability by mitigating sensitivity to scale, heteroskedasticity, and heavy-tailed noise. Within this framework, the use of random forests serves as a complementary stabilizing component: the inherent resilience of random forests to local perturbations and model

mis-specification enables reliable out-of-bag error estimation under evolving data-generating processes. As a result, **OnlinePermOOB** provides a practically effective mechanism for detecting performance-relevant distribution shift in streaming environments, while maintaining consistent detection behavior even in the presence of gradual or nonlinear drift - no matter whether there will be overlap between the neighboring batches of data observations.

Another significant advantage is the proposed method’s low computational cost relative to almost all of the existing online distribution shift detection methods: it only requires one well-fitted model with the very light computation cost in the downstream online testing procedures. Thus from the computational aspect, the streaming extension of the testing procedure would not be a bottleneck as long as the real-time inference is enabled. Additionally we want to mention again that the parallelization procedure can significantly enhance the computational scalability by 70% as demonstrated in Figure 5 where in the production environment, the batch-wise testing procedure can be efficiently conducted every 20-30 minutes which will further enhance the applicability of the procedure. We refer readers to Appendix A for more detailed discussions.

A potential limitation for the proposed procedure lies in the fact that it may require complete refresh/restart after a significant distribution shift is detected where the new reference model should be fit and the empirical p-values should be re-calculated for the downstream update of the distribution shift detection procedure.

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A Production Considerations

The proposed procedure enjoys enormous flexibilities include but not limited to the component model choice, on-line anomaly detection method choice as well as the feature extraction method. As demonstrated in Figure 6, the modality-dependent feature extraction method enjoys highly flexibility even in the same domain - taking the video data for example, the Video Glue([61]), ViViT([1]) and the VideoLlama([62]) are those available for the representation extractions. Similarly, various pre-trained embeddings can be leveraged to deal with text data including Qwen([63]), Gemma([51]) and Llama3([2]) and among a lot of different versions. We want to emphasize that in the current state-of-art landscape, a comprehensive pre-trained feature extractor would be essential for not only the robustness of the

model fitting but also important for the efficient & valid testing procedure - both of which are well supported by our proposed framework.

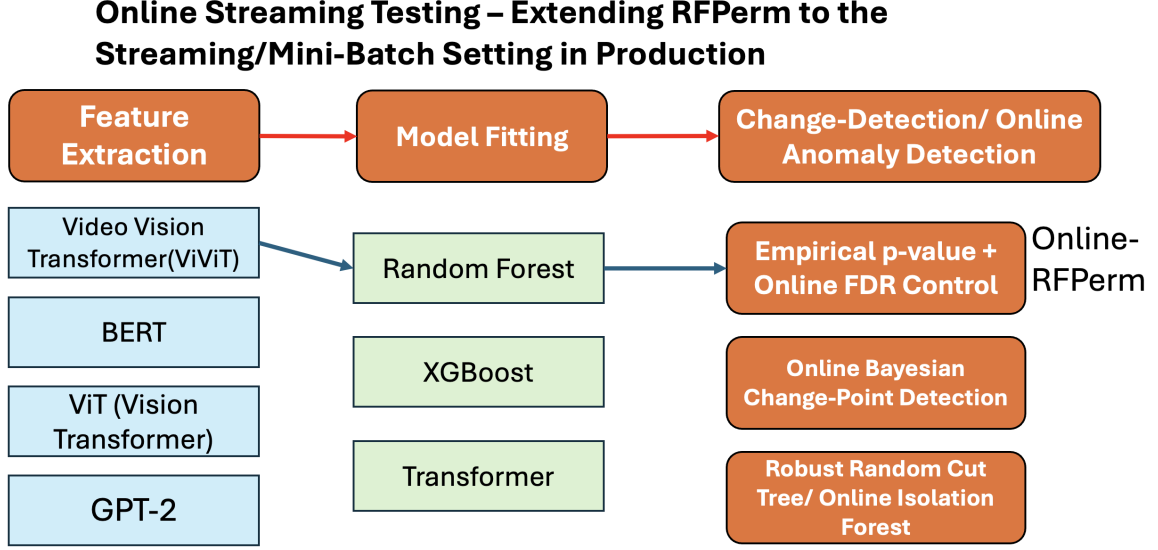


Figure 6: Illustration for the overall pipeline of the streaming testing procedure

One of the fundamental challenges for the production deployed machine learning model is **distribution shift evolves continuously over time**. It may come from behavioral, seasonal or the user mixture shift, a model that performed well when pushed into the production environment will probably degrade after that. Traditional monitoring procedures including tracking the accuracy, F1-score or the loss function will give people an idea of the performance is dropping but won't be able to tell people when the data shift is becoming significantly enough that required to retrain the model.

The **OnlinePermOOB** procedure addresses this gap with an online hypothesis testing framework that continuously monitors the degradation and shift of model performance for a deployed model. For each incoming batch or observation, an empirical p-value-based procedure is employed by comparing the current batch error against the existing pool of prior errors. An online FDR procedure incorporating **ADDIS** and **SAFFRON**, along with their asynchronous extensions, is adapted under a significance level of $\alpha = 0.05$.

A.1 Model Selection

The choice of the component model - the predictive model whose performance degradation is continuously monitored - will directly influence both detection sensitivity and computational burden in production. In the system, the model registry supports Random Forest, XGBoost, MLP, and Transformer-based architectures, allowing the practitioner to select the most appropriate model for the specific data modality and operational constraints. For the tabular data with moderate dimensionality, random forest or XGBoost is what people may want to pursue - especially random forest with its resilience and robustness to hyperparameter tuning, build-in OOB error evaluators and the near-real-time inference cost enables it to be the best choice. On the other hand, for data with other modalities like text, image or video with text captions, the embedding can be extracted first with the corresponding architecture - for instance, vision transformer([31]) for images and video vision transformer([1]) for video data operates on top of the extracted high-level representations. This flexibility preserves the modality-agnostic nature of the **OnlinePermOOB** framework ensuring the component model remains fit properly. The model registry with only requiring **fit** and **predict** nature enables seamless swapping of component models without altering the core detection logic, allowing people to adapt to evolving production requirements with minimal engineering effort. Specifically, we want to emphasize that the random forest model does play its role in isolating the effect of out-of-distribution itself, without requiring repeatedly training with very light computational cost.

A.2 Online Extension Procedure

In practical deployment settings, the procedure is highly efficient in terms of computation. The only requirement is the inference for each incoming observation. We adapt component models ranging from Random Forest and XGBoost to MLP. For tree-based models, near-real-time inference in the streaming setting is feasible even on CPU, with

latencies below 0.2 seconds — the p-value computation itself is even faster given straightforward implementation. A key advantage of leveraging Random Forest as the component model is not only its robustness in prediction and under high-dimensional covariates, but also its ability to achieve competitive performance with a less complex model structure (with $mtry = \sqrt{p}$ for classification and $mtry = p/3$ for regression settings).

Critically, upon detecting a significant shift and subsequently retraining the reference model, the testing cycle must undergo a **full state reset**. This reset clears the historical MSE pool, reinitializes the burn-in period, and resets the online FDR algorithm’s testing wealth. This step is mandatory to restore the exchangeability assumption underlying the permutation p-values and to ensure that false discovery rate control remains theoretically valid under the new reference model.

A.3 Static Batch Testing

In practical deployment settings, the various versions of tests under a permute-then-refit procedure can be significantly accelerated through parallel computing using **Joblib** in Python or the **foreach** and **parallel** packages in R. As demonstrated in Figure 5, a 70% computational cost saving can be achieved using 4 out of 8 CPU cores. The speedup proportion follows Amdahl’s law, given by $Speedup = \frac{1}{(1-p) + \frac{p}{N}}$, where the parallelizable proportion of the program is approximately 80% under the permutation procedure. In a production deployment environment, at least 8 physical CPU cores, 32GB of RAM, and 500GB of SSD storage are recommended. The 8-core minimum ensures that the parallel permutation computation achieves a meaningful speedup while leaving capacity for other production workloads. The 32GB memory budget accommodates the component model storage, the batch data, and the worker processes. Storage is primarily allocated for historical data and model checkpoints, with 500GB recommended to provide sufficient headroom.

A.4 Deployment Workflow & State Transitions

The whole running process is linear but includes a critical branching patch for the resets - under the initial stage of the deployment, the system loads the reference model and dataset, computes E_{ref} and enters the burn-in phase. During the burn-in stage the next 5 (by default) batches are consumed solely for the data collection with no decision made and no p-value collected. Once the burn-in stage is finished. Once the burn-in cycle is completed, the system switches in the full decision making mode. For each of the incoming batch of data (either stream or batch), it computes T statistic, calculating the empirical p-value, adapts the decision through ADDIS or SAFFRON and updates its internal state. If the decision is "not reject", the batch’s T is added to the history of the values and the system continues to the next batch/stream. If the decision is "reject", the system triggers the retraining pipeline, waits for confirmation that the new model is deployed and then executes the full state reset procedure into the next cycle.

A.5 Reliability

If one individual batch processing cycle exceeds 30 minutes, the system must gracefully degrade - skip the detection for that batch, log the timeout event and reuse the most recently valid p-value. It efficiently prevent the monitoring pipeline from stalling due to exhausted resources contention or abnormally large data-sizes.

Failure of workers including out-of-memory errors, must be handled by restarting the affected worker with the reduced permutation count. A fallback n_{perm} of 100 is recommended. The system would be capable of recovering without manual intervention. All of the reference data, model artifacts with the checkpoint states must be persisted to durable storage (S3 or equivalent) with versioning enabled. This ensures that the rollback to an older version of model or state is always possible.

A.6 Maintainability

This system would be capable of being integrated with a model registry supporting various types of the built-in model types (Random Forests, XGBoost, MLP and Transformers) with customer adapters. This can be achieved through a consistent interface requiring only a fit and predict method. All of the hyperparameters, ADDIS threshold for discarding, burn-in sizes and the number of permutations as well as the parallelization settings - should be configurable through environmental variables or a central configuration file without requiring code changes. All logs must be emitted in JSON format and the key metrics including p-values, α_t , rejection flags and the latency, must be exposed to real-time monitoring dashboards.

A.7 Success Metrics and Risks

This monitoring system is evaluated against the following primary metrics. First, detection delay is defined as the average number of batches from the actual onset of a significant distribution shift to the first rejection. The false

alarm rate — the proportion of rejections under no shift — should be controlled at or below the target FDR of 5%. Computation time is monitored as the P95 batch processing duration, which must remain under 25–30 minutes. Additionally, reset correctness is verified through logs, ensuring that all internal states are cleared after every retraining cycle without exception.

Several risks are potentially identified, if the retraining operation failed, the subsequent p-value procedure would become invalid. The mitigation procedure would be increasing the burn-in periods to a slightly larger size as well as adapting the hyperparameter of the online FDR procedure in **ADDIS** and **SAFFRON**. On the other hand, if the system experience too many discards(p value lies above the discarding threshold λ), it will lose the ability to reject. A periodic wealth reset after cycles would be desirable for ensuring the robustness of the system preserving the capability of making rejections.

B Applicability across domains

In the experiments, we apply the developed hypothesis testing procedures on a wide variety of types of datasets ranging from tabular data, image data, text data, motion planning data to video data as well as the audio data - demonstrated their efficiency for both of the batch testing procedure and the streaming testing procedure - making the developed distribution shift testing procedure generalizable to different domains as long as the well-leveraged representation(ranging from image embedding, text embedding to audio embeddings) are derived from a comprehensively pre-trained Information Extraction Model(either through LLM or self-supervised training modules).

C Real-Data Application Results

We present the simulation results(False Alarm Rate Control) and the selected real-data application results here for each of the dataset we have leveraged to benchmark in this section. Table 1 presents the datasets, feature extraction methods and the modality of the datasets. The feature extraction pipelines and notebooks are presented below for the feature extraction and data processing procedures.

- Audio Embedding Extraction:
https://colab.research.google.com/drive/1L40ck3VwhmU1_0oiQhhtUak_KGMPTohV#scrollTo=Wn6M2NBC2rUa.
- Image Embedding Extraction:
<https://colab.research.google.com/drive/1MH0lZRno8bQetxjMNDeoRn31sI5tipj4#scrollTo=RXHhe3KUhtTht>
- ChronoBerg Data Manipulation Scripts:
<https://colab.research.google.com/drive/1-gvoK9pNUK3EPuS1eA9jqV1KYdCGg1zH#scrollTo=iWRIS61JAK10>
- Text Embedding Extraction
<https://colab.research.google.com/drive/1lYQFzql9wtXdpDvEh2QQ7q7shlFMjCv>
- The Walmart M5 reformulated data after Feature Engineering across each of the shop.
<https://drive.google.com/file/d/1F0gYdNNTYN4Ztj8HdR2KI9t7G1hixHad/view?usp=sharing>
- Video Embedding Extraction:
https://colab.research.google.com/drive/1InFg_qWdiS0aGMRVSy00oB5uN92n3sSd#scrollTo=hhyJ_TWvqlK6
- Waymo Moving Trajectory Data:
<https://colab.research.google.com/drive/168X4qmnCG9hS4WKkaBI7q-j44pLZv-M>
- Point Cloud Extraction Pipeline:
<https://colab.research.google.com/drive/1R-NvLj-HwKHvBwKcycVRPZtaYXj0XbeH#scrollTo=1LI62GoYS2QJ>

Table 1: Dataset Evaluation Procedure for the Testing Problem

Modality	Datasets	Pre-Processing / Feature Extraction	Type
Tabular	Interstate ([14])	Predict Traffic Volume in the next day	Streaming
Tabular	Walmart M5 ([38])	Predict sales in the next day with 30 features in the lagged days manually built.	Streaming
Tabular	Air Quality([52])	Predict air quality in the next day	Streaming
Tabular	NYC Taxi ([41])	Predict fare amount from the pickup and dropoff locations	Streaming
Tabular	Stock Emotion Data ([18])	Predict the next period return based on the emotion in this stage	Streaming
Image	MDS-A ([24])	[31] vision transformer	Batch & Streaming
Image	FMoW ([6])	[31] and embedding for feature extraction	Batch & Streaming
Image	Wilds ([16])	[31] and embeddings for feature extractions	Batch & Streaming
Text Data	ChronoBerg ([13])	Various pre-trained LLMs for Feature Extraction – [63], [30], [53], [35]	Batch & Streaming
Text Data	SemEval([22], [8])	Various pre-trained LLMs for Feature Extraction – [63], [30], [53], [35]	Batch
Text Data	Bias Detection([43],[15])	Various pre-trained LLMs for Feature Extraction – [63], [30], [53], [35]	Batch
Text Data	Back-Translation([50], [40])	Various pre-trained LLMs for Feature Extraction – [63], [30], [53], [35] with [23] model	Batch
Video Data	YouTube([54])	ViViT, VideoLlama and VideoGlue([1], [62])	Batch & Streaming
Video Data	MSR-VTT Dataset ([57])	ViViT, VideoLlama and VideoGlue([1], [62])	Batch & Streaming
Audio Data	[56]	Audio Embedding CLAP ([59])	Batch & Streaming
Moving Trajectory	[44]	Predict the acceleration for the next stage	Streaming
AV Moving Data	[42]	Predict the speed of the autonomous vehicles for the next second	Streaming

Table 2: Fine Tune Baseline False Alarm Rate For the Null DGP

	0	1	2	3	4	5	6	7	8
dgp	linear	linear	linear	mars	mars	mars	rn	rn	rn
models	mlp	rf	xgb	mlp	rf	xgb	mlp	rf	xgb
BOCPDrate	0.015	0.012	0.016	0.015	0.015	0.014	0.001	0.007	0.003
EWMArate	0.004	0.006	0.005	0.004	0.004	0.004	0.003	0.003	0.003
TSrate	0.015	0.012	0.016	0.015	0.015	0.014	0.001	0.007	0.003
ADWINrate	0	0	0	0	0	0	0	0	0
HDDMArate	0	0	0	0	0	0	0	0	0
ECDDWTrate	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
HalfSpaceTreesrate	0.172	0.172	0.172	0.172	0.172	0.172	0.172	0.172	0.172
martingalerate	0	0	0	0	0	0	0	0	0
martingale2rate	0	0	0	0	0	0	0	0	0
fixrate	0.019	0.01	0.04	0.049	0.038	0.027	0.01	0.036	0.01
addisrate	0.009	0.01	0.01	0.009	0.01	0.009	0.01	0.01	0.012
saffronrate	0.009	0.01	0.01	0.009	0.01	0.009	0.01	0.01	0.01
DDMrate	0.019	0.062	0.053	0.015	0.003	0.04	0.011	0.013	0.027
STEPDrate	0	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
HDDMArate	0	0	0	0	0	0	0	0	0

Table 3: Fine Tune Baseline False Alarm Rate For the Null DGP - EWMA

	0	1	2	3	4	5	6	7	8
dgp	linear	linear	linear	mars	mars	mars	rn	rn	rn
models	mlp	rf	xgb	mlp	rf	xgb	mlp	rf	xgb
BOCPDrate	0.01	0.007	0.005	0.004	0.004	0.006	0.005	0.003	0
EWMArate	0.002	0.003	0.002	0.002	0.001	0.002	0.002	0.001	0
TSrate	0.01	0.007	0.005	0.004	0.004	0.006	0.005	0.003	0
ADWINrate	0	0	0	0	0	0	0	0	0
HDDMArate	0	0	0	0	0	0	0	0	0
ECDDWTrate	0.001	0.001	0.001	0	0	0	0.001	0.001	0
HalfSpaceTreesrate	0.113	0.105	0.061	0.065	0.054	0.081	0.103	0.082	0.012
martingalerate	0.131	0.123	0.079	0.083	0.072	0.099	0.121	0.101	0.03
martingale2rate	0.13	0.122	0.078	0.081	0.071	0.097	0.12	0.099	0.029
fixrate	0.033	0.037	0.042	0.033	0.026	0.035	0.039	0.03	0.012
addisrate	0.033	0.031	0.02	0.021	0.018	0.025	0.031	0.025	0.008
saffronrate	0.033	0.031	0.02	0.021	0.018	0.025	0.031	0.025	0.008
DDMrate	0.001	0.032	0.028	0.002	0.001	0.01	0.03	0.008	0.003
STEPDrate	0	0	0	0	0	0	0	0.001	0
HDDMArate	0	0	0	0	0	0	0	0	0

Table 4: NYC-Taxi Data hour = 15 starting, Batch 1

Methods	mean	std	DetectionRate	q25	q75
0 BOCPDSUM	883.09	332.56	1	704.5	1123
1 BOCPDbcpd2	91.16	53.87	1	61	101
2 BOCPDbcpd1	68.01	23.65	1	54.5	70.5
3 BOCPDbcpd3	1641.76	2294.84	0.95	317	2086.5
4 EWMA3	1072.67	212.29	1	941.5	1219
5 EWMA3	239.16	261.25	1	81.5	252
6 EWMA2	238.11	261.28	1	80.5	251
7 EWMA1	226	253.8	1	79.5	242.5
8 TSSUM	251.15	288.74	1	163	179
9 TS3			0		
10TS2			0		
11TS1			0		
12ADWINSUM	16.68	4.59	1	14	20
13ADWIN3			0		
14ADWIN2			0		
15ADWIN1	631.49	539.93	1	274	702
16HDMMASUM	12.42	3.53	1	10	15
17HDMMA3			0		
18HDMMA2			0		
19HDMMA1	758.28	651.33	1	320	992
20ECDDWTSUM	92.51	31.7	1	72	106
21ECDDWT3	1959.97	1487.34	1	800	2738
22ECDDWT2	1836.66	1380.56	1	799	2391.5
23HalfSpaceTreesSUM	11255.71	1083.65	1	10800	10800
24HalfSpaceTrees1	201	0	1	201	201
25HalfSpaceTrees2	202	0	1	202	202
26HalfSpaceTrees3	203	0	1	203	203
27martingaleSUM	1062.89	3731.12	1	0	0
28martingale2SUM	1061.97	3727.1	1	0	0
29martingale1	5.17	3.43	0.08	3	7.75
30martingale2	6.5	3.39	0.08	4	9.25
31martingale3	7.5	3.39	0.08	5	10.25
32martingale21	17.9	10.76	0.13	9.5	24.25
33martingale22	20.62	14.77	0.1	11.75	23
34martingale23	22.75	14.48	0.1	12.75	27.75
35fixSUM	812.42	781.4	1	557	618
36fix1	42.44	25.33	1	26	55.5
37fix2	338.95	394.48	1	93	467.5
38fix3	3763.51	2976.48	0.9	1324	5831.5
39addisSUM	496.68	870.78	1	231.5	273.5
40addis1	133.61	77.79	1	83	171
41addis2	1292.33	1480.52	1	498.5	1666
42addis3	3437.35	3673.1	0.22	340	6038
43saffronSUM	266.85	936.76	1	0	0
44saffron1			0		
45saffron2			0		
46saffron3			0		
47DDMSUM	3425.01	3883.18	1	1.5	5428.5
48DDM1	2470.1	3096.54	0.76	58	5725.75
49DDM2	2536.19	3096.24	0.75	64.5	5825.5
50DDM3	2626.98	3115.82	0.72	70	5961
51STEPDSUM	99.28	65.16	1	50.5	138
52STEPD1	1673.22	1346.65	1	568.5	2357.5
53STEPD2	1908.67	1497.69	1	643	2561
54STEPD3	2365.19	1778.99	1	877	3300
55HDDMASUM	0.46	0.62	1	0	1
56HDDMA1	7265.26	1952.98	0.39	6226.5	7982
57HDDMA2			0		
58HDDMA3			0		
59ECDDWT1	338.33	258.72	1	165	494.5

Table 5: NYC-Taxi Data hour = 15 starting, Batch 1 - EWMA

Methods	mean	std	DetectionRate	q25	q75
0 pickupdate			0		
1 timeslot			0		
2 model			0		
3 nrows	9930.63624.890.99			10000	10000
4 refbatchsize	9730.63624.890.99			9800	9800
5 PH1	61.97	31.62	0.96	35.75	77.25
6 PH2	63.48	31.53	0.95	37.5	79
7 PH3	65.54	32.89	0.95	40	80.5
8 BOCPDSUM	13.59	5.61	0.99	10	17
9 BOCPDbcpd2	85.46	34.27	0.97	58.25	99
10BOCPDbcpd1	66.4	22.96	0.99	53	70.75
11BOCPDbcpd3	111.72	45.11	0.22	71.25	147.5
12EWMA3	14.78	15.18	0.99	1	23.75
13EWMA3	110.63	44.06	0.65	72	144.75
14EWMA2	107.96	44.05	0.67	70	143
15EWMA1	102.6	43.45	0.73	67	138.5
16TSSUM	13.59	5.61	0.99	10	17
17TS3			0		
18TS2			0		
19TS1			0		
20ADWINSUM	0.11	0.32	0.99	0	0
21ADWIN3			0		
22ADWIN2			0		
23ADWIN1	145.06	40.02	0.11	114.5	183
24HDMMASUM	0.07	0.25	0.99	0	0
25HDMMA3			0		
26HDMMA2			0		
27HDMMA1	171.64	32.86	0.07	160	192
28ECDDWTSUM	0.8	1.45	0.99	0	1
29ECDDWT3	124.29	53.33	0.04	79.5	177.5
30ECDDWT2	115.73	54.68	0.07	71.5	171.5
31ECDDWT1	116.5	47.63	0.39	81.5	165.5
32HalfSpaceTreesSUM0		0	0.99	0	0
33HalfSpaceTrees1			0		
34HalfSpaceTrees2			0		
35HalfSpaceTrees3			0		
36martingaleSUM	0	0	0.99	0	0
37martingale2SUM	0.56	2.97	0.99	0	0
38martingale1			0		
39martingale2			0		
40martingale3			0		
41martingale21	21.7	4.14	0.06	18.25	25.25
42martingale22	22.86	4.34	0.04	19.5	26
43martingale23	24.29	4.39	0.04	20.5	27.5
44fixSUM	8.48	2.76	0.99	7	10
45fix1	42.57	21.98	0.99	26	53.75
46fix2	98.38	50.03	0.57	58	139
47fix3	128.75	62.05	0.02	85.25	179.5
48addisSUM	8.48	2.76	0.99	7	10
49addis1	94.66	35.58	0.97	67.25	114.75
50addis2	124.46	41.93	0.28	84.75	158.5
51addis3			0		
52saffronSUM	8.48	2.76	0.99	7	10
53saffron1			0		
54saffron2			0		
55saffron3			0		
56DDMSUM	25.92	49.28	0.99	0	15.75
57DDM1	80.86	44.7	0.44	48	90.5
58DDM2	80.74	41.65	0.4	49	92.5
59DDM3	82.78	42.62	0.39	49.75	94.25
60STEPDSUM	0.42	2.28	0.99	0	0
61STEPD1	130.67	42.92	0.06	99	166
62STEPD2	141.67	39.18	0.06	101	168
63STEPD3	145	40.54	0.04	124	170
64HDDMASUM	0.01	0.08	0.99	0	0
65HDDMA1	81	0	0.01	81	81
66HDDMA2			0		
67HDDMA3			0		

Table 6: M5 Data XGBoost Model Delay Statistic - Exponentially Weighted Moving Average(EWMA)

	Method	mean	std	detectionrate	q25	q75
0	BOCPDSUM	81.751	33.957	1.0	54.0	105.0
1	BOCPDbcpd2	121.265	100.16	1.0	65.0	139.0
2	BOCPDbcpd1	79.751	53.246	1.0	53.0	82.0
3	BOCPDbcpd3	480.351	376.321	0.488	167.5	763.0
4	EWMA3SUM	59.07	45.609	1.0	31.0	76.0
5	EWMA3	319.614	287.667	0.961	108.0	423.0
6	EWMA2	288.052	265.315	0.976	95.0	384.0
7	EWMA1	232.748	220.971	0.987	86.0	287.5
8	TSSUM	81.751	33.957	1.0	54.0	105.0
9	TS3			0.0		
10	TS2			0.0		
11	TS1			0.0		
12	ADWINSUM	4.272	4.79	1.0	1.0	7.0
13	ADWIN3			0.0		
14	ADWIN2			0.0		
15	ADWIN1	408.106	349.362	0.798	141.0	595.25
16	HDMMASUM	3.767	3.997	1.0	1.0	6.0
17	HDMMA3			0.0		
18	HDMMA2			0.0		
19	HDMMA1	430.179	350.824	0.803	160.0	608.0
20	ECDDWTSUM	48.151	42.754	1.0	15.0	69.0
21	ECDDWT3	371.884	314.666	0.861	149.0	496.75
22	ECDDWT2	347.698	312.803	0.905	133.75	450.25
23	HalfSpaceTreesSUM	1152.611	58.887	1.0	1159.0	1180.0
24	HalfSpaceTrees1	201.254	5.766	1.0	201.0	201.0
25	HalfSpaceTrees2	202.442	8.914	1.0	202.0	202.0
26	HalfSpaceTrees3	203.442	8.914	1.0	203.0	203.0
27	martingaleSUM	946.089	560.842	1.0	317.0	1360.0
28	martingale2SUM	254.441	459.21	1.0	0.0	251.0
29	martingale1	58.869	155.89	0.831	8.0	31.0
30	martingale2	63.031	157.567	0.825	11.0	34.0
31	martingale3	66.868	160.685	0.82	13.0	39.0
32	martingale21	73.363	135.077	0.403	20.0	65.75
33	martingale22	77.518	136.931	0.391	23.0	75.0
34	martingale23	86.191	150.066	0.382	26.0	84.0
35	fixSUM	204.816	59.953	1.0	172.0	244.0
36	fix1	8.353	21.254	1.0	1.0	8.0
37	fix2	80.27	135.682	0.977	18.0	80.0
38	fix3	324.718	328.283	0.789	80.0	475.5
39	addisSUM	204.816	59.953	1.0	172.0	244.0
40	addis1	11.656	22.104	1.0	4.0	13.0
41	addis2	75.514	105.659	0.995	19.0	86.25
42	addis3	365.245	340.477	0.803	104.0	516.0
43	saffronSUM	204.816	59.953	1.0	172.0	244.0
44	saffron1	144.512	167.499	0.121	50.0	171.0
45	saffron2	330.481	266.401	0.053	166.25	395.25
46	saffron3			0.0		
47	DDMSUM	539.458	533.114	1.0	16.0	1165.0
48	DDM1	212.431	259.369	0.792	58.0	233.0
49	DDM2	216.897	261.86	0.785	60.0	235.75
50	DDM3	221.481	262.628	0.782	62.0	237.0
51	STEPDSUM	85.881	94.551	1.0	16.0	126.0
52	STEPD1	358.592	337.052	0.862	110.0	478.0
53	STEPD2	362.56	332.777	0.849	113.5	485.5
54	STEPD3	375.813	341.833	0.837	120.5	512.0
55	HDDMASUM	1.478	1.85	1.0	0.0	3.0
56	HDDMA1	445.84	354.735	0.552	176.0	672.0
57	HDDMA2			0.0		
58	HDDMA3			0.0		
59	ECDDWT1	212.911	209.365	0.995	84.0	244.0

Table 7: M5 Data XGBoost Model Delay Statistic

	mean	std	missing _{ratio}	q25	q75
BOCPDSUM	81.76	33.94	1.0	54.0	105.0
BOCPDbcpd2	121.21	100.12	1.0	65.0	139.0
BOCPDbcpd1	79.74	53.22	1.0	53.0	82.0
BOCPDbcpd3	480.25	375.95	0.49	168.0	759.0
EWMA3	59.2	45.78	1.0	31.0	76.0
EWMA3	319.37	287.62	0.96	108.0	422.75
EWMA2	287.84	265.27	0.98	95.0	384.0
EWMA1	232.59	220.92	0.99	86.0	287.0
TSSUM	81.76	33.94	1.0	54.0	105.0
TS3			0.0		
TS2			0.0		
TS1			0.0		
ADWINSUM	4.27	4.79	1.0	1.0	7.0
ADWIN3			0.0		
ADWIN2			0.0		
ADWIN1	408.14	349.15	0.8	141.0	595.0
HDMMASUM	3.77	4.0	1.0	1.0	6.0
HDMMA3			0.0		
HDMMA2			0.0		
HDMMA1	430.2	350.61	0.8	160.0	608.0
ECDDWTSUM	48.11	42.75	1.0	15.0	69.0
ECDDWT3	371.88	314.67	0.86	149.0	496.75
ECDDWT2	348.44	313.44	0.9	134.0	451.0
HalfSpaceTreesSUM	1152.34	59.21	1.0	1159.0	1180.0
HalfSpaceTrees1	201.25	5.76	1.0	201.0	201.0
HalfSpaceTrees2	202.44	8.91	1.0	202.0	202.0
HalfSpaceTrees3	203.44	8.91	1.0	203.0	203.0
martingaleSUM	172.56	374.59	1.0	0.0	100.25
martingale2SUM	160.74	352.94	1.0	0.0	105.5
martingale1	55.52	94.32	0.47	17.0	48.0
martingale2	58.33	96.31	0.45	18.0	52.0
martingale3	62.26	97.57	0.44	19.0	56.5
martingale21	32.44	79.31	0.66	15.0	24.0
martingale22	36.14	83.39	0.62	16.0	26.0
martingale23	38.42	86.85	0.59	17.0	28.0
fixSUM	84.24	93.59	1.0	54.0	87.0
fix1	39.16	31.05	1.0	22.0	45.0
fix2	212.62	259.4	0.98	39.0	273.0
fix3	415.6	396.17	0.65	83.0	667.0
addisSUM	84.24	93.59	1.0	54.0	87.0
addis1	188.21	149.74	0.97	92.0	230.0
addis2	515.69	360.53	0.72	190.0	808.0
addis3	606.03	365.48	0.27	292.0	905.0
saffronSUM	84.24	93.59	1.0	54.0	87.0
saffron1	209.74	90.62	0.09	124.0	283.5
saffron2	270.63	76.61	0.06	235.0	308.0
saffron3	295.09	75.97	0.03	253.25	324.5
DDMSUM	538.93	533.12	1.0	16.0	1163.75
DDM1	212.43	259.37	0.79	58.0	233.0
DDM2	216.9	261.86	0.78	60.0	235.75
DDM3	221.48	262.63	0.78	62.0	237.0
STEPDSUM	85.81	94.53	1.0	16.0	126.0
STEPD1	358.69	336.87	0.86	110.25	477.0
STEPD2	362.65	332.6	0.85	113.75	485.25
STEPD3	375.91	341.64	0.84	120.75	512.0
HDDMASUM	1.48	1.85	1.0	0.0	3.0
HDDMA1	445.84	354.74	0.55	176.0	672.0
HDDMA2			0.0		
HDDMA3			0.0		
ECDDWT1	213.04	209.3	1.0	84.0	244.0

Table 8: M5 Data Random Forest Model Delay Statistic - Exponentially Weighted Moving Average

	Method	mean	std	detectionrate	q25	q75
0	BOCPDSUM	81.751	33.957	1.0	54.0	105.0
1	BOCPDbcpd2	121.265	100.16	1.0	65.0	139.0
2	BOCPDbcpd1	79.751	53.246	1.0	53.0	82.0
3	BOCPDbcpd3	480.351	376.321	0.488	167.5	763.0
4	EWMA3SUM	59.07	45.609	1.0	31.0	76.0
5	EWMA3	319.614	287.667	0.961	108.0	423.0
6	EWMA2	288.052	265.315	0.976	95.0	384.0
7	EWMA1	232.748	220.971	0.987	86.0	287.5
8	TSSUM	81.751	33.957	1.0	54.0	105.0
9	TS3			0.0		
10	TS2			0.0		
11	TS1			0.0		
12	ADWINSUM	4.272	4.79	1.0	1.0	7.0
13	ADWIN3			0.0		
14	ADWIN2			0.0		
15	ADWIN1	408.106	349.362	0.798	141.0	595.25
16	HDMMASUM	3.767	3.997	1.0	1.0	6.0
17	HDMMA3			0.0		
18	HDMMA2			0.0		
19	HDMMA1	430.179	350.824	0.803	160.0	608.0
20	ECDDWTSUM	48.151	42.754	1.0	15.0	69.0
21	ECDDWT3	371.884	314.666	0.861	149.0	496.75
22	ECDDWT2	347.698	312.803	0.905	133.75	450.25
23	HalfSpaceTreesSUM	1152.611	58.887	1.0	1159.0	1180.0
24	HalfSpaceTrees1	201.254	5.766	1.0	201.0	201.0
25	HalfSpaceTrees2	202.442	8.914	1.0	202.0	202.0
26	HalfSpaceTrees3	203.442	8.914	1.0	203.0	203.0
27	martingaleSUM	946.089	560.842	1.0	317.0	1360.0
28	martingale2SUM	254.441	459.21	1.0	0.0	251.0
29	martingale1	58.869	155.89	0.831	8.0	31.0
30	martingale2	63.031	157.567	0.825	11.0	34.0
31	martingale3	66.868	160.685	0.82	13.0	39.0
32	martingale21	73.363	135.077	0.403	20.0	65.75
33	martingale22	77.518	136.931	0.391	23.0	75.0
34	martingale23	86.191	150.066	0.382	26.0	84.0
35	fixSUM	204.816	59.953	1.0	172.0	244.0
36	fix1	8.353	21.254	1.0	1.0	8.0
37	fix2	80.27	135.682	0.977	18.0	80.0
38	fix3	324.718	328.283	0.789	80.0	475.5
39	addisSUM	204.816	59.953	1.0	172.0	244.0
40	addis1	11.656	22.104	1.0	4.0	13.0
41	addis2	75.514	105.659	0.995	19.0	86.25
42	addis3	365.245	340.477	0.803	104.0	516.0
43	saffronSUM	204.816	59.953	1.0	172.0	244.0
44	saffron1	144.512	167.499	0.121	50.0	171.0
45	saffron2	330.481	266.401	0.053	166.25	395.25
46	saffron3			0.0		
47	DDMSUM	539.458	533.114	1.0	16.0	1165.0
48	DDM1	212.431	259.369	0.792	58.0	233.0
49	DDM2	216.897	261.86	0.785	60.0	235.75
50	DDM3	221.481	262.628	0.782	62.0	237.0
51	STEPDSUM	85.881	94.551	1.0	16.0	126.0
52	STEPD1	358.592	337.052	0.862	110.0	478.0
53	STEPD2	362.56	332.777	0.849	113.5	485.5
54	STEPD3	375.813	341.833	0.837	120.5	512.0
55	HDDMASUM	1.478	1.85	1.0	0.0	3.0
56	HDDMA1	445.84	354.735	0.552	176.0	672.0
57	HDDMA2			0.0		
58	HDDMA3			0.0		
59	ECDDWT1	212.911	209.365	0.995	84.0	244.0

Table 9: M5 Data Random Forest average Delay Statistic

	mean	std	missing _{ratio}	q25	q75
BOCPDSUM	78.52	34.1	1.0	52.0	102.0
BOCPDbcpd2	130.84	128.56	1.0	65.0	144.0
BOCPDbcpd1	81.99	56.51	1.0	54.0	85.0
BOCPDbcpd3	448.81	356.3	0.47	156.0	673.0
EWMA3SUM	58.35	36.75	1.0	31.0	78.75
EWMA3	323.48	299.53	0.97	108.75	418.75
EWMA2	284.74	273.04	0.98	95.0	364.25
EWMA1	242.04	239.63	0.99	86.0	298.0
TSSUM	78.52	34.1	1.0	52.0	102.0
TS3			0.0		
TS2			0.0		
TS1			0.0		
ADWINSUM	3.24	3.9	1.0	1.0	5.0
ADWIN3			0.0		
ADWIN2			0.0		
ADWIN1	457.38	350.21	0.77	181.0	687.0
HDMMASUM	2.89	3.14	1.0	1.0	4.0
HDMMA3			0.0		
HDMMA2			0.0		
HDMMA1	477.67	353.19	0.78	192.0	704.0
ECDDWTSUM	43.3	44.91	1.0	11.0	60.0
ECDDWT3	408.98	330.19	0.81	174.0	543.0
ECDDWT2	386.03	338.16	0.85	148.0	509.75
HalfSpaceTreesSUM	1154.26	56.06	1.0	1166.0	1180.0
HalfSpaceTrees1	201.0	0.0	1.0	201.0	201.0
HalfSpaceTrees2	202.0	0.0	1.0	202.0	202.0
HalfSpaceTrees3	203.0	0.0	1.0	203.0	203.0
martingaleSUM	54.94	154.69	1.0	0.0	27.0
martingale2SUM	44.7	107.6	1.0	0.0	35.0
martingale1	67.14	126.82	0.37	17.0	53.0
martingale2	71.44	129.91	0.35	18.0	63.75
martingale3	76.72	133.11	0.33	19.0	73.0
martingale21	34.93	72.06	0.58	15.0	23.0
martingale22	37.74	74.24	0.55	16.0	26.0
martingale23	40.55	78.06	0.52	17.0	27.0
fixSUM	67.03	26.61	1.0	50.0	79.0
fix1	41.84	35.28	1.0	22.0	48.0
fix2	269.94	305.26	0.97	51.0	366.0
fix3	458.53	407.99	0.48	127.0	765.0
addisSUM	67.03	26.61	1.0	50.0	79.0
addis1	165.43	116.78	1.0	85.0	204.0
addis2	549.07	383.83	0.72	204.0	868.5
addis3	593.0	396.01	0.21	207.0	912.0
saffronSUM	67.03	26.61	1.0	50.0	79.0
saffron1	177.36	122.35	0.02	93.25	219.0
saffron2	352.33	218.12	0.0	226.5	419.5
saffron3	353.33	218.12	0.0	227.5	420.5
DDMSUM	513.82	524.44	1.0	4.0	1108.75
DDM1	231.16	278.19	0.77	57.0	262.0
DDM2	236.63	280.63	0.76	60.5	266.0
DDM3	239.65	280.05	0.75	64.0	267.25
STEPDSUM	76.64	94.63	1.0	4.25	113.0
STEPD1	394.17	357.68	0.81	119.0	551.5
STEPD2	402.9	357.92	0.79	124.0	570.5
STEPD3	414.17	364.88	0.77	129.0	594.0
HDDMASUM	1.13	1.74	1.0	0.0	2.0
HDDMA1	462.58	354.27	0.46	182.25	670.25
HDDMA2			0.0		
HDDMA3			0.0		
ECDDWT1	241.85	236.87	0.99	90.75	280.0

Table 10: MDS-A dataset Random Forest under EWMA

	Method	mean	std	detectionrate	q25	q75
0	BOCPDSUM	0.89	0.32	0.0	1.0	1.0
1	BOCPDbcpd2			0.0		
2	BOCPDbcpd1			0.0		
3	BOCPDbcpd3			0.0		
4	EWMA3SUM	0.0	0.0	0.0	0.0	0.0
5	EWMA3			0.0		
6	EWMA2			0.0		
7	EWMA1			0.0		
8	TSSUM	0.89	0.32	0.0	1.0	1.0
9	TS3			0.0		
10	TS2			0.0		
11	TS1			0.0		
12	ADWINSUM	0.0	0.0	0.0	0.0	0.0
13	ADWIN3			0.0		
14	ADWIN2			0.0		
15	ADWIN1			0.0		
16	HDMMASUM	0.0	0.0	0.0	0.0	0.0
17	HDMMA3			0.0		
18	HDMMA2			0.0		
19	HDMMA1			0.0		
20	ECDDWTSUM	0.08	0.28	0.0	0.0	0.0
21	ECDDWT3			0.0		
22	ECDDWT2			0.0		
23	HalfSpaceTreesSUM	18.0	27.53	0.0	0.0	34.0
24	HalfSpaceTrees1	201.0	0.0	0.0	201.0	201.0
25	HalfSpaceTrees2	202.0	0.0	0.0	202.0	202.0
26	HalfSpaceTrees3	203.0	0.0	0.0	203.0	203.0
27	martingaleSUM	136.72	86.69	0.0	50.5	230.25
28	martingale2SUM	123.19	89.44	0.0	44.5	214.5
29	martingale1	6.69	5.18	0.0	2.0	9.0
30	martingale2	8.03	5.33	0.0	3.0	10.0
31	martingale3	9.47	5.67	0.0	4.0	12.0
32	martingale21	15.8	10.44	0.0	8.5	22.5
33	martingale22	17.91	10.24	0.0	10.25	25.75
34	martingale23	20.85	10.62	0.0	12.0	27.0
35	fixSUM	35.56	21.53	0.0	14.75	59.75
36	fix1	1.78	1.29	0.0	1.0	2.0
37	fix2	13.7	14.0	0.0	2.0	20.0
38	fix3	70.48	82.08	0.0	11.0	114.0
39	addisSUM	35.56	21.53	0.0	14.75	59.75
40	addis1	4.03	2.81	0.0	2.0	5.0
41	addis2	20.68	25.07	0.0	5.25	21.0
42	addis3	68.41	88.64	0.0	9.0	70.0
43	saffronSUM	35.56	21.53	0.0	14.75	59.75
44	saffron1			0.0		
45	saffron2			0.0		
46	saffron3			0.0		
47	DDMSUM	8.5	38.03	0.0	0.0	0.0
48	DDM1	58.2	24.22	0.0	39.0	67.0
49	DDM2	59.2	24.22	0.0	40.0	68.0
50	DDM3	59.75	27.94	0.0	39.25	75.5
51	STEPDSUM	0.0	0.0	0.0	0.0	0.0
52	STEPD1			0.0		
53	STEPD2			0.0		
54	STEPD3			0.0		
55	HDDMASUM	0.0	0.0	0.0	0.0	0.0
56	HDDMA1			0.0		
57	HDDMA2			0.0		
58	HDDMA3			0.0		
59	ECDDWT1	91.0	38.43	0.0	69.5	106.5
60	PH1	39.18	15.71	0.0	30.0	40.0
61	PH2	44.25	22.06	0.0	31.0	47.25
62	PH3	53.12	31.75	0.0	32.0	60.5

Table 11: MDS-A dataset MLP model under EWMA

	Method	mean	std	detectionrate	q25	q75
0	BOCPDSUM	3.69	6.27	0.0	1.0	1.0
1	BOCPDbcpd2	78.67	21.46	0.0	63.75	83.0
2	BOCPDbcpd1	69.58	22.27	0.0	56.25	76.5
3	BOCPDbcpd3	78.0	25.46	0.0	69.0	87.0
4	EWMA3	1.87	4.47	0.0	0.0	0.0
5	EWMA3	125.44	51.58	0.0	80.0	158.0
6	EWMA2	124.44	51.58	0.0	79.0	157.0
7	EWMA1	123.44	51.58	0.0	78.0	156.0
8	TSSUM	3.69	6.27	0.0	1.0	1.0
9	TS3			0.0		
10	TS2			0.0		
11	TS1			0.0		
12	ADWINSUM	0.0	0.0	0.0	0.0	0.0
13	ADWIN3			0.0		
14	ADWIN2			0.0		
15	ADWIN1			0.0		
16	HDMMASUM	0.0	0.0	0.0	0.0	0.0
17	HDMMA3			0.0		
18	HDMMA2			0.0		
19	HDMMA1			0.0		
20	ECDDWTSUM	0.33	0.85	0.0	0.0	0.0
21	ECDDWT3	104.0	7.07	0.0	101.5	106.5
22	ECDDWT2	103.0	7.07	0.0	100.5	105.5
23	HalfSpaceTreesSUM	18.0	27.4	0.0	0.0	34.0
24	HalfSpaceTrees1	201.0	0.0	0.0	201.0	201.0
25	HalfSpaceTrees2	202.0	0.0	0.0	202.0	202.0
26	HalfSpaceTrees3	203.0	0.0	0.0	203.0	203.0
27	martingaleSUM	137.26	86.06	0.0	49.5	229.75
28	martingale2SUM	124.3	88.24	0.0	43.5	214.0
29	martingale1	6.26	4.89	0.0	2.0	8.75
30	martingale2	7.48	5.02	0.0	3.0	9.75
31	martingale3	9.02	5.21	0.0	4.0	11.75
32	martingale21	14.87	9.58	0.0	7.75	22.0
33	martingale22	17.2	9.15	0.0	10.5	23.0
34	martingale23	19.92	10.1	0.0	12.0	26.0
35	fixSUM	35.41	21.33	0.0	15.0	59.0
36	fix1	1.85	1.32	0.0	1.0	2.75
37	fix2	13.02	13.75	0.0	2.0	18.5
38	fix3	76.42	78.33	0.0	11.0	121.0
39	addisSUM	35.41	21.33	0.0	15.0	59.0
40	addis1	3.87	2.61	0.0	2.0	5.0
41	addis2	18.79	22.14	0.0	4.0	23.25
42	addis3	77.41	86.58	0.0	9.5	125.5
43	saffronSUM	35.41	21.33	0.0	15.0	59.0
44	saffron1			0.0		
45	saffron2			0.0		
46	saffron3			0.0		
47	DDMSUM	10.72	39.35	0.0	0.0	0.0
48	DDM1	56.89	22.17	0.0	39.0	67.0
49	DDM2	58.67	22.21	0.0	40.0	68.0
50	DDM3	61.57	25.63	0.0	38.5	80.5
51	STEPDSUM	0.04	0.27	0.0	0.0	0.0
52	STEPD1	218.0		0.0	218.0	218.0
53	STEPD2	219.0		0.0	219.0	219.0
54	STEPD3			0.0		
55	HDDMASUM	0.0	0.0	0.0	0.0	0.0
56	HDDMA1			0.0		
57	HDDMA2			0.0		
58	HDDMA3			0.0		
59	ECDDWT1	119.56	53.32	0.0	79.0	134.0
60	PH1	39.96	16.31	0.0	30.0	40.5
61	PH2	44.02	21.08	0.0	31.0	51.75
62	PH3	50.06	28.13	0.0	32.0	59.75

Table 12: MDS-A dataset: RF Model

	0	1	0	1	0	1
ModelType	convxnetRF	vitRF	convxnetmlp	vitmlp	convxnetxgb	vitxgb
PH1	104	53	62	51	60	35
PH2	105	204	63	52	61	36
PH3	106	205	64	53	62	37
BOCPDSUM	1	1	19	27	1	1
BOCPDbcpd2			83.0	156.0		
BOCPDbcpd1			61.0	51.0		
BOCPDbcpd3						
EWMA3	9	0	14	1	0	0
EWMA2	222.0		128.0			
EWMA1	221.0		127.0			
TSSUM	220.0		126.0			
TS3	4	2	3	7	3	4
TS2						
TS1						
ADWINSUM	0	0	0	1	0	0
ADWIN3						
ADWIN2						
ADWIN1				119.0		
HDMMASUM	0	0	0	0	0	0
HDMMA3						
HDMMA2						
HDMMA1						
ECDDWT3	0	0	0	5	0	0
ECDDWT2				115.0		
ECDDWT1				78.0		
HalfSpaceTreesSUM	54	54	54	54	54	54
HalfSpaceTrees1	201	201	201	201	201	201
HalfSpaceTrees2	202	202	202	202	202	202
HalfSpaceTrees3	203	203	203	203	203	203
martingaleSUM	0	0	0	0	0	0
martingale2SUM	0	0	0	0	0	0
martingale1						
martingale2						
martingale3						
martingale21						
martingale22						
martingale23						
fixSUM	15	11	11	18	15	12
fix1	49	85	28	45	32	34
fix2		228.0	126.0	46.0		
fix3						
addisSUM	11	9	5	6	5	5
addis1	49	85	61	113	50	115
addis2		228.0	126.0	114.0		
addis3						
saffronSUM	0	0	0	0	0	0
saffron1						
saffron2						
saffron3						
DDMSUM	0	0	0	155	162	3
DDM1				78.0	86.0	42.0
DDM2				79.0	87.0	43.0
DDM3				80.0	88.0	44.0
STEPDSUM	0	0	0	1	0	0
STEPD1				140.0		
STEPD2						
STEPD3						
HDDMASUM	0	0	0	0	0	0
HDDMA1						
HDDMA2						
HDDMA3						

Table 13: Audio Streaming Results - random forest

	mean	std	detectionRate	q25
BOCPDSUM	13.109	12.364	1.0	3.0
BOCPDbcpd2	521.013	423.662	0.155	135.75
BOCPDbcpd1	68.674	23.182	1.0	59.0
BOCPDbcpd3			0.0	
EWMA SUM	87.636	32.446	1.0	49.0
EWMA3	159.562	236.86	1.0	31.0
EWMA2	143.942	210.685	1.0	30.0
EWMA1	33.672	5.252	1.0	29.0
TSSUM	13.109	12.364	1.0	3.0
TS3			0.0	
TS2			0.0	
TS1			0.0	
ADWINSUM	12.357	13.279	1.0	2.0
ADWIN3			0.0	
ADWIN2			0.0	
ADWIN1	32.231	3.582	1.0	31.0
HDMMA SUM	10.041	10.016	1.0	2.0
HDMMA3			0.0	
HDMMA2			0.0	
HDMMA1	32.186	2.435	1.0	32.0
ECDDWT SUM	161.463	209.874	1.0	8.0
ECDDWT3	211.019	154.504	1.0	71.0
ECDDWT2	210.019	154.504	1.0	70.0
HalfSpaceTrees SUM	66.667	94.372	1.0	0.0
HalfSpaceTrees1	201.0	0.0	0.333	201.0
HalfSpaceTrees2	202.0	0.0	0.333	202.0
HalfSpaceTrees3	203.0	0.0	0.333	203.0
martingale SUM	570.818	660.562	1.0	74.0
martingale2 SUM	573.122	663.032	1.0	73.0
martingale1	11.554	10.487	1.0	2.0
martingale2	12.734	10.845	1.0	3.0
martingale3	13.94	11.211	1.0	4.0
martingale21	9.839	6.764	1.0	3.0
martingale22	10.86	6.797	1.0	4.0
martingale23	11.86	6.797	1.0	5.0
fix SUM	324.103	396.128	1.0	59.0
fix1	11.329	10.431	1.0	1.0
fix2	14.368	13.029	1.0	2.0
fix3	16.076	13.578	1.0	3.0
addis SUM	85.787	134.567	1.0	0.0
addis1	58.519	268.585	0.516	1.0
addis2	46.525	225.017	0.51	2.0
addis3	23.31	96.687	0.5	3.0
saffron SUM	0.0	0.0	1.0	0.0
saffron1			0.0	
saffron2			0.0	
saffron3			0.0	
DDMSUM	374.647	511.255	1.0	8.0
DDM1	209.019	154.504	1.0	69.0
DDM2	210.019	154.504	1.0	70.0
DDM3	211.019	154.504	1.0	71.0
STEPD SUM	519.056	666.13	1.0	11.0
STEPD1	60.0	0.0	1.0	60.0
STEPD2	61.0	0.0	1.0	61.0
STEPD3	62.0	0.0	1.0	62.0
HDDMA SUM	1.176	0.438	1.0	1.0
HDDMA1	221.684	164.896	1.0	72.0
HDDMA2			0.0	
HDDMA3			0.0	
ECDDWT1	209.019	154.504	1.0	69.0

Table 14: Audio Streaming Results - XGBoost

	mean	std	detectionRate	q25
BOCPDSUM	11.109	10.042	1.0	3.0
BOCPDbcpd2	769.826	414.545	0.089	432.25
BOCPDbcpd1	73.585	24.046	1.0	58.0
BOCPDbcpd3			0.0	
EWMA3SUM	80.748	32.37	1.0	49.0
EWMA3	99.054	192.91	0.996	31.0
EWMA2	78.124	143.206	1.0	30.0
EWMA1	57.407	92.286	1.0	29.0
TSSUM	11.109	10.042	1.0	3.0
TS3			0.0	
TS2			0.0	
TS1			0.0	
ADWINSUM	10.872	11.186	1.0	2.0
ADWIN3			0.0	
ADWIN2			0.0	
ADWIN1	44.372	19.067	1.0	31.0
HDMMASUM	8.771	8.223	1.0	2.0
HDMMA3			0.0	
HDMMA2			0.0	
HDMMA1	49.736	26.586	1.0	32.0
ECDDWTSUM	164.523	213.827	1.0	8.0
ECDDWT3	176.035	106.223	1.0	71.0
ECDDWT2	175.035	106.223	1.0	70.0
HalfSpaceTreesSUM	66.667	94.372	1.0	0.0
HalfSpaceTrees1	201.0	0.0	0.333	201.0
HalfSpaceTrees2	202.0	0.0	0.333	202.0
HalfSpaceTrees3	203.0	0.0	0.333	203.0
martingaleSUM	555.839	641.496	1.0	73.0
martingale2SUM	556.744	642.342	1.0	71.0
martingale1	27.132	40.962	1.0	3.0
martingale2	28.153	41.004	1.0	4.0
martingale3	29.223	41.127	1.0	5.0
martingale21	18.837	29.597	1.0	5.0
martingale22	21.01	31.171	1.0	6.0
martingale23	22.75	32.366	1.0	7.0
fixSUM	279.572	332.233	1.0	49.0
fix1	13.826	14.003	1.0	1.0
fix2	16.899	12.69	1.0	6.0
fix3	32.703	40.48	1.0	7.0
addisSUM	82.014	127.749	1.0	0.0
addis1	27.085	41.571	0.665	2.0
addis2	41.032	60.436	0.661	6.0
addis3	56.66	123.606	0.587	7.0
saffronSUM	0.0	0.0	1.0	0.0
saffron1			0.0	
saffron2			0.0	
saffron3			0.0	
DDMSUM	409.64	560.802	1.0	8.0
DDM1	174.027	106.23	1.0	69.0
DDM2	175.027	106.23	1.0	70.0
DDM3	176.027	106.23	1.0	71.0
STEPDSUM	485.043	618.002	1.0	11.0
STEPD1	60.0	0.0	1.0	60.0
STEPD2	61.0	0.0	1.0	61.0
STEPD3	62.0	0.0	1.0	62.0
HDDMASUM	1.333	0.472	1.0	1.0
HDDMA1	222.453	165.939	1.0	72.0
HDDMA2			0.0	
HDDMA3			0.0	
ECDDWT1	174.027	106.23	1.0	69.0

Table 15: Audio Streaming Results - MLP

	mean	std	detectionRate	q25
BOCPDSUM	22.01	29.989	1.0	1.0
BOCPDbcpd2	661.468	402.834	0.24	331.0
BOCPDbcpd1	70.426	18.009	0.628	58.0
BOCPDbcpd3	661.8	283.405	0.01	490.0
EWMA3SUM	459.527	545.724	1.0	49.0
EWMA3	73.291	97.387	1.0	31.0
EWMA2	72.291	97.387	1.0	30.0
EWMA1	71.291	97.387	1.0	29.0
TSSUM	22.01	29.989	1.0	1.0
TS3			0.0	
TS2			0.0	
TS1			0.0	
ADWINSUM	18.236	21.586	1.0	2.0
ADWIN3			0.0	
ADWIN2			0.0	
ADWIN1	31.0	0.0	1.0	31.0
HDMMASUM	17.593	20.676	1.0	2.0
HDMMA3			0.0	
HDMMA2			0.0	
HDMMA1	32.0	0.0	1.0	32.0
ECDDWT3SUM	156.025	203.908	1.0	8.0
ECDDWT3	220.63	166.029	1.0	71.0
ECDDWT2	219.63	166.029	1.0	70.0
HalfSpaceTreesSUM	66.667	94.372	1.0	0.0
HalfSpaceTrees1	201.0	0.0	0.333	201.0
HalfSpaceTrees2	202.0	0.0	0.333	202.0
HalfSpaceTrees3	203.0	0.0	0.333	203.0
martingaleSUM	573.415	663.846	1.0	74.0
martingale2SUM	573.785	663.647	1.0	73.0
martingale1	9.742	7.56	1.0	2.0
martingale2	10.754	7.57	1.0	3.0
martingale3	11.76	7.576	1.0	4.0
martingale21	9.378	5.908	1.0	3.0
martingale22	10.382	5.913	1.0	4.0
martingale23	11.388	5.908	1.0	5.0
fixSUM	491.537	569.077	1.0	75.0
fix1	11.093	10.043	1.0	1.0
fix2	12.36	10.113	1.0	2.0
fix3	13.812	10.147	1.0	3.0
addisSUM	142.729	301.407	1.0	0.0
addis1	395.176	523.788	0.461	1.0
addis2	412.688	543.294	0.459	2.0
addis3	433.937	570.554	0.459	3.0
saffronSUM	0.0	0.0	1.0	0.0
saffron1			0.0	
saffron2			0.0	
saffron3			0.0	
DDMSUM	365.037	499.766	1.0	8.0
DDM1	218.63	166.029	1.0	69.0
DDM2	219.63	166.029	1.0	70.0
DDM3	220.63	166.029	1.0	71.0
STEPDSUM	520.236	665.736	1.0	11.0
STEPD1	60.0	0.0	1.0	60.0
STEPD2	61.0	0.0	1.0	61.0
STEPD3	62.0	0.0	1.0	62.0
HDDMASUM	1.0	0.0	1.0	1.0
HDDMA1	221.926	166.442	1.0	72.0
HDDMA2			0.0	
HDDMA3			0.0	
ECDDWT1	218.63	166.029	1.0	69.0

Table 16: M5 Dataset: Random Forest Summarization

	mean	std	missing _{ratio}	q25	q75
BOCPDSUM	78.52	34.1	1.0	52.0	102.0
BOCPDbcpd2	130.84	128.56	1.0	65.0	144.0
BOCPDbcpd1	81.99	56.51	1.0	54.0	85.0
BOCPDbcpd3	448.81	356.3	0.47	156.0	673.0
EWMASUM	58.35	36.75	1.0	31.0	78.75
EWMA3	323.48	299.53	0.97	108.75	418.75
EWMA2	284.74	273.04	0.98	95.0	364.25
EWMA1	242.04	239.63	0.99	86.0	298.0
TSSUM	78.52	34.1	1.0	52.0	102.0
TS3			0.0		
TS2			0.0		
TS1			0.0		
ADWINSUM	3.24	3.9	1.0	1.0	5.0
ADWIN3			0.0		
ADWIN2			0.0		
ADWIN1	457.38	350.21	0.77	181.0	687.0
HDMMASUM	2.89	3.14	1.0	1.0	4.0
HDMMA3			0.0		
HDMMA2			0.0		
HDMMA1	477.67	353.19	0.78	192.0	704.0
ECDDWTSUM	43.3	44.91	1.0	11.0	60.0
ECDDWT3	408.98	330.19	0.81	174.0	543.0
ECDDWT2	386.03	338.16	0.85	148.0	509.75
HalfSpaceTreesSUM	1154.26	56.06	1.0	1166.0	1180.0
HalfSpaceTrees1	201.0	0.0	1.0	201.0	201.0
HalfSpaceTrees2	202.0	0.0	1.0	202.0	202.0
HalfSpaceTrees3	203.0	0.0	1.0	203.0	203.0
martingaleSUM	54.94	154.69	1.0	0.0	27.0
martingale2SUM	44.7	107.6	1.0	0.0	35.0
martingale1	67.14	126.82	0.37	17.0	53.0
martingale2	71.44	129.91	0.35	18.0	63.75
martingale3	76.72	133.11	0.33	19.0	73.0
martingale21	34.93	72.06	0.58	15.0	23.0
martingale22	37.74	74.24	0.55	16.0	26.0
martingale23	40.55	78.06	0.52	17.0	27.0
fixSUM	67.03	26.61	1.0	50.0	79.0
fix1	41.84	35.28	1.0	22.0	48.0
fix2	269.94	305.26	0.97	51.0	366.0
fix3	458.53	407.99	0.48	127.0	765.0
addisSUM	67.03	26.61	1.0	50.0	79.0
addis1	165.43	116.78	1.0	85.0	204.0
addis2	549.07	383.83	0.72	204.0	868.5
addis3	593.0	396.01	0.21	207.0	912.0
saffronSUM	67.03	26.61	1.0	50.0	79.0
saffron1	177.36	122.35	0.02	93.25	219.0
saffron2	352.33	218.12	0.0	226.5	419.5
saffron3	353.33	218.12	0.0	227.5	420.5
DDMSUM	513.82	524.44	1.0	4.0	1108.75
DDM1	231.16	278.19	0.77	57.0	262.0
DDM2	236.63	280.63	0.76	60.5	266.0
DDM3	239.65	280.05	0.75	64.0	267.25
STEPDSUM	76.64	94.63	1.0	4.25	113.0
STEPD1	394.17	357.68	0.81	119.0	551.5
STEPD2	402.9	357.92	0.79	124.0	570.5
STEPD3	414.17	364.88	0.77	129.0	594.0
HDDMASUM	1.13	1.74	1.0	0.0	2.0
HDDMA1	462.58	354.27	0.46	182.25	670.25
HDDMA2			0.0		
HDDMA3			0.0		
ECDDWT1	241.85	236.87	0.99	90.75	280.0

Table 17: M5 Dataset: Random Forest Summarization via the EWMA procedure

	Method	mean	std	detectionrate	q25	q75
0	BOCPDSUM	80.14	34.06	1.0	53.0	104.0
1	BOCPDbcpd2	126.05	115.32	1.0	65.0	142.0
2	BOCPDbcpd1	80.87	54.9	1.0	54.0	84.0
3	BOCPDbcpd3	464.79	366.73	0.48	160.0	718.5
4	EWMASUM	58.71	41.41	1.0	31.0	78.0
5	EWMA3	321.56	293.61	0.96	108.0	422.0
6	EWMA2	286.39	269.16	0.98	95.0	373.0
7	EWMA1	237.4	230.5	0.99	86.0	292.0
8	TSSUM	80.14	34.06	1.0	53.0	104.0
9	TS3			0.0		
10	TS2			0.0		
11	TS1			0.0		
12	ADWINSUM	3.75	4.4	1.0	1.0	6.0
13	ADWIN3			0.0		
14	ADWIN2			0.0		
15	ADWIN1	432.23	350.54	0.78	155.0	640.0
16	HDMMASUM	3.33	3.62	1.0	1.0	5.0
17	HDMMA3			0.0		
18	HDMMA2			0.0		
19	HDMMA1	453.61	352.68	0.79	160.0	672.0
20	ECDDWTSUM	45.72	43.9	1.0	13.0	65.0
21	ECDDWT3	389.82	322.7	0.83	159.75	527.0
22	ECDDWT2	366.33	325.85	0.88	142.0	472.0
23	HalfSpaceTreesSUM	1153.36	57.81	1.0	1166.0	1180.0
24	HalfSpaceTrees1	201.13	4.08	1.0	201.0	201.0
25	HalfSpaceTrees2	202.22	6.3	1.0	202.0	202.0
26	HalfSpaceTrees3	203.22	6.3	1.0	203.0	203.0
27	martingaleSUM	1073.39	473.53	1.0	1081.0	1360.0
28	martingale2SUM	261.6	461.99	1.0	0.0	279.0
29	martingale1	53.71	134.24	0.9	8.0	31.0
30	martingale2	57.49	136.38	0.9	11.0	35.0
31	martingale3	60.8	138.24	0.9	14.0	41.0
32	martingale21	71.11	134.58	0.41	20.0	69.0
33	martingale22	77.35	137.47	0.4	23.0	77.0
34	martingale23	86.41	146.47	0.39	27.0	85.0
35	fixSUM	216.92	51.46	1.0	190.0	248.0
36	fix1	6.8	15.59	1.0	1.0	7.0
37	fix2	65.3	110.97	0.99	17.0	71.0
38	fix3	293.2	305.23	0.84	72.0	415.0
39	addisSUM	216.92	51.46	1.0	190.0	248.0
40	addis1	9.96	16.17	1.0	4.0	12.0
41	addis2	65.65	93.92	1.0	18.0	79.0
42	addis3	335.5	324.15	0.84	86.0	476.0
43	saffronSUM	216.92	51.46	1.0	190.0	248.0
44	saffron1	137.18	162.75	0.07	49.25	167.75
45	saffron2	330.48	266.4	0.03	166.25	395.25
46	saffron3			0.0		
47	DDMSUM	526.63	528.82	1.0	11.0	1146.5
48	DDM1	221.65	268.88	0.78	57.25	245.0
49	DDM2	226.59	271.34	0.77	60.0	248.0
50	DDM3	230.41	271.39	0.77	64.0	253.0
51	STEPDSUM	81.26	94.68	1.0	9.0	118.5
52	STEPD1	375.87	347.58	0.84	114.0	511.0
53	STEPD2	382.05	345.64	0.82	119.0	520.0
54	STEPD3	394.17	353.46	0.8	124.0	540.25
55	HDDMASUM	1.3	1.8	1.0	0.0	2.0
56	HDDMA1	453.44	354.45	0.5	179.0	672.0
57	HDDMA2			0.0		
58	HDDMA3			0.0		
59	ECDDWT1	227.35	223.93	0.99	87.0	266.0

Table 18: StockEmo Dataset: Random Forest Summarization

	ref0	ref1000	ref0set2	ref1000set2	method
0	3.0	3.0	1.0	1.0	BOCPDSUM
1	7226.0	6226.0			BOCPDbcpd2
2	7225.0	6225.0			BOCPDbcpd1
3					BOCPDbcpd3
4	293.0	185.0	0.0	0.0	EWMA3SUM
5	794.0	3192.0			EWMA3
6	793.0	3187.0			EWMA2
7	792.0	3186.0			EWMA1
8	3.0	3.0	1.0	1.0	TSSUM
9					TS3
10					TS2
11					TS1
12	1.0	0.0	0.0	0.0	ADWINSUM
13					ADWIN3
14					ADWIN2
15	3574.0				ADWIN1
16	1.0	0.0	0.0	0.0	HDMMASUM
17					HDMMA3
18					HDMMA2
19	3040.0				HDMMA1
20	352.0	648.0	23.0	27.0	ECDDWTSUM
21	250.0	77.0	46.0	283.0	ECDDWT3
22	249.0	76.0	45.0	282.0	ECDDWT2
23	8737.0	7737.0	694.0	594.0	HalfSpaceTreesSUM
24	201.0	201.0	201.0	201.0	HalfSpaceTrees1
25	202.0	202.0	202.0	202.0	HalfSpaceTrees2
26	203.0	203.0	203.0	203.0	HalfSpaceTrees3
27	108.0	0.0	95.0	0.0	martingaleSUM
28	10.0	0.0	83.0	0.0	martingale2SUM
29	547.0		33.0		martingale1
30	548.0		34.0		martingale2
31	549.0		35.0		martingale3
32	23.0		26.0		martingale21
33	26.0		27.0		martingale22
34	27.0		28.0		martingale23
35	284.0	377.0	35.0	29.0	fixSUM
36	22.0	32.0	23.0	156.0	fix1
37	228.0	76.0	24.0	279.0	fix2
38	229.0	188.0	25.0	321.0	fix3
39	284.0	377.0	35.0	29.0	addisSUM
40	180.0	72.0	419.0	230.0	addis1
41	228.0	76.0	420.0	279.0	addis2
42	229.0	1146.0	421.0	321.0	addis3
43	284.0	377.0	35.0	29.0	saffronSUM
44					saffron1
45					saffron2
46					saffron3
47	8690.0	7893.0	138.0	54.0	DDMSUM
48	248.0	45.0	43.0	331.0	DDM1
49	249.0	46.0	44.0	332.0	DDM2
50	250.0	47.0	45.0	333.0	DDM3
51	802.0	1071.0	77.0	46.0	STEPDSUM
52	60.0	86.0	60.0	327.0	STEPD1
53	61.0	87.0	61.0	328.0	STEPD2
54	62.0	88.0	62.0	329.0	STEPD3
55	7.0	20.0	2.0	1.0	HDDMASUM
56	451.0	617.0	48.0	323.0	HDDMA1
57					HDDMA2
58					HDDMA3
59	248.0	75.0	44.0	116.0	ECDDWT1

Table 19: StockEmotion Dataset: Random Forest Summarization via the EWMA procedure

	stat	mean	std	q25	q50	q75	detectionrate
0	BOCPDSUM	7.5	17.49	1	1	3	1
1	BOCPDbcpd2	4703.167	3177.968	2383	6226	6976	0.333
2	BOCPDbcpd1	4702.167	3177.968	2382	6225	6975	0.333
3	BOCPDbcpd3						0
4	EWMAΣ	85.944	130.709	0	0	179	1
5	EWMA3	1824.667	1292.461	867.5	1778	3011	0.333
6	EWMA2	1778	1325.185	800	1644	3007	0.333
7	EWMA1	1434.5	1374.376	528.75	790.5	2587.5	0.333
8	TSSUM	7.5	17.49	1	1	3	1
9	TS3						0
10	TS2						0
11	TS1						0
12	ADWINSUM	0.111	0.323	0	0	0	1
13	ADWIN3						0
14	ADWIN2						0
15	ADWIN1	3568	8.485	3565	3568	3571	0.111
16	HDMMASUM	0.111	0.323	0	0	0	1
17	HDMMA3						0
18	HDMMA2						0
19	HDMMA1	3040	0	3040	3040	3040	0.111
20	ECDDWTSUM	137.778	233.474	8.5	24.5	91.75	1
21	ECDDWT3	386.944	538.366	164	215	283	1
22	ECDDWT2	340.333	498.571	163	214	282	1
23	HalfSpaceTreesSUM	3034.333	3801.414	247	644	7737	1
24	HalfSpaceTrees1	201	0	201	201	201	1
25	HalfSpaceTrees2	202	0	202	202	202	1
26	HalfSpaceTrees3	203	0	203	203	203	1
27	martingaleΣ	3153.444	3708.737	436.5	836	7673	1
28	martingale2Σ	2276.111	3391.699	389	771	873.5	1
29	martingale1	24	67.935	3	6.5	12.5	1
30	martingale2	25.944	67.707	4	11	14	1
31	martingale3	27.722	67.506	6	13	17.25	1
32	martingale21	17.5	10.875	9.25	17	23	0.889
33	martingale22	24	13.948	15.25	23	27	0.889
34	martingale23	27	13.337	21	25	31	0.889
35	fixΣ	715.611	835.476	122	239.5	1181.25	1
36	fix1	5.5	13.849	1	1	3	1
37	fix2	23.278	38.051	4	10	12.75	1
38	fix3	168.444	317.358	29.25	55.5	104.75	1
39	addisΣ	715.611	835.476	122	239.5	1181.25	1
40	addis1	8.444	13.272	3	4	8	1
41	addis2	28.056	36.724	9	13	24.75	1
42	addis3	677.667	1632.41	29.25	60.5	215.5	1
43	saffronΣ	715.611	835.476	122	239.5	1181.25	1
44	saffron1						0
45	saffron2						0
46	saffron3						0
47	DDMSUM	2031.333	3482.206	0	111	1652.75	1
48	DDM1	182	153.317	44.5	146.5	328.75	0.667
49	DDM2	183	153.317	45.5	147.5	329.75	0.667
50	DDM3	185.667	155.251	46.5	148.5	330.75	0.667
51	STEPDSUM	250.167	393.574	19.25	44	178	1
52	STEPD1	212.25	243.479	61.5	90.5	326	0.889
53	STEPD2	360	685.312	68	96	327.5	0.833
54	STEPD3	361	685.312	69	97	328.5	0.833
55	HDDMASUM	4.333	6.979	1	1.5	2	1
56	HDDMA1	673.588	1169.043	164	323	451	0.944
57	HDDMA2						0
58	HDDMA3						0
59	ECDDWT1	175	134.304	85.75	162	213	1

Table 20: Air Quality Dataset: Random Forest Summarization

	method	ref0	ref1000	ref2000	ref0 _{set2}	ref1000 _{set2}	ref2000 _{set2}
0	BOCPDSUM	110.0	110.0	110.0	219.0	219.0	219.0
1	BOCPDbcpd2	159.0	159.0	159.0	58.0	58.0	58.0
2	BOCPDbcpd1	78.0	78.0	78.0	57.0	57.0	57.0
3	BOCPDbcpd3				2723.0	2723.0	2723.0
4	EWMA SUM	483.0	483.0	483.0	392.0	392.0	392.0
5	EWMA3	36.0	36.0	36.0	362.0	362.0	362.0
6	EWMA2	35.0	35.0	35.0	361.0	361.0	361.0
7	EWMA1	34.0	34.0	34.0	360.0	360.0	360.0
8	TSSUM	110.0	110.0	110.0	219.0	219.0	219.0
9	TS3						
10	TS2						
11	TS1						
12	ADWINSUM	56.0	56.0	56.0	72.0	72.0	72.0
13	ADWIN3						
14	ADWIN2						
15	ADWIN1	41.0	41.0	41.0	520.0	520.0	520.0
16	HDMMASUM	52.0	52.0	52.0	37.0	37.0	37.0
17	HDMMA3						
18	HDMMA2						
19	HDMMA1	64.0	64.0	64.0	544.0	544.0	544.0
20	ECDDWTSUM	927.0	927.0	927.0	1245.0	1245.0	1245.0
21	ECDDWT3	36.0	36.0	36.0	278.0	278.0	278.0
22	ECDDWT2	35.0	35.0	35.0	277.0	277.0	277.0
23	HalfSpaceTreesSUM	7156.0	7156.0	7156.0	6156.0	6156.0	6156.0
24	HalfSpaceTrees1	201.0	201.0	201.0	201.0	201.0	201.0
25	HalfSpaceTrees2	202.0	202.0	202.0	202.0	202.0	202.0
26	HalfSpaceTrees3	203.0	203.0	203.0	203.0	203.0	203.0
27	martingaleSUM	65.0	65.0	65.0	530.0	530.0	530.0
28	martingale2SUM	64.0	64.0	64.0	0.0	0.0	0.0
29	martingale1	35.0	35.0	35.0	1250.0	1250.0	1250.0
30	martingale2	36.0	36.0	36.0	1251.0	1251.0	1251.0
31	martingale3	37.0	37.0	37.0	1252.0	1252.0	1252.0
32	martingale21	27.0	27.0	27.0			
33	martingale22	28.0	28.0	28.0			
34	martingale23	36.0	36.0	36.0			
35	fixSUM	469.0	469.0	469.0	253.0	253.0	253.0
36	fix1	24.0	24.0	24.0	33.0	33.0	33.0
37	fix2	25.0	25.0	25.0	514.0	514.0	514.0
38	fix3	26.0	26.0	26.0	515.0	515.0	515.0
39	addisSUM	469.0	469.0	469.0	253.0	253.0	253.0
40	addis1	225.0	225.0	225.0	105.0	105.0	105.0
41	addis2	1517.0	1517.0	1517.0	517.0	517.0	517.0
42	addis3	1518.0	1518.0	1518.0	518.0	518.0	518.0
43	saffronSUM	469.0	469.0	469.0	253.0	253.0	253.0
44	saffron1						
45	saffron2						
46	saffron3						
47	DDMSUM	7324.0	7324.0	7324.0	5834.0	5834.0	5834.0
48	DDM1	33.0	33.0	33.0	523.0	523.0	523.0
49	DDM2	34.0	34.0	34.0	524.0	524.0	524.0
50	DDM3	35.0	35.0	35.0	525.0	525.0	525.0
51	STEPDSUM	1396.0	1396.0	1396.0	4414.0	4414.0	4414.0
52	STEPD1	60.0	60.0	60.0	297.0	297.0	297.0
53	STEPD2	61.0	61.0	61.0	298.0	298.0	298.0
54	STEPD3	62.0	62.0	62.0	299.0	299.0	299.0
55	HDDMASUM	19.0	19.0	19.0	10.0	10.0	10.0
56	HDDMA1	36.0	36.0	36.0	279.0	279.0	279.0
57	HDDMA2						
58	HDDMA3						
59	ECDDWT1	34.0	34.0	34.0	276.0	276.0	276.0

Table 21: Air Quality Dataset: Random Forest Summarization via the EWMA procedure

stat	mean	std	detection _{rate}	q25	q50	q75
BOCPDSUM	45.5	52.35	1	8.25	12.5	94.75
BOCPDbcpd2	226.09	196.66	0.61	80.5	153	318.5
BOCPDbcpd1	101.11	41.43	1	76	81.5	119.75
BOCPDbcpd3	3334		0.06	3334	3334	3334
EWMA SUM	177.11	225.3	1	11.75	46.5	397.75
EWMA3	166.77	128.25	0.72	51	144	196
EWMA2	159.4	96.51	0.83	66.5	167	216.5
EWMA1	151.8	86.15	0.83	65.5	166	213
TSSUM	45.5	52.35	1	8.25	12.5	94.75
TS3			0			
TS2			0			
TS1			0			
ADWINSUM	34.11	39.09	1	6	10	77.25
ADWIN3			0			
ADWIN2			0			
ADWIN1	85.61	39.75	1	51.75	86	98.75
HDMMASUM	26.61	30.85	1	3.5	8	63.5
HDMMA3			0			
HDMMA2			0			
HDMMA1	113.78	65.14	1	64	112	128
ECDDWTSUM	284.72	404.7	1	24.5	46.5	466
ECDDWT3	167.17	123.14	1	88.25	117.5	191.75
ECDDWT2	150.5	120.31	1	80.25	112.5	154
HalfSpaceTreesSUM	2803.28	3527.98	1	218	586	7157
HalfSpaceTrees1	201	0	1	201	201	201
HalfSpaceTrees2	202	0	1	202	202	202
HalfSpaceTrees3	203	0	1	203	203	203
martingaleSUM	3005.44	3541.16	1	408.5	777	7347.25
martingale2SUM	2994.61	3538.4	1	405.5	765.5	7340
martingale1	6	3.9	1	3	4	8
martingale2	7.28	4.07	1	4	5	10.5
martingale3	8.61	3.9	1	6	8	11.5
martingale21	14.89	10.74	1	9	12	15.75
martingale22	16.56	10.84	1	12	14	17.5
martingale23	18.89	11.8	1	13	17	19.75
fixSUM	893.44	1070.26	1	122	214	2113.75
fix1	1.78	1.31	1	1	1	2
fix2	6.5	4.33	1	3	5	10.25
fix3	52.56	64.73	1	12.25	18.5	79.5
addisSUM	893.44	1070.26	1	122	214	2113.75
addis1	6	3.38	1	3	5.5	9
addis2	9.5	7.13	1	4.25	7.5	12
addis3	63.83	73.17	1	14	22	92
saffronSUM	893.44	1070.26	1	122	214	2113.75
saffron1			0			
saffron2			0			
saffron3			0			
DDMSUM	2456.11	2992.5	1	305	545.5	6011.5
DDM1	169.33	200.39	1	48	92	189.75
DDM2	170.33	200.39	1	49	93	190.75
DDM3	171.33	200.39	1	50	94	191.75
STEPDSUM	481.72	680.03	1	46.25	75	953.25
STEPD1	128.59	82.57	0.94	60	106	193
STEPD2	129.71	82.54	0.94	61	109	194
STEPD3	130.71	82.54	0.94	62	110	195
HDDMASUM	7.61	9.49	1	1	2	15.5
HDDMA1	288.06	254.85	0.94	116	195	419
HDDMA2			0			
HDDMA3			0			

Table 22: Interstate Loading Data Delay

	method	ref0xgb	ref0rf	ref0mlp	ref1000xgb	ref1000rf	ref1000mlp
0	BOCPDSUM	19.0	7.0	1.0	7.0	11.0	1.0
1	BOCPDbcpd2	136.0				445.0	
2	BOCPDbcpd1	84.0	223.0		71.0	138.0	
3	BOCPDbcpd3						
4	EWMA SUM	42.0	22.0	2.0	18.0	5.0	2.0
5	EWMA3	46.0	428.0		617.0		
6	EWMA2	45.0	427.0		616.0		
7	EWMA1	44.0	426.0	470.0	475.0	780.0	370.0
8	TSSUM	19.0	7.0	1.0	7.0	11.0	1.0
9	TS3						
10	TS2						
11	TS1						
12	ADWIN SUM	5.0	1.0	1.0	3.0	1.0	0.0
13	ADWIN3						
14	ADWIN2						
15	ADWIN1	138.0	644.0	872.0	284.0	788.0	
16	HDMMA SUM	2.0	1.0	1.0	3.0	1.0	0.0
17	HDMMA3						
18	HDMMA2						
19	HDMMA1	160.0	672.0	864.0	288.0		
20	ECDDWT SUM	19.0	22.0	6.0	55.0	11.0	8.0
21	ECDDWT3	88.0	616.0		38.0	479.0	
22	ECDDWT2	87.0	615.0		37.0	38.0	
23	HalfSpaceTrees SUM	700.0	700.0	700.0	600.0	600.0	600.0
24	HalfSpaceTrees1	201.0	201.0	201.0	201.0	201.0	201.0
25	HalfSpaceTrees2	202.0	202.0	202.0	202.0	202.0	202.0
26	HalfSpaceTrees3	203.0	203.0	203.0	203.0	203.0	203.0
27	martingale SUM	895.0	892.0	897.0	785.0	795.0	787.0
28	martingale2 SUM	891.0	852.0	889.0	766.0	790.0	774.0
29	martingale1	5.0	8.0	3.0	13.0	5.0	13.0
30	martingale2	6.0	9.0	4.0	17.0	6.0	14.0
31	martingale3	7.0	10.0	5.0	18.0	7.0	15.0
32	martingale21	5.0	48.0	11.0	34.0	8.0	16.0
33	martingale22	6.0	49.0	12.0	35.0	12.0	21.0
34	martingale23	7.0	50.0	13.0	36.0	13.0	36.0
35	fix SUM	221.0	216.0	219.0	210.0	192.0	192.0
36	fix1	1.0	2.0	1.0	7.0	1.0	6.0
37	fix2	13.0	9.0	11.0	8.0	13.0	7.0
38	fix3	34.0	254.0	87.0	160.0	91.0	8.0
39	addis SUM	221.0	216.0	219.0	210.0	192.0	192.0
40	addis1	5.0	2.0	3.0	7.0	5.0	8.0
41	addis2	13.0	9.0	16.0	8.0	16.0	16.0
42	addis3	34.0	291.0	87.0	161.0	279.0	123.0
43	saffron SUM	221.0	216.0	219.0	210.0	192.0	192.0
44	saffron1						
45	saffron2						
46	saffron3						
47	DDM SUM	488.0	862.0	326.0	766.0	64.0	456.0
48	DDM1	39.0	39.0	575.0	35.0	36.0	344.0
49	DDM2	40.0	40.0	576.0	36.0	37.0	345.0
50	DDM3	41.0	41.0	577.0	37.0	38.0	346.0
51	STEPD SUM	16.0	55.0	5.0	113.0	22.0	9.0
52	STEPD1	154.0	617.0	595.0	60.0	326.0	330.0
53	STEPD2	155.0	618.0	596.0	61.0	327.0	331.0
54	STEPD3	156.0	619.0	597.0	62.0	328.0	332.0
55	HDDMA SUM	2.0	1.0	0.0	2.0	0.0	0.0
56	HDDMA1	89.0	670.0		38.0		
57	HDDMA2						
58	HDDMA3						
59	ECDDWT1	42.0	84.0	121.0	36.0	37.0	198.0

Table 23: Interstate Loading Data - EWMA

	method	ref0xgb	ref0rf	ref0mlp	ref1000xgb	ref1000rf	ref1000mlp
0	BOCPDSUM	19	7	1	7	11	1
1	BOCPDbcpd2	136.0				445.0	
2	BOCPDbcpd1	84.0	223.0		71.0	138.0	
3	BOCPDbcpd3						
4	EWMA SUM	42	22	2	18	5	2
5	EWMA3	46.0	428.0		617.0		
6	EWMA2	45.0	427.0		616.0		
7	EWMA1	44	426	470	475	780	370
8	TSSUM	19	7	1	7	11	1
9	TS3						
10	TS2						
11	TS1						
12	ADWINSUM	5	1	1	3	1	0
13	ADWIN3						
14	ADWIN2						
15	ADWIN1	138.0	644.0	872.0	284.0	788.0	
16	HDMMASUM	2	1	1	3	1	0
17	HDMMA3						
18	HDMMA2						
19	HDMMA1	160.0	672.0	864.0	288.0		
20	ECDDWT SUM	19	22	6	55	11	8
21	ECDDWT3	88.0	616.0		38.0	479.0	
22	ECDDWT2	87.0	615.0		37.0	38.0	
23	HalfSpaceTreesSUM	700	700	700	600	600	600
24	HalfSpaceTrees1	201	201	201	201	201	201
25	HalfSpaceTrees2	202	202	202	202	202	202
26	HalfSpaceTrees3	203	203	203	203	203	203
27	martingaleSUM	0	0	0	0	0	0
28	martingale2SUM	0	0	5	0	0	0
29	martingale1						
30	martingale2						
31	martingale3						
32	martingale21			22.0			
33	martingale22			23.0			
34	martingale23			24.0			
35	fixSUM	24	40	59	50	44	46
36	fix1	38	38	56	35	35	77
37	fix2	39.0	138.0	117.0	36.0	36.0	
38	fix3	137.0			37.0	37.0	
39	addisSUM	24	40	59	50	44	46
40	addis1	135	137	117	300	305	195
41	addis2	138.0	138.0		389.0	733.0	
42	addis3						
43	saffronSUM	24	40	59	50	44	46
44	saffron1						
45	saffron2						
46	saffron3						
47	DDMSUM	488	862	326	766	64	456
48	DDM1	39	39	575	35	36	344
49	DDM2	40	40	576	36	37	345
50	DDM3	41	41	577	37	38	346
51	STEPDSUM	16	55	5	113	22	9
52	STEPD1	154	617	595	60	326	330
53	STEPD2	155	618	596	61	327	331
54	STEPD3	156	619	597	62	328	332
55	HDDMASUM	2	1	0	2	0	0
56	HDDMA1	89.0	670.0		38.0		
57	HDDMA2						
58	HDDMA3						
59	ECDDWT1	42	84	121	36	37	198

Table 24: NYC-Taxi Data XGBoost Delay Statistic - starting from 7am

	Methods	mean	std	DetectionRate	q25	q75
0	BOCPDSUM	889.36	313.81	1.0	685.5	1094.5
1	BOCPDbcpd2	86.14	36.63	1.0	61.0	102.0
2	BOCPDbcpd1	67.29	25.55	1.0	54.0	70.0
3	BOCPDbcpd3	1359.38	2068.43	0.9299999999999999	173.5	1428.75
4	EWMA3SUM	1009.8	223.97	1.0	887.5	1136.0
5	EWMA3	227.08	316.86	1.0	80.0	244.5
6	EWMA2	221.51	316.63	1.0	78.5	239.0
7	EWMA1	210.03	305.34	1.0	75.0	210.5
8	TSSUM	172.24	21.85	1.0	163.0	178.0
9	TS3			0.0		
10	TS2			0.0		
11	TS1			0.0		
12	ADWINSUM	16.39	4.57	1.0	14.0	19.0
13	ADWIN3			0.0		
14	ADWIN2			0.0		
15	ADWIN1	770.87	785.73	1.0	302.0	957.5
16	HDMMASUM	12.3	3.88	1.0	10.0	15.0
17	HDMMA3			0.0		
18	HDMMA2			0.0		
19	HDMMA1	1001.33	960.98	1.0	416.0	1248.0
20	ECDDWT3SUM	78.22	21.93	1.0	64.0	89.0
21	ECDDWT3	2222.82	1878.76	1.0	786.25	3142.0
22	ECDDWT2	1937.99	1665.85	1.0	685.25	2660.0
23	HalfSpaceTreesSUM	11030.98	1246.8	1.0	10800.0	10800.0
24	HalfSpaceTrees1	201.0	0.0	1.0	201.0	201.0
25	HalfSpaceTrees2	202.0	0.0	1.0	202.0	202.0
26	HalfSpaceTrees3	203.0	0.0	1.0	203.0	203.0
27	martingaleSUM	0.04	0.63	1.0	0.0	0.0
28	martingale2SUM	0.57	3.05	1.0	0.0	0.0
29	martingale1	32.0	11.31	0.010000000000000009	28.0	36.0
30	martingale2	25.0		0.0	25.0	25.0
31	martingale3	26.0		0.0	26.0	26.0
32	martingale21	22.3	5.16	0.069999999999999995	18.0	25.5
33	martingale22	24.18	6.78	0.0500000000000000044	19.0	29.0
34	martingale23	26.17	7.02	0.0400000000000000036	20.75	32.25
35	fixSUM	578.95	75.04	1.0	546.0	593.0
36	fix1	43.88	23.67	1.0	26.0	55.5
37	fix2	389.6	358.62	1.0	110.5	564.5
38	fix3	3998.59	3113.81	0.87	1369.5	6139.0
39	addisSUM	256.7	33.73	1.0	242.0	268.0
40	addis1	142.14	65.16	1.0	91.0	184.0
41	addis2	1538.55	1465.83	0.99	511.0	2078.0
42	addis3	5474.22	3153.71	0.209999999999999996	3097.75	8110.5
43	saffronSUM	0.0	0.0	1.0	0.0	0.0
44	saffron1			0.0		
45	saffron2			0.0		
46	saffron3			0.0		
47	DDMSUM	2416.99	3802.1	1.0	0.0	3782.5
48	DDM1	2288.02	3492.39	0.62	63.0	4797.0
49	DDM2	2314.12	3495.04	0.61	67.75	4809.75
50	DDM3	2341.97	3506.14	0.61	70.25	4838.25
51	STEPDSUM	71.06	43.22	1.0	42.0	91.0
52	STEPD1	1807.87	1496.5	1.0	747.0	2537.0
53	STEPD2	2126.84	1820.41	1.0	863.25	3056.25
54	STEPD3	2416.78	2034.59	0.99	966.0	3466.5
55	HDDMASUM	0.11	0.34	1.0	0.0	0.0
56	HDDMA1	6755.06	4343.22	0.099999999999999998	3335.25	10869.5
57	HDDMA2			0.0		
58	HDDMA3			0.0		
59	ECDDWT1	362.33	297.7	1.0	154.0	464.5

Table 25: NYC-Taxi Data XGBoost Delay Statistic - starting from 10am

	Methods	mean	std	DetectionRate	q25	q75
0	BOCPDSUM	885.52	325.33	1.0	668.5	1073.5
1	BOCPDbcpd2	89.9	41.73	1.0	62.0	103.0
2	BOCPDbcpd1	66.95	22.93	1.0	53.0	72.0
3	BOCPDbcpd3	1532.88	2327.15	0.92	203.0	1686.5
4	EWMA3SUM	1053.53	251.7	1.0	931.0	1196.0
5	EWMA3	194.26	204.38	1.0	79.5	226.5
6	EWMA2	184.53	198.81	1.0	78.0	213.0
7	EWMA1	179.14	198.08	1.0	77.0	198.0
8	TSSUM	170.89	24.21	1.0	161.0	179.0
9	TS3			0.0		
10	TS2			0.0		
11	TS1			0.0		
12	ADWINSUM	15.73	4.46	1.0	13.0	18.5
13	ADWIN3			0.0		
14	ADWIN2			0.0		
15	ADWIN1	657.51	620.42	1.0	286.0	782.5
16	HDMMASUM	11.77	3.66	1.0	10.0	14.0
17	HDMMA3			0.0		
18	HDMMA2			0.0		
19	HDMMA1	879.65	863.66	0.99	384.0	1056.0
20	ECDDWTSUM	89.63	28.83	1.0	69.0	105.5
21	ECDDWT3	2013.49	1812.23	0.99	647.0	2758.0
22	ECDDWT2	1743.17	1499.37	0.99	563.0	2414.0
23	HalfSpaceTreesSUM	11028.93	1454.26	1.0	10800.0	10800.0
24	HalfSpaceTrees1	201.0	0.0	1.0	201.0	201.0
25	HalfSpaceTrees2	202.0	0.0	1.0	202.0	202.0
26	HalfSpaceTrees3	203.0	0.0	1.0	203.0	203.0
27	martingaleSUM	0.0	0.0	1.0	0.0	0.0
28	martingale2SUM	0.18	0.98	1.0	0.0	0.0
29	martingale1			0.0		
30	martingale2			0.0		
31	martingale3			0.0		
32	martingale21	24.62	7.41	0.050000000000000044	18.0	29.0
33	martingale22	24.14	7.01	0.020000000000000018	19.0	28.0
34	martingale23	28.5	7.18	0.020000000000000018	23.25	30.75
35	fixSUM	589.33	86.75	1.0	551.0	615.0
36	fix1	43.0	24.06	1.0	26.5	51.0
37	fix2	354.4	346.79	1.0	124.5	475.5
38	fix3	4070.52	2993.3	0.85	1529.75	6272.75
39	addisSUM	249.7	36.06	1.0	232.0	266.0
40	addis1	139.99	61.49	1.0	92.5	174.5
41	addis2	1439.05	1274.44	1.0	540.0	1952.5
42	addis3	5424.5	3181.8	0.22999999999999998	2271.0	7902.25
43	saffronSUM	0.0	0.0	1.0	0.0	0.0
44	saffron1			0.0		
45	saffron2			0.0		
46	saffron3			0.0		
47	DDMSUM	3399.43	4022.3	1.0	0.0	5608.5
48	DDM1	2419.52	3145.96	0.69	67.0	5663.0
49	DDM2	2448.05	3142.28	0.69	68.75	5688.25
50	DDM3	2506.94	3157.28	0.6699999999999999	76.5	5791.5
51	STEPDSUM	94.22	62.05	1.0	49.0	126.0
52	STEPD1	1591.72	1489.63	1.0	522.75	2216.75
53	STEPD2	1887.1	1722.23	0.99	570.0	2599.0
54	STEPD3	2198.25	1898.24	0.99	672.5	3065.0
55	HDDMASUM	0.43	0.56	1.0	0.0	1.0
56	HDDMA1	7181.74	2101.88	0.39	6527.0	8246.0
57	HDDMA2			0.0		
58	HDDMA3			0.0		
59	ECDDWT1	327.05	245.35	1.0	152.0	457.5

Table 26: NYC-Taxi Data MLP Model Delay Statistic - starting from 3am

	Methods	mean	std	DetectionRate	q25	q75
0	BOCPDSUM	402.07	438.02	0.66	68.0	749.0
1	BOCPDbcpd2	99.67	53.28	0.66	67.0	114.0
2	BOCPDbcpd1	67.65	19.27	0.66	55.0	73.0
3	BOCPDbcpd3	1342.62	1988.3	0.41	236.0	1290.75
4	EWMA3	444.29	439.91	0.66	76.0	884.0
5	EWMA3	228.64	230.6	0.66	89.0	279.25
6	EWMA2	216.65	223.35	0.66	85.0	262.0
7	EWMA1	206.92	222.69	0.66	82.0	244.0
8	TSSUM	87.82	76.44	0.66	22.0	168.0
9	TS3			0.0		
10	TS2			0.0		
11	TS1			0.0		
12	ADWINSUM	7.54	7.41	0.66	1.0	15.0
13	ADWIN3			0.0		
14	ADWIN2			0.0		
15	ADWIN1	653.63	562.27	0.61	301.5	830.25
16	HDMMASUM	5.74	5.58	0.66	1.0	11.0
17	HDMMA3			0.0		
18	HDMMA2			0.0		
19	HDMMA1	802.11	703.56	0.59	352.0	992.0
20	ECDDWTSUM	42.35	49.68	0.66	5.0	71.0
21	ECDDWT3	1904.02	2226.09	0.43	449.25	2237.5
22	ECDDWT2	1446.88	1561.33	0.47	334.25	1758.0
23	HalfSpaceTreesSUM	5486.2	4980.7	0.66	1200.0	10800.0
24	HalfSpaceTrees1	201.0	0.0	0.66	201.0	201.0
25	HalfSpaceTrees2	202.0	0.0	0.66	202.0	202.0
26	HalfSpaceTrees3	203.0	0.0	0.66	203.0	203.0
27	martingaleSUM	1.76	19.62	0.66	0.0	0.0
28	martingale2SUM	1.68	14.8	0.66	0.0	0.0
29	martingale1	224.23	152.39	0.01	39.0	323.0
30	martingale2	251.33	137.47	0.01	194.75	327.75
31	martingale3	291.4	118.35	0.01	248.25	337.0
32	martingale21	72.51	123.83	0.04	21.5	31.0
33	martingale22	88.96	139.32	0.03	22.0	37.0
34	martingale23	95.48	143.58	0.03	23.0	40.0
35	fixSUM	258.85	241.0	0.66	52.0	508.0
36	fix1	41.88	23.65	0.66	26.0	49.0
37	fix2	387.64	373.89	0.62	109.0	529.0
38	fix3	2971.73	2978.66	0.28	781.0	4342.5
39	addisSUM	110.41	98.44	0.66	25.0	211.0
40	addis1	130.89	60.99	0.66	88.0	160.0
41	addis2	1090.62	1159.53	0.45	344.5	1321.5
42	addis3	3274.2	3009.09	0.06	1192.75	4897.0
43	saffronSUM	0.1	2.45	0.66	0.0	0.0
44	saffron1			0.0		
45	saffron2			0.0		
46	saffron3			0.0		
47	DDMSUM	1779.78	3449.21	0.66	0.0	1308.0
48	DDM1	1215.28	2589.38	0.37	57.0	975.0
49	DDM2	1295.36	2680.22	0.37	61.0	1046.5
50	DDM3	1333.28	2711.16	0.36	65.0	1129.25
51	STEPDSUM	60.23	101.79	0.66	0.0	72.0
52	STEPD1	1556.52	1983.72	0.47	307.25	1618.0
53	STEPD2	1810.23	2226.61	0.46	327.25	2121.5
54	STEPD3	2007.61	2431.17	0.45	389.0	2281.0
55	HDDMASUM	0.23	0.52	0.66	0.0	0.0
56	HDDMA1	2910.7	2794.24	0.12	1548.0	2713.0
57	HDDMA2			0.0		
58	HDDMA3			0.0		
59	ECDDWT1	416.29	339.63	0.64	164.0	620.5

Table 27: NYC-Taxi Data MLP Model Delay Statistic - starting from 11am

	Methods	mean	std	DetectionRate	q25	q75
0	BOCPDSUM	449.07	453.47	1.0	94.0	848.0
1	BOCPDbcpd2	91.26	38.75	1.0	63.0	108.0
2	BOCPDbcpd1	67.03	22.59	1.0	54.0	71.0
3	BOCPDbcpd3	998.55	1574.13	0.74	219.0	1068.0
4	EWMA3	495.76	487.68	1.0	85.0	990.0
5	EWMA3	218.98	264.68	1.0	86.25	241.0
6	EWMA2	207.88	260.01	1.0	81.25	233.5
7	EWMA1	198.23	251.06	1.0	78.25	218.0
8	TSSUM	87.0	76.77	1.0	21.0	168.0
9	TS3			0.0		
10	TS2			0.0		
11	TS1			0.0		
12	ADWINSUM	8.21	8.13	1.0	1.0	16.0
13	ADWIN3			0.0		
14	ADWIN2			0.0		
15	ADWIN1	666.61	611.04	0.93	309.0	862.0
16	HDMMASUM	6.01	6.24	1.0	1.0	12.0
17	HDMMA3			0.0		
18	HDMMA2			0.0		
19	HDMMA1	838.34	720.64	0.85	416.0	1048.0
20	ECDDWTSUM	39.48	38.02	1.0	8.0	73.0
21	ECDDWT3	1613.69	1626.07	0.72	560.5	2107.5
22	ECDDWT2	1404.88	1397.45	0.75	484.0	1801.5
23	HalfSpaceTreesSUM	5485.88	4990.68	1.0	1200.0	10800.0
24	HalfSpaceTrees1	201.0	0.0	1.0	201.0	201.0
25	HalfSpaceTrees2	202.0	0.0	1.0	202.0	202.0
26	HalfSpaceTrees3	203.0	0.0	1.0	203.0	203.0
27	martingaleSUM	0.01	0.2	1.0	0.0	0.0
28	martingale2SUM	0.42	2.43	1.0	0.0	0.0
29	martingale1	47.5	36.06	0.0	34.75	60.25
30	martingale2	51.5	40.31	0.0	37.25	65.75
31	martingale3			0.0		
32	martingale21	25.05	7.27	0.07	19.0	31.0
33	martingale22	27.59	10.02	0.04	20.0	35.5
34	martingale23	29.79	10.86	0.03	21.0	36.5
35	fixSUM	292.66	259.49	1.0	69.0	564.0
36	fix1	44.2	23.18	1.0	26.0	56.0
37	fix2	390.2	340.55	0.99	134.0	566.0
38	fix3	3358.86	3038.82	0.46	895.25	5338.25
39	addisSUM	131.81	113.32	1.0	34.0	250.0
40	addis1	131.38	59.52	1.0	87.0	162.0
41	addis2	1155.38	1193.08	0.79	458.0	1297.0
42	addis3	5067.13	3299.96	0.1	2200.5	7676.5
43	saffronSUM	0.12	2.98	1.0	0.0	0.0
44	saffron1			0.0		
45	saffron2			0.0		
46	saffron3			0.0		
47	DDMSUM	1210.2	2741.01	1.0	0.0	862.0
48	DDM1	1440.29	2839.15	0.55	60.0	878.5
49	DDM2	1464.65	2857.21	0.54	63.0	937.0
50	DDM3	1496.07	2877.75	0.53	68.0	994.0
51	STEPDSUM	35.27	43.63	1.0	2.0	55.0
52	STEPD1	1334.64	1284.55	0.79	529.0	1605.5
53	STEPD2	1586.92	1617.88	0.74	587.5	2025.25
54	STEPD3	1813.63	1870.64	0.7	649.25	2276.5
55	HDDMASUM	0.06	0.26	1.0	0.0	0.0
56	HDDMA1	5997.12	4660.53	0.05	1150.0	10836.0
57	HDDMA2			0.0		
58	HDDMA3			0.0		
59	ECDDWT1	357.44	276.68	1.0	157.75	470.0

Table 28: False Alarm Rate for the Video Data Streaming(Sliding Window Asynchronous) - Un-weighted Procedure

mean	sd	method
0.03	0.0336	BOCPDSUM
0.0119	0.0111	EWMASUM
0.03	0.0336	TSSUM
0.0001	0.0002	ADWINSUM
0.0001	0.0001	HDMMASUM
0.0028	0.0024	ECDDWTSUM
0.475	0.4167	HalfSpaceTreesSUM
0	0	martingaleSUM
0	0.0001	martingale2SUM
0.0249	0.021	fixSUM
0.0114	0.0092	addisSUM
0.0086	0.0032	saffronSUM
0.0768	0.213	DDMSUM
0.0022	0.0023	STEPDSUM
0	0	HDDMASUM

Table 29: Asynchronous Detection Delay & Count for the Video Data(all videos/methods summarizations) - **addisSUM** is the asynchronous testing procedure with total number of detections, first detection, first 2 consecutive detections and first 3 consecutive detections

	method	mean	sd
0	BOCPDSUM	18.35	16.23
1	BOCPDbcpd2	1052.0	1132.52
2	BOCPDbcpd1	110.45	42.78
3	BOCPDbcpd3		
4	EWMA SUM	133.42	164.2
5	EWMA3	191.64	167.65
6	EWMA2	152.31	134.96
7	EWMA1	135.59	122.77
8	TSSUM	18.35	16.23
9	TS3		
10	TS2		
11	TS1		
12	ADWINSUM	27.64	28.44
13	ADWIN3		
14	ADWIN2		
15	ADWIN1	54.53	23.75
16	HDMMASUM	21.44	22.54
17	HDMMA3		
18	HDMMA2		
19	HDMMA1	66.33	28.79
20	ECDDWTSUM	320.71	295.73
21	ECDDWT3	285.18	92.92
22	ECDDWT2	284.18	92.92
23	HalfSpaceTreesSUM	642.16	650.05
24	HalfSpaceTrees1	201.0	0.0
25	HalfSpaceTrees2	202.0	0.0
26	HalfSpaceTrees3	203.0	0.0
27	martingaleSUM	1357.05	1320.77
28	martingale2SUM	1313.87	1283.48
29	martingale1	32.25	77.09
30	martingale2	47.38	124.95
31	martingale3	48.71	124.93
32	martingale21	29.65	81.4
33	martingale22	33.6	83.21
34	martingale23	36.16	83.33
35	fixSUM	551.25	593.39
36	fix1	13.96	23.91
37	fix2	20.71	26.84
38	fix3	40.8	39.96
39	addisSUM	340.15	491.31
40	addis1	35.35	55.66
41	addis2	41.31	55.41
42	addis3	68.17	71.49
43	DDMSUM	1128.69	1392.34
44	DDM1	283.18	92.92
45	DDM2	284.18	92.92
46	DDM3	285.18	92.92
47	STEPDSUM	1106.71	1042.4
48	STEPD1	61.02	5.29
49	STEPD2	62.02	5.29
50	STEPD3	63.02	5.29
51	HDDMASUM	3.6	3.28
52	HDDMA1	328.42	99.03
53	HDDMA2		
54	HDDMA3		
55	ECDDWT1	283.18	92.92

Table 30: Benchmark Comparison for the Video Data Batch Testing Procedures, three subsets of the dataframes of the video embeddings.

	df1df2	df2df3	df1df3
xu16	0.06	0.12	0.08
pan14	1.0	1.0	1.0
chen10	0.0	0.02	0.06
chen14	0.1	0.16	0.08
c2st	0.06	0.08	0.04
sri	0.06	0.04	0.08
skk	0.62	0.68	0.74
zwl	0.0	0.0	0.0
zwlm	0.0	0.02	0.04
bai96	0.1	0.14	0.08
rrperm	1.0	1.0	1.0
drperm	1.0	1.0	1.0
miles16	0.12	0.16	0.14
kmmd	0.14	0.08	0.22
rfperm	1.0	1.0	1.0
xgbperm	1.0	1.0	1.0

Table 31: Asynchronous Detection Delay & Count for the Video Data(all videos/methods summarizations) - **addisSUM** is the asynchronous testing procedure with total number of detections, first detection, first 2 consecutive detections and first 3 consecutive detections

	method	mean	sd
0	BOCPDSUM	18.35	16.23
1	BOCPDbcpd2	1052.0	1132.52
2	BOCPDbcpd1	110.45	42.78
3	BOCPDbcpd3		
4	EWMA3	133.42	164.2
5	EWMA3	191.64	167.65
6	EWMA2	152.31	134.96
7	EWMA1	135.59	122.77
8	TSSUM	18.35	16.23
9	TS3		
10	TS2		
11	TS1		
12	ADWINSUM	27.64	28.44
13	ADWIN3		
14	ADWIN2		
15	ADWIN1	54.53	23.75
16	HDMMASUM	21.44	22.54
17	HDMMA3		
18	HDMMA2		
19	HDMMA1	66.33	28.79
20	ECDDWTSUM	320.71	295.73
21	ECDDWT3	285.18	92.92
22	ECDDWT2	284.18	92.92
23	HalfSpaceTreesSUM	642.16	650.05
24	HalfSpaceTrees1	201.0	0.0
25	HalfSpaceTrees2	202.0	0.0
26	HalfSpaceTrees3	203.0	0.0
27	martingaleSUM	1357.05	1320.77
28	martingale2SUM	1313.87	1283.48
29	martingale1	32.25	77.09
30	martingale2	47.38	124.95
31	martingale3	48.71	124.93
32	martingale21	29.65	81.4
33	martingale22	33.6	83.21
34	martingale23	36.16	83.33
35	fixSUM	551.25	593.39
36	fix1	13.96	23.91
37	fix2	20.71	26.84
38	fix3	40.8	39.96
39	addisSUM	340.15	491.31
40	addis1	35.35	55.66
41	addis2	41.31	55.41
42	addis3	68.17	71.49
43	DDMSUM	1128.69	1392.34
44	DDM1	283.18	92.92
45	DDM2	284.18	92.92
46	DDM3	285.18	92.92
47	STEPDSUM	1106.71	1042.4
48	STEPD1	61.02	5.29
49	STEPD2	62.02	5.29
50	STEPD3	63.02	5.29
51	HDDMASUM	3.6	3.28
52	HDDMA1	328.42	99.03
53	HDDMA2		
54	HDDMA3		
55	ECDDWT1	283.18	92.92

Table 32: False Alarm Rate for the Weighted Procedure in the Video Sliding Window Asynchronous Testing

mean	sd	method
0.0291	0.0351	BOCPDSUM
0.0112	0.0111	EWMASUM
0.0291	0.0351	TSSUM
0.0001	0.0002	ADWINSUM
0.0001	0.0001	HDMMASUM
0.0027	0.0025	ECDDWTSUM
0.475	0.4107	HalfSpaceTreesSUM
0.4942	0.4108	martingaleSUM
0.4929	0.4107	martingale2SUM
0.1244	0.1031	fixSUM
0.0342	0.0479	addisSUM
0.0279	0.0146	saffronSUM
0.0842	0.2317	DDMSUM
0.0021	0.0025	STEPDSUM
0	0	HDDMASUM

Table 33: False Alarm Rate for the Unweighted Procedure in the Video Sliding Window Asynchronous Testing

mean	sd	method
0.0291	0.0351	BOCPDSUM
0.0112	0.0111	EWMASUM
0.0291	0.0351	TSSUM
0.0001	0.0002	ADWINSUM
0.0001	0.0001	HDMMASUM
0.0027	0.0025	ECDDWTSUM
0.475	0.4107	HalfSpaceTreesSUM
0.4942	0.4108	martingaleSUM
0.4929	0.4107	martingale2SUM
0.1244	0.1031	fixSUM
0.0342	0.0479	addisSUM
0.0279	0.0146	saffronSUM
0.0842	0.2317	DDMSUM
0.0021	0.0025	STEPDSUM
0	0	HDDMASUM

Table 34: False Alarm Rate for the Unweighted Procedure in the Video Sliding Window Asynchronous Testing

mean	sd	method
0.0291	0.0351	BOCPDSUM
0.0112	0.0111	EWMASUM
0.0291	0.0351	TSSUM
0.0001	0.0002	ADWINSUM
0.0001	0.0001	HDMMASUM
0.0027	0.0025	ECDDWTSUM
0.475	0.4107	HalfSpaceTreesSUM
0.4942	0.4108	martingaleSUM
0.4929	0.4107	martingale2SUM
0.1244	0.1031	fixSUM
0.0342	0.0479	addisSUM
0.0279	0.0146	saffronSUM
0.0842	0.2317	DDMSUM
0.0021	0.0025	STEPDSUM
0	0	HDDMASUM

Table 35: Waymo Dataset - MLP Regressor EWMA

	mean	std	detection _{rate}	q25	q75	median
BOCPDSUM	142.83	207.44	1.0	11.0	265.0	24.0
BOCPDbcpd2	129.73	96.57	0.93	68.0	155.0	98.0
BOCPDbcpd1	96.87	66.55	0.99	59.0	108.0	75.0
BOCPDbcpd3	796.92	1131.04	0.43	133.0	921.0	332.5
EWMA3SUM	190.95	328.7	1.0	5.0	288.0	17.0
EWMA3	275.47	350.15	0.77	107.5	324.5	184.0
EWMA2	257.69	337.02	0.83	102.0	295.0	173.0
EWMA1	241.68	321.07	0.88	94.0	276.5	160.0
TSSUM	142.83	207.44	1.0	11.0	265.0	24.0
TS3			0.0			
TS2			0.0			
TS1			0.0			
ADWINSUM	1.48	1.5	1.0	0.0	2.0	1.0
ADWIN3			0.0			
ADWIN2			0.0			
ADWIN1	920.99	1174.11	0.74	213.0	1190.5	410.5
HDMMASUM	1.36	1.36	1.0	0.0	2.0	1.0
HDMMA3			0.0			
HDMMA2			0.0			
HDMMA1	980.01	1224.79	0.73	224.0	1312.0	416.0
ECDDWT3SUM	20.74	33.7	1.0	1.0	28.0	9.0
ECDDWT3	1202.34	1535.69	0.47	212.0	1586.0	495.0
ECDDWT2	1109.91	1422.65	0.56	178.0	1488.0	440.0
HalfSpaceTreesSUM	2090.4	2824.27	1.0	158.0	4269.5	355.0
HalfSpaceTrees1	201.0	0.0	1.0	201.0	201.0	201.0
HalfSpaceTrees2	202.0	0.0	0.99	202.0	202.0	202.0
HalfSpaceTrees3	203.0	0.0	0.99	203.0	203.0	203.0
martingaleSUM	16.73	124.35	1.0	0.0	0.0	0.0
martingale2SUM	16.35	174.72	1.0	0.0	0.0	0.0
martingale1	199.8	341.07	0.11	79.0	251.0	136.0
martingale2	209.37	350.36	0.1	88.0	256.0	147.0
martingale3	217.22	355.34	0.1	96.75	258.25	160.0
martingale21	98.56	118.23	0.13	25.0	125.0	40.0
martingale22	113.81	123.74	0.11	30.0	151.5	51.0
martingale23	119.7	124.78	0.11	33.0	156.0	58.0
fixSUM	122.03	153.02	1.0	17.25	214.0	38.0
fix1	45.67	32.33	1.0	25.0	53.0	36.0
fix2	368.68	513.09	0.75	98.0	430.75	210.0
fix3	2137.48	2226.65	0.26	310.0	3638.25	1276.0
addisSUM	107.43	141.04	1.0	15.0	167.0	35.0
addis1	129.23	93.52	0.97	73.0	161.0	107.0
addis2	1296.96	1549.66	0.53	246.0	1901.0	561.0
addis3	2072.28	2291.32	0.05	310.0	3430.0	816.0
saffronSUM	94.43	143.55	1.0	3.0	117.75	25.0
saffron1	43.0		0.0	43.0	43.0	43.0
saffron2			0.0			
saffron3			0.0			
DDMSUM	640.56	1498.97	1.0	0.0	359.0	69.0
DDM1	541.41	1218.55	0.61	60.0	318.0	121.0
DDM2	547.19	1196.59	0.6	63.0	327.5	125.5
DDM3	556.43	1203.52	0.6	65.0	341.0	131.0
STEPDSUM	38.12	76.45	1.0	0.0	46.0	11.0
STEPD1	943.91	1455.39	0.68	102.0	1063.0	303.0
STEPD2	1024.93	1529.09	0.65	112.0	1236.0	322.0
STEPD3	1075.9	1595.88	0.63	113.0	1374.0	337.0
HDDMASUM	0.24	0.52	1.0	0.0	0.0	0.0
HDDMA1	1763.33	2005.19	0.21	380.0	2724.5	648.5
HDDMA2			0.0			
HDDMA3			0.0			
ECDDWT1	320.59	414.03	0.77	105.5	379.0	199.0

Table 36: Waymo Dataset - RF Regressor

	mean	std	detection _{rate}	q25	q75	median
BOCPDSUM	119.55	160.62	1.0	15.0	210.75	31.0
BOCPDbcpd2	127.02	105.42	0.94	67.0	148.0	96.0
BOCPDbcpd1	92.86	79.76	1.0	57.0	101.0	70.0
BOCPDbcpd3	928.05	1269.32	0.53	152.0	1156.0	343.0
EWMA3	191.05	328.66	1.0	5.0	287.75	18.0
EWMA2	277.15	354.87	0.77	105.0	330.25	180.0
EWMA1	260.46	343.03	0.83	99.0	298.0	167.0
TSSUM	244.75	327.16	0.88	93.0	283.5	158.0
TS3	119.55	160.62	1.0	15.0	210.75	31.0
TS2			0.0			
TS1			0.0			
ADWINSUM	1.4	1.25	1.0	1.0	2.0	1.0
ADWIN3			0.0			
ADWIN2			0.0			
ADWIN1	987.41	1313.71	0.78	198.0	1300.0	376.0
HDMMASUM	1.31	1.15	1.0	1.0	2.0	1.0
HDMMA3			0.0			
HDMMA2			0.0			
HDMMA1	994.61	1293.03	0.77	192.0	1328.0	384.0
ECDDWTSUM	20.77	33.67	1.0	1.0	27.75	9.0
ECDDWT3	1190.06	1535.21	0.48	208.25	1579.75	460.0
ECDDWT2	1090.98	1416.93	0.57	180.5	1455.0	402.0
HalfSpaceTreesSUM	2090.54	2824.47	1.0	158.0	4269.5	355.0
HalfSpaceTrees1	201.0	0.0	1.0	201.0	201.0	201.0
HalfSpaceTrees2	202.0	0.0	0.99	202.0	202.0	202.0
HalfSpaceTrees3	203.0	0.0	0.99	203.0	203.0	203.0
martingaleSUM	2281.25	2825.18	1.0	342.0	4460.75	541.5
martingale2SUM	2235.78	2806.54	1.0	314.5	4432.75	514.0
martingale1	8.9	6.0	1.0	4.0	13.0	8.0
martingale2	10.6	6.38	1.0	5.0	15.0	10.0
martingale3	12.15	6.5	1.0	6.0	17.0	12.0
martingale21	22.71	15.57	1.0	12.0	30.5	20.0
martingale22	25.81	18.05	1.0	15.0	33.0	23.0
martingale23	28.72	19.66	1.0	17.0	37.0	26.0
fixSUM	565.29	694.41	1.0	86.25	1117.5	136.0
fix1	2.73	2.27	1.0	1.0	4.0	2.0
fix2	22.41	23.66	1.0	6.0	31.0	15.0
fix3	135.82	131.23	0.88	39.0	192.0	100.0
addisSUM	529.03	652.88	1.0	80.0	1063.5	131.5
addis1	5.57	3.39	1.0	3.0	7.0	5.0
addis2	41.47	41.4	1.0	12.0	57.0	28.0
addis3	244.18	251.98	0.74	70.75	332.25	174.5
saffronSUM	431.52	646.84	1.0	51.0	998.5	108.0
saffron1	46.0	21.63	0.0	34.0	55.0	40.0
saffron2			0.0			
saffron3			0.0			
DDMSUM	642.88	1498.29	1.0	0.0	364.0	74.0
DDM1	532.35	1208.42	0.63	62.0	299.0	116.0
DDM2	536.11	1183.52	0.62	63.75	313.5	123.5
DDM3	547.79	1193.15	0.61	66.75	321.75	129.5
STEPDSUM	37.96	76.19	1.0	0.0	46.75	11.0
STEPD1	939.78	1453.72	0.68	107.5	1063.0	296.0
STEPD2	1022.76	1530.88	0.65	110.5	1244.0	312.0
STEPD3	1085.93	1606.15	0.62	115.5	1391.5	321.0
HDDMASUM	0.26	0.53	1.0	0.0	0.0	0.0
HDDMA1	1674.01	1983.14	0.23	321.25	2592.5	541.5
HDDMA2			0.0			
HDDMA3			0.0			
ECDDWT1	317.5	414.19	0.77	103.0	380.25	195.5

Table 37: Waymo Dataset - XGB Regressor

	mean	std	detection _{rate}	q25	q75	median
BOCPDSUM	58.58	92.65	1.0	4.0	84.0	12.0
BOCPDbcpd2	272.73	339.54	0.83	95.0	316.25	166.0
BOCPDbcpd1	232.5	285.56	0.89	82.0	265.0	143.0
BOCPDbcpd3	1634.62	1749.7	0.29	261.5	2728.0	904.5
EWMA3	248.24	413.55	1.0	4.0	393.0	22.0
EWMA2	319.73	370.86	0.77	126.0	359.0	217.0
EWMA1	307.91	364.4	0.8	123.0	343.75	205.5
TSSUM	301.28	360.94	0.82	121.0	333.0	197.0
TS3	58.58	92.65	1.0	4.0	84.0	12.0
TS2			0.0			
TS1			0.0			
ADWINSUM	0.82	1.18	1.0	0.0	1.0	0.0
ADWIN3			0.0			
ADWIN2			0.0			
ADWIN1	1595.32	1574.45	0.47	353.0	2602.0	949.0
HDMMASUM	0.81	1.13	1.0	0.0	1.0	0.0
HDMMA3			0.0			
HDMMA2			0.0			
HDMMA1	1575.61	1499.84	0.48	352.0	2592.0	992.0
ECDDWTSUM	17.0	21.71	1.0	1.0	27.0	10.0
ECDDWT3	1295.11	1573.83	0.47	215.0	2004.5	493.5
ECDDWT2	1154.56	1455.85	0.55	178.0	1746.0	413.0
HalfSpaceTreesSUM	2073.47	2811.78	1.0	158.0	4247.0	355.0
HalfSpaceTrees1	201.0	0.0	1.0	201.0	201.0	201.0
HalfSpaceTrees2	202.0	0.0	0.99	202.0	202.0	202.0
HalfSpaceTrees3	203.0	0.0	0.99	203.0	203.0	203.0
martingaleSUM	1286.02	2671.93	1.0	0.0	135.0	0.0
martingale2SUM	1414.7	2777.6	1.0	0.0	185.0	0.0
martingale1	262.74	832.5	0.32	26.0	151.0	53.0
martingale2	262.92	834.21	0.32	27.0	152.0	55.0
martingale3	266.36	838.29	0.32	28.0	156.0	56.0
martingale21	52.08	196.51	0.33	17.0	35.25	22.0
martingale22	58.43	213.46	0.32	18.0	38.0	24.0
martingale23	61.13	215.72	0.32	19.0	40.0	26.0
fixSUM	132.55	319.56	1.0	18.0	223.0	38.0
fix1	44.49	35.39	0.99	25.0	50.0	34.0
fix2	294.88	359.46	0.75	86.75	369.25	192.0
fix3	1816.26	2085.35	0.23	182.0	3193.5	742.0
addisSUM	107.31	311.11	1.0	12.0	158.0	28.0
addis1	245.22	515.53	0.85	70.0	212.0	106.0
addis2	640.23	1042.64	0.27	166.0	504.0	288.0
addis3	541.39	1075.13	0.02	163.5	382.5	208.0
saffronSUM	103.14	312.04	1.0	4.0	141.0	25.0
saffron1	1839.22	1904.41	0.04	207.0	2930.5	1545.0
saffron2	1879.44	1574.41	0.03	264.0	3085.0	1835.0
saffron3	2370.19	1457.62	0.02	1412.5	3459.25	2238.5
DDMSUM	524.18	1342.54	1.0	0.0	353.0	80.0
DDM1	459.5	1097.04	0.62	52.0	224.0	96.0
DDM2	472.89	1109.53	0.61	54.0	230.5	100.0
DDM3	479.11	1116.17	0.6	56.0	233.0	102.0
STEPDSUM	31.94	55.47	1.0	0.0	39.0	10.0
STEPD1	877.77	1316.13	0.68	104.0	1033.0	285.0
STEPD2	937.03	1395.52	0.65	109.0	1146.5	288.5
STEPD3	1014.96	1493.79	0.63	112.0	1290.0	304.0
HDDMASUM	0.18	0.4	1.0	0.0	0.0	0.0
HDDMA1	1412.59	1861.14	0.17	274.0	2292.5	457.0
HDDMA2			0.0			
HDDMA3			0.0			
ECDDWT1	275.45	269.31	0.79	104.0	352.0	197.0

Table 38: Waymo Dataset - MLP Regressor

	mean	std	detection _{rate}	q25	q75	median
BOCPDSUM	186.85	256.72	1.0	15.0	382.0	32.0
BOCPDbcpd2	108.6	96.91	0.98	64.0	122.0	84.0
BOCPDbcpd1	86.48	54.2	1.0	56.0	93.0	67.0
BOCPDbcpd3	419.04	639.82	0.74	113.0	394.75	197.0
EWMA3	152.77	241.69	1.0	10.0	214.0	38.0
EWMA3	199.71	217.8	0.85	89.0	243.75	146.0
EWMA2	188.12	209.52	0.88	86.0	225.25	138.5
EWMA1	176.57	199.34	0.91	83.0	208.0	130.0
TSSUM	186.85	256.72	1.0	15.0	382.0	32.0
TS3			0.0			
TS2			0.0			
TS1			0.0			
ADWINSUM	1.97	2.64	1.0	0.0	3.0	1.0
ADWIN3			0.0			
ADWIN2			0.0			
ADWIN1	735.2	849.01	0.64	251.75	850.5	388.0
HDMMASUM	1.66	2.22	1.0	0.0	2.0	1.0
HDMMA3			0.0			
HDMMA2			0.0			
HDMMA1	851.1	921.33	0.6	288.0	1088.0	480.0
ECDDWTSUM	20.71	32.63	1.0	1.0	27.0	7.0
ECDDWT3	1149.03	1324.07	0.48	249.75	1641.5	526.0
ECDDWT2	1058.42	1284.83	0.53	200.0	1488.5	473.0
HalfSpaceTreesSUM	2090.61	2824.3	1.0	158.0	4274.0	355.0
HalfSpaceTrees1	201.0	0.0	1.0	201.0	201.0	201.0
HalfSpaceTrees2	202.0	0.0	0.99	202.0	202.0	202.0
HalfSpaceTrees3	203.0	0.0	0.99	203.0	203.0	203.0
martingaleSUM	6.39	33.08	1.0	0.0	0.0	0.0
martingale2SUM	4.48	25.05	1.0	0.0	0.0	0.0
martingale1	201.35	130.28	0.07	97.75	253.5	183.0
martingale2	208.91	131.78	0.06	102.5	261.25	188.0
martingale3	211.89	131.88	0.06	104.5	265.5	192.0
martingale21	101.88	113.22	0.08	24.0	138.0	42.0
martingale22	121.17	125.67	0.07	26.5	170.5	55.0
martingale23	133.89	129.76	0.06	32.25	202.25	95.0
fixSUM	120.35	150.79	1.0	17.0	229.0	33.0
fix1	42.79	27.02	1.0	25.0	50.0	35.0
fix2	302.07	348.63	0.74	91.0	380.0	195.5
fix3	1941.81	1883.71	0.26	316.0	3144.0	1337.0
addisSUM	105.92	138.74	1.0	14.0	191.0	29.0
addis1	111.69	61.47	0.99	72.0	141.0	100.0
addis2	1194.6	1344.21	0.48	276.0	1624.0	679.0
addis3	2762.27	2218.45	0.06	559.5	4103.0	2908.0
saffronSUM	92.1	140.9	1.0	5.0	135.0	22.0
saffron1	31.0		0.0	31.0	31.0	31.0
saffron2			0.0			
saffron3			0.0			
DDMSUM	543.96	1349.64	1.0	0.0	329.0	32.0
DDM1	601.64	1296.98	0.59	52.5	301.5	105.0
DDM2	622.37	1326.71	0.58	55.0	318.0	109.0
DDM3	639.92	1342.64	0.57	57.0	325.75	114.0
STEPDSUM	28.99	57.96	1.0	0.0	34.0	6.0
STEPD1	1010.62	1327.26	0.62	184.0	1286.75	389.5
STEPD2	1101.77	1464.11	0.58	190.0	1471.5	396.5
STEPD3	1154.68	1509.9	0.55	196.0	1649.5	407.0
HDDMASUM	0.22	0.68	1.0	0.0	0.0	0.0
HDDMA1	2197.63	2172.07	0.16	429.25	3581.0	1232.5
HDDMA2			0.0			
HDDMA3			0.0			
ECDDWT1	267.99	256.53	0.79	106.0	338.75	196.0

Table 39: Waymo Dataset - MLP Regressor

	mean	std	detectionrate	p25	p50	p75
BOCPDSUM	28.917	52.245	1.0	0.0	7.0	31.25
BOCPDbcpd2	98.484	44.053	0.5740740740740741	66.0	88.0	126.0
BOCPDbcpd1	65.088	12.763	0.6296296296296297	55.0	65.0	71.0
BOCPDbcpd3	568.217	598.59	0.21296296296296297	70.0	137.0	1272.0
EWMA3SUM	23.667	50.733	1.0	0.0	2.0	22.5
EWMA3	261.907	318.121	0.39814814814814814	113.5	137.0	201.0
EWMA2	236.0	311.858	0.4351851851851852	77.0	121.0	175.5
EWMA1	240.0	309.603	0.4537037037037037	76.0	120.0	167.0
TSSUM	28.917	52.245	1.0	0.0	7.0	31.25
TS3			0.0			
TS2			0.0			
TS1			0.0			
ADWINSUM	1.704	2.93	1.0	0.0	0.0	2.0
ADWIN3			0.0			
ADWIN2			0.0			
ADWIN1	143.5	76.028	0.4444444444444444	87.0	108.5	196.25
HDMMASUM	1.546	2.57	1.0	0.0	0.0	2.0
HDMMA3			0.0			
HDMMA2			0.0			
HDMMA1	152.96	78.992	0.46296296296296297	96.0	144.0	192.0
ECDDWTSUM	77.472	236.344	1.0	0.0	1.0	28.0
ECDDWT3	151.022	83.84	0.42592592592592593	87.0	134.5	189.25
ECDDWT2	129.918	71.345	0.4537037037037037	84.0	103.0	173.0
HalfSpaceTreesSUM	370.667	791.535	1.0	0.0	0.0	157.0
HalfSpaceTrees1	201.0	0.0	0.4166666666666667	201.0	201.0	201.0
HalfSpaceTrees2	202.0	0.0	0.4166666666666667	202.0	202.0	202.0
HalfSpaceTrees3	203.0	0.0	0.4166666666666667	203.0	203.0	203.0
martingaleSUM	342.472	722.973	1.0	0.0	57.0	261.5
martingale2SUM	322.343	714.542	1.0	0.0	34.0	245.5
martingale1	38.883	194.32	0.7129629629629629	4.0	10.0	16.0
martingale2	40.474	195.557	0.7037037037037037	5.0	11.0	17.25
martingale3	43.479	199.372	0.6759259259259259	6.0	13.0	19.0
martingale21	62.91	283.237	0.6203703703703703	11.0	22.0	39.0
martingale22	66.922	289.596	0.5925925925925926	14.0	23.0	42.0
martingale23	69.25	289.39	0.5925925925925926	19.5	24.0	43.0
fixSUM	102.852	200.406	1.0	5.0	25.0	79.0
fix1	14.77	23.561	0.9259259259259259	1.0	4.0	22.0
fix2	27.434	36.762	0.7685185185185185	6.0	14.0	33.0
fix3	84.721	176.457	0.5648148148148148	21.0	44.0	87.0
addisSUM	73.704	143.564	1.0	4.0	15.0	61.5
addis1	14.426	34.566	1.0	3.0	6.0	7.0
addis2	59.04	187.655	0.6944444444444444	6.0	15.0	29.0
addis3	156.782	303.245	0.5092592592592593	27.0	64.0	130.0
saffronSUM	0.019	0.135	1.0	0.0	0.0	0.0
saffron1	18.0	1.414	0.018518518518518517	17.5	18.0	18.5
saffron2			0.0			
saffron3			0.0			
DDMSUM	267.491	632.284	1.0	0.0	0.0	168.0
DDM1	164.043	232.714	0.4351851851851852	65.0	84.0	135.0
DDM2	165.681	232.742	0.4351851851851852	66.0	85.0	136.0
DDM3	184.553	235.407	0.4351851851851852	68.5	127.0	185.0
STEPDSUM	104.556	290.101	1.0	0.0	0.0	50.0
STEPD1	131.623	109.873	0.49074074074074076	60.0	73.0	166.0
STEPD2	134.833	114.955	0.4444444444444444	61.0	67.5	170.0
STEPD3	135.833	114.955	0.4444444444444444	62.0	68.5	171.0
HDDMASUM	0.694	1.603	1.0	0.0	0.0	1.0
HDDMA1	177.242	92.936	0.30555555555555556	87.0	188.0	235.0
HDDMA2			0.0			
HDDMA3			0.0			
ECDDWT1	110.87	64.449	0.5	65.0	102.0	135.0

Table 40: Waymo Dataset - MLP Regressor

	mean	std	detectionrate	q25	q75
BOCPDSUM	76.615	125.771	1.0	3.0	122.25
BOCPDbcpd2	107.756	65.5	0.7	64.75	127.0
BOCPDbcpd1	78.614	34.143	0.863	56.0	86.0
BOCPDbcpd3	582.704	866.063	0.295	106.25	772.0
EWMA3SUM	66.435	187.266	1.0	1.0	45.0
EWMA3	301.597	390.371	0.516	91.0	321.5
EWMA2	276.318	370.092	0.58	85.0	287.75
EWMA1	245.237	348.315	0.666	77.25	248.75
TSSUM	76.615	125.771	1.0	3.0	122.25
TS3			0.0		
TS2			0.0		
TS1			0.0		
ADWINSUM	0.89	1.014	1.0	0.0	1.0
ADWIN3			0.0		
ADWIN2			0.0		
ADWIN1	655.737	787.365	0.576	130.0	949.0
HDMMASUM	0.842	0.979	1.0	0.0	1.0
HDMMA3			0.0		
HDMMA2			0.0		
HDMMA1	678.937	775.414	0.56	128.0	1024.0
ECDDWTSUM	12.159	34.384	1.0	0.0	12.0
ECDDWT3	861.508	1154.31	0.316	140.25	1078.25
ECDDWT2	736.269	989.528	0.384	110.0	960.25
HalfSpaceTreesSUM	906.475	1660.713	1.0	0.0	1159.75
HalfSpaceTrees1	201.0	0.0	0.587	201.0	201.0
HalfSpaceTrees2	202.0	0.0	0.582	202.0	202.0
HalfSpaceTrees3	203.0	0.0	0.58	203.0	203.0
martingaleSUM	12.228	53.824	1.0	0.0	0.0
martingale2SUM	10.742	35.965	1.0	0.0	0.0
martingale1	115.65	293.957	0.183	39.0	111.25
martingale2	120.101	305.386	0.168	40.0	116.0
martingale3	125.699	312.307	0.161	41.0	120.5
martingale21	54.456	99.273	0.223	22.5	46.0
martingale22	59.825	102.678	0.209	24.0	53.5
martingale23	62.587	104.805	0.202	25.0	56.0
fixSUM	58.213	92.011	1.0	5.0	71.0
fix1	45.111	34.812	0.93	24.0	50.0
fix2	300.879	505.496	0.616	55.0	339.25
fix3	990.97	1442.576	0.176	82.0	1608.0
addisSUM	58.213	92.011	1.0	5.0	71.0
addis1	122.733	85.934	0.758	73.0	138.0
addis2	933.5	1097.426	0.334	182.75	1403.25
addis3	1028.593	1167.381	0.035	154.5	1985.0
saffronSUM	58.213	92.011	1.0	5.0	71.0
saffron1			0.0		
saffron2			0.0		
saffron3			0.0		
DDMSUM	282.381	729.77	1.0	0.0	145.75
DDM1	343.986	809.189	0.548	50.0	216.25
DDM2	352.244	813.342	0.546	53.0	233.5
DDM3	360.791	823.212	0.531	55.0	237.0
STEPDSUM	24.875	79.286	1.0	0.0	19.75
STEPD1	555.945	863.813	0.501	66.75	649.25
STEPD2	603.874	929.056	0.475	69.5	723.25
STEPD3	635.73	995.929	0.445	71.0	769.0
HDDMASUM	0.18	0.439	1.0	0.0	0.0
HDDMA1	1023.521	1101.89	0.158	221.0	1522.0
HDDMA2			0.0		
HDDMA3			0.0		
ECDDWT1	279.194	383.458	0.64	78.25	335.0

Table 41: Waymo Dataset - MLP Regressor

	mean	std	detectionrate	p25	p50	p75
BOCPDSUM	10.917	20.744	1.0	0.0	2.0	5.0
BOCPDbcpd2	299.296	222.564	0.25	124.0	228.0	409.5
BOCPDbcpd1	118.085	52.079	0.5462962962962963	74.5	115.0	132.0
BOCPDbcpd3	360.0	0.0	0.018518518518518517	360.0	360.0	360.0
EWMA3SUM	13.833	20.438	1.0	0.0	1.0	19.0
EWMA3	301.979	254.806	0.4351851851851852	95.0	145.0	506.5
EWMA2	243.362	248.939	0.4351851851851852	74.5	134.0	305.5
EWMA1	200.915	236.928	0.4351851851851852	73.0	93.0	182.5
TSSUM	10.917	20.744	1.0	0.0	2.0	5.0
TS3			0.0			
TS2			0.0			
TS1			0.0			
ADWINSUM	1.176	2.004	1.0	0.0	0.0	1.0
ADWIN3			0.0			
ADWIN2			0.0			
ADWIN1	165.0	88.483	0.42592592592592593	104.75	150.5	178.0
HDMMASUM	1.13	1.924	1.0	0.0	0.0	1.0
HDMMA3			0.0			
HDMMA2			0.0			
HDMMA1	168.348	92.851	0.42592592592592593	104.0	160.0	160.0
ECDDWTSUM	9.139	14.47	1.0	0.0	1.0	11.25
ECDDWT3	131.043	87.925	0.42592592592592593	81.0	98.0	157.25
ECDDWT2	121.3	84.685	0.46296296296296297	52.0	95.0	145.0
HalfSpaceTreesSUM	373.417	799.548	1.0	0.0	0.0	157.0
HalfSpaceTrees1	201.0	0.0	0.4166666666666667	201.0	201.0	201.0
HalfSpaceTrees2	202.0	0.0	0.4166666666666667	202.0	202.0	202.0
HalfSpaceTrees3	203.0	0.0	0.4166666666666667	203.0	203.0	203.0
martingaleSUM	330.537	720.643	1.0	0.0	46.0	261.0
martingale2SUM	320.852	718.508	1.0	0.0	28.0	251.75
martingale1	21.224	51.259	0.7037037037037037	3.0	7.0	13.0
martingale2	22.613	51.79	0.6944444444444444	4.0	8.0	13.5
martingale3	25.662	54.47	0.6851851851851852	5.0	9.0	16.75
martingale21	26.671	31.465	0.6481481481481481	9.0	18.0	32.0
martingale22	30.0	31.699	0.6388888888888888	12.0	20.0	36.0
martingale23	32.758	31.914	0.6111111111111112	15.0	25.0	39.0
fixSUM	103.185	222.632	1.0	3.0	23.5	78.0
fix1	13.621	22.347	0.8796296296296297	1.0	4.0	17.5
fix2	21.494	31.736	0.75	4.0	10.0	22.0
fix3	64.182	67.187	0.6111111111111112	17.0	43.5	99.25
addisSUM	73.315	155.366	1.0	2.0	16.5	55.25
addis1	19.142	67.181	0.9814814814814815	3.0	5.5	6.0
addis2	42.234	101.579	0.7129629629629629	5.0	12.0	31.0
addis3	73.847	80.01	0.5462962962962963	16.5	43.0	104.5
saffronSUM	0.019	0.135	1.0	0.0	0.0	0.0
saffron1	28.5	6.364	0.018518518518518517	26.25	28.5	30.75
saffron2			0.0			
saffron3			0.0			
DDMSUM	389.306	832.353	1.0	0.0	7.5	165.75
DDM1	89.0	84.163	0.5277777777777778	44.0	69.0	93.0
DDM2	91.036	84.562	0.5185185185185185	45.75	72.0	94.0
DDM3	94.339	84.764	0.5185185185185185	46.75	73.0	110.0
STEPDSUM	24.694	51.397	1.0	0.0	0.0	17.75
STEPD1	112.83	90.294	0.4351851851851852	60.0	71.0	108.5
STEPD2	114.66	91.953	0.4351851851851852	61.0	72.0	109.5
STEPD3	116.0	91.9	0.4351851851851852	62.0	73.0	114.5
HDDMASUM	0.287	0.454	1.0	0.0	0.0	1.0
HDDMA1	109.097	61.225	0.28703703703703703	54.0	92.0	136.0
HDDMA2			0.0			
HDDMA3			0.0			
ECDDWT1	96.534	78.69	0.5370370370370371	45.0	74.5	102.5

Table 42: Waymo Dataset - MLP Regressor

	ADDSRate	SAFFRONRate	PIXRate
lam0.1tau0.4W0.0	1.9341563786008233e-06	1.9341563786008233e-06	0.047601604938271605
lam0.1tau0.4W0.01	1.9341563786008233e-06	1.9341563786008233e-06	0.047601604938271605
lam0.1tau0.4W0.02	1.9341563786008233e-06	1.9341563786008233e-06	0.047601604938271605
lam0.1tau0.4W0.03	1.9341563786008233e-06	1.9341563786008233e-06	0.047601604938271605
lam0.1tau0.4W0.04	1.9341563786008233e-06	1.9341563786008233e-06	0.047601604938271605
lam0.1tau0.5W0.0	5.041152263374485e-06	5.041152263374485e-06	0.047601604938271605
lam0.1tau0.5W0.01	5.041152263374485e-06	5.041152263374485e-06	0.047601604938271605
lam0.1tau0.5W0.02	5.349794238683128e-06	5.349794238683128e-06	0.047601604938271605
lam0.1tau0.5W0.03	5.3497942386831275e-06	5.3497942386831275e-06	0.047601604938271605
lam0.1tau0.5W0.04	5.3497942386831275e-06	5.3497942386831275e-06	0.047601604938271605
lam0.1tau0.6W0.0	0.0	0.0	0.047601604938271605
lam0.1tau0.6W0.01	0.0	0.0	0.047601604938271605
lam0.1tau0.6W0.02	0.0	0.0	0.047601604938271605
lam0.1tau0.6W0.03	0.0	0.0	0.047601604938271605
lam0.1tau0.6W0.04	0.0	0.0	0.047601604938271605
lam0.15tau0.4W0.0	1.2551440329218106e-06	1.2551440329218106e-06	0.047601604938271605
lam0.15tau0.4W0.01	1.2551440329218106e-06	1.2551440329218106e-06	0.047601604938271605
lam0.15tau0.4W0.02	1.2551440329218106e-06	1.2551440329218106e-06	0.047601604938271605
lam0.15tau0.4W0.03	1.2551440329218106e-06	1.2551440329218106e-06	0.047601604938271605
lam0.15tau0.4W0.04	1.2551440329218106e-06	1.2551440329218106e-06	0.047601604938271605
lam0.15tau0.5W0.0	1.2757201646090535e-06	1.2757201646090535e-06	0.047601604938271605
lam0.15tau0.5W0.01	1.2757201646090535e-06	1.2757201646090535e-06	0.047601604938271605
lam0.15tau0.5W0.02	1.2757201646090535e-06	1.2757201646090535e-06	0.047601604938271605
lam0.15tau0.5W0.03	1.2757201646090535e-06	1.2757201646090535e-06	0.047601604938271605
lam0.15tau0.5W0.04	1.2757201646090535e-06	1.2757201646090535e-06	0.047601604938271605
lam0.15tau0.6W0.0	0.0	0.0	0.047601604938271605
lam0.15tau0.6W0.01	0.0	0.0	0.047601604938271605
lam0.15tau0.6W0.02	0.0	0.0	0.047601604938271605
lam0.15tau0.6W0.03	0.0	0.0	0.047601604938271605
lam0.15tau0.6W0.04	0.0	0.0	0.047601604938271605
lam0.2tau0.4W0.0	2.469135802469136e-07	2.469135802469136e-07	0.047601604938271605
lam0.2tau0.4W0.01	2.469135802469136e-07	2.469135802469136e-07	0.047601604938271605
lam0.2tau0.4W0.02	2.469135802469136e-07	2.469135802469136e-07	0.047601604938271605
lam0.2tau0.4W0.03	2.469135802469136e-07	2.469135802469136e-07	0.047601604938271605
lam0.2tau0.4W0.04	2.469135802469136e-07	2.469135802469136e-07	0.047601604938271605
lam0.2tau0.5W0.0	2.0576131687242797e-07	2.0576131687242797e-07	0.047601604938271605
lam0.2tau0.5W0.01	2.0576131687242797e-07	2.0576131687242797e-07	0.047601604938271605
lam0.2tau0.5W0.02	2.05761316872428e-07	2.05761316872428e-07	0.047601604938271605
lam0.2tau0.5W0.03	2.0576131687242797e-07	2.0576131687242797e-07	0.047601604938271605
lam0.2tau0.5W0.04	2.0576131687242797e-07	2.0576131687242797e-07	0.047601604938271605
lam0.2tau0.6W0.0	0.0	0.0	0.047601604938271605
lam0.2tau0.6W0.01	0.0	0.0	0.047601604938271605
lam0.2tau0.6W0.02	0.0	0.0	0.047601604938271605
lam0.2tau0.6W0.03	0.0	0.0	0.047601604938271605
lam0.2tau0.6W0.04	0.0	0.0	0.047601604938271605
lam0.25tau0.4W0.0	3.7860082304526754e-06	3.7860082304526754e-06	0.047601604938271605
lam0.25tau0.4W0.01	3.7860082304526754e-06	3.7860082304526754e-06	0.047601604938271605
lam0.25tau0.4W0.02	3.7860082304526754e-06	3.7860082304526754e-06	0.047601604938271605
lam0.25tau0.4W0.03	3.786008230452674e-06	3.786008230452674e-06	0.047601604938271605
lam0.25tau0.4W0.04	3.786008230452674e-06	3.786008230452674e-06	0.047601604938271605
lam0.25tau0.5W0.0	2.0576131687242797e-07	2.0576131687242797e-07	0.047601604938271605
lam0.25tau0.5W0.01	2.0576131687242797e-07	2.0576131687242797e-07	0.047601604938271605
lam0.25tau0.5W0.02	2.05761316872428e-07	2.05761316872428e-07	0.047601604938271605
lam0.25tau0.5W0.03	2.0576131687242797e-07	2.0576131687242797e-07	0.047601604938271605
lam0.25tau0.5W0.04	2.0576131687242797e-07	2.0576131687242797e-07	0.047601604938271605
lam0.25tau0.6W0.0	0.0	0.0	0.047601604938271605
lam0.25tau0.6W0.01	0.0	0.0	0.047601604938271605
lam0.25tau0.6W0.02	0.0	0.0	0.047601604938271605
lam0.25tau0.6W0.03	0.0	0.0	0.047601604938271605
lam0.25tau0.6W0.04	0.0	0.0	0.047601604938271605
lam0.3tau0.4W0.0	5.144032921810699e-07	5.144032921810699e-07	0.047601604938271605
lam0.3tau0.4W0.01	5.144032921810699e-07	5.144032921810699e-07	0.047601604938271605
lam0.3tau0.4W0.02	5.144032921810699e-07	5.144032921810699e-07	0.047601604938271605
lam0.3tau0.4W0.03	5.144032921810699e-07	5.144032921810699e-07	0.047601604938271605
lam0.3tau0.4W0.04	5.144032921810699e-07	5.144032921810699e-07	0.047601604938271605
lam0.3tau0.5W0.0	0.0	0.0	0.047601604938271605
lam0.3tau0.5W0.01	0.0	0.0	0.047601604938271605
lam0.3tau0.5W0.02	0.0	0.0	0.047601604938271605
lam0.3tau0.5W0.03	0.0	0.0	0.047601604938271605
lam0.3tau0.5W0.04	0.0	0.0	0.047601604938271605
lam0.3tau0.6W0.0	0.0	0.0	0.047601604938271605
lam0.3tau0.6W0.01	0.0	0.0	0.047601604938271605
lam0.3tau0.6W0.02	0.0	0.0	0.047601604938271605
lam0.3tau0.6W0.03	0.0	0.0	0.047601604938271605
lam0.3tau0.6W0.04	0.0	0.0	0.047601604938271605
lam0.35000			

Table 43: False Alarm Rate - Online FDR Tuning - Unweighed

	ADDISRate	SAFFRONRate	FIXRate
lam0.1tau0.4W0.0	1.93E-06	1.93E-06	0.997530864
lam0.1tau0.4W0.01	1.93E-06	1.93E-06	0.997530864
lam0.1tau0.4W0.02	1.93E-06	1.93E-06	0.997530864
lam0.1tau0.4W0.03	1.93E-06	1.93E-06	0.997530864
lam0.1tau0.4W0.04	1.93E-06	1.93E-06	0.997530864
lam0.1tau0.5W0.0	5.04E-06	5.04E-06	0.997530864
lam0.1tau0.5W0.01	5.04E-06	5.04E-06	0.997530864
lam0.1tau0.5W0.02	5.35E-06	5.35E-06	0.997530864
lam0.1tau0.5W0.03	5.35E-06	5.35E-06	0.997530864
lam0.1tau0.5W0.04	5.35E-06	5.35E-06	0.997530864
lam0.1tau0.6W0.0	0	0	0.997530864
lam0.1tau0.6W0.01	5.04E-06	5.04E-06	0.997530864
lam0.1tau0.6W0.02	5.35E-06	5.35E-06	0.997530864
lam0.1tau0.6W0.03	5.35E-06	5.35E-06	0.997530864
lam0.1tau0.6W0.04	5.35E-06	5.35E-06	0.997530864
lam0.15tau0.4W0.0	1.26E-06	1.26E-06	0.997530864
lam0.15tau0.4W0.01	1.26E-06	1.26E-06	0.997530864
lam0.15tau0.4W0.02	1.26E-06	1.26E-06	0.997530864
lam0.15tau0.4W0.03	1.26E-06	1.26E-06	0.997530864
lam0.15tau0.4W0.04	1.26E-06	1.26E-06	0.997530864
lam0.15tau0.5W0.0	1.28E-06	1.28E-06	0.997530864
lam0.15tau0.5W0.01	1.28E-06	1.28E-06	0.997530864
lam0.15tau0.5W0.02	1.28E-06	1.28E-06	0.997530864
lam0.15tau0.5W0.03	1.28E-06	1.28E-06	0.997530864
lam0.15tau0.5W0.04	1.28E-06	1.28E-06	0.997530864
lam0.15tau0.6W0.0	0	0	0.997530864
lam0.15tau0.6W0.01	1.28E-06	1.28E-06	0.997530864
lam0.15tau0.6W0.02	1.28E-06	1.28E-06	0.997530864
lam0.15tau0.6W0.03	1.28E-06	1.28E-06	0.997530864
lam0.15tau0.6W0.04	1.28E-06	1.28E-06	0.997530864
lam0.2tau0.4W0.0	2.47E-07	2.47E-07	0.997530864
lam0.2tau0.4W0.01	2.47E-07	2.47E-07	0.997530864
lam0.2tau0.4W0.02	2.47E-07	2.47E-07	0.997530864
lam0.2tau0.4W0.03	2.47E-07	2.47E-07	0.997530864
lam0.2tau0.4W0.04	2.47E-07	2.47E-07	0.997530864
lam0.2tau0.5W0.0	2.06E-07	2.06E-07	0.997530864
lam0.2tau0.5W0.01	2.06E-07	2.06E-07	0.997530864
lam0.2tau0.5W0.02	2.06E-07	2.06E-07	0.997530864
lam0.2tau0.5W0.03	2.06E-07	2.06E-07	0.997530864
lam0.2tau0.5W0.04	2.06E-07	2.06E-07	0.997530864
lam0.2tau0.6W0.0	0	0	0.997530864
lam0.2tau0.6W0.01	2.06E-07	2.06E-07	0.997530864
lam0.2tau0.6W0.02	2.06E-07	2.06E-07	0.997530864
lam0.2tau0.6W0.03	0	0	0.997530864
lam0.2tau0.6W0.04	0	0	0.997530864
lam0.25tau0.4W0.0	3.79E-06	3.79E-06	0.997530864
lam0.25tau0.4W0.01	3.79E-06	3.79E-06	0.997530864
lam0.25tau0.4W0.02	3.79E-06	3.79E-06	0.997530864
lam0.25tau0.4W0.03	3.79E-06	3.79E-06	0.997530864
lam0.25tau0.4W0.04	3.79E-06	3.79E-06	0.997530864
lam0.25tau0.5W0.0	2.06E-07	2.06E-07	0.997530864
lam0.25tau0.5W0.01	2.06E-07	2.06E-07	0.997530864
lam0.25tau0.5W0.02	2.06E-07	2.06E-07	0.997530864
lam0.25tau0.5W0.03	2.06E-07	2.06E-07	0.997530864
lam0.25tau0.5W0.04	2.06E-07	2.06E-07	0.997530864
lam0.25tau0.6W0.0	0	0	0.997530864
lam0.25tau0.6W0.01	2.06E-07	2.06E-07	0.997530864
lam0.25tau0.6W0.02	2.06E-07	2.06E-07	0.997530864
lam0.25tau0.6W0.03	0	0	0.997530864
lam0.25tau0.6W0.04	0	0	0.997530864
lam0.3tau0.4W0.0	5.14E-07	5.14E-07	0.997530864
lam0.3tau0.4W0.01	5.14E-07	5.14E-07	0.997530864
lam0.3tau0.4W0.02	5.14E-07	5.14E-07	0.997530864
lam0.3tau0.4W0.03	5.14E-07	5.14E-07	0.997530864
lam0.3tau0.4W0.04	5.14E-07	5.14E-07	0.997530864
lam0.3tau0.5W0.0	0	0	0.997530864
lam0.3tau0.5W0.01	0	0	0.997530864
lam0.3tau0.5W0.02	0	0	0.997530864
lam0.3tau0.5W0.03	0	0	0.997530864
lam0.3tau0.5W0.04	0	0	0.997530864
lam0.3tau0.6W0.0	5.14E-07	5.14E-07	0.997530864
lam0.3tau0.6W0.01	5.14E-07	5.14E-07	0.997530864
lam0.3tau0.6W0.02	5.14E-07	5.14E-07	0.997530864
lam0.3tau0.6W0.03	0	0	0.997530864
lam0.3tau0.6W0.04	0	0	0.997530864
lam0.35000000000000001tau0.4W0.0	0	0	0.997530864
lam0.35000000000000001tau0.4W0.01	0	0	0.997530864
lam0.35000000000000001tau0.4W0.02	0	0	0.997530864
lam0.35000000000000001tau0.4W0.03	5.14E-07	5.14E-07	0.997530864
lam0.35000000000000001tau0.4W0.04	5.14E-07	5.14E-07	0.997530864
lam0.35000000000000001tau0.5W0.0	5.14E-07	5.14E-07	0.997530864
lam0.35000000000000001tau0.5W0.01	0	0	0.997530864
lam0.35000000000000001tau0.5W0.02	0	0	0.997530864
lam0.35000000000000001tau0.5W0.03	0	0	0.997530864
lam0.35000000000000001tau0.5W0.04	0	0	0.997530864
lam0.35000000000000001tau0.6W0.0	0	0	0.997530864
lam0.35000000000000001tau0.6W0.01	0	0	0.997530864
lam0.35000000000000001tau0.6W0.02	0	0	0.997530864
lam0.35000000000000001tau0.6W0.03	0	0	0.997530864
lam0.35000000000000001tau0.6W0.04	0	0	0.997530864
lam0.4tau0.4W0.0	0	0	0.997530864
lam0.4tau0.4W0.01	0	0	0.997530864
lam0.4tau0.4W0.02	0	0	0.997530864
lam0.4tau0.4W0.03	0	0	0.997530864
lam0.4tau0.4W0.04	0	0	0.997530864
lam0.4tau0.5W0.0	0	0	0.997530864
lam0.4tau0.5W0.01	0	0	0.997530864
lam0.4tau0.5W0.02	0	0	0.997530864
lam0.4tau0.5W0.03	0	0	0.997530864
lam0.4tau0.5W0.04	0	0	0.997530864
lam0.4tau0.6W0.0	0	0	0.997530864
lam0.4tau0.6W0.01	0	0	0.997530864
lam0.4tau0.6W0.02	0	0	0.997530864
lam0.4tau0.6W0.03	0	0	0.997530864
lam0.4tau0.6W0.04	0	0	0.997530864

Table 44: False Alarm Rate - Online FDR Tuning - EWMA

	ADDISRate	SAFFRONRate	FIXRate
lam0.1tau0.4W0.0	1.9341563786008233e-06	1.9341563786008233e-06	0.047601694938271605
lam0.1tau0.4W0.01	1.9341563786008233e-06	1.9341563786008233e-06	0.047601694938271605
lam0.1tau0.4W0.02	1.9341563786008233e-06	1.9341563786008233e-06	0.047601694938271605
lam0.1tau0.4W0.03	1.9341563786008233e-06	1.9341563786008233e-06	0.047601694938271605
lam0.1tau0.4W0.04	1.9341563786008233e-06	1.9341563786008233e-06	0.047601694938271605
lam0.1tau0.4W0.05	5.04115226374485e-06	5.04115226374485e-06	0.047601694938271605
lam0.1tau0.5W0.01	5.04115226374485e-06	5.04115226374485e-06	0.047601694938271605
lam0.1tau0.5W0.02	5.349794238683128e-06	5.349794238683128e-06	0.047601694938271605
lam0.1tau0.5W0.03	5.3497942386831275e-06	5.3497942386831275e-06	0.047601694938271605
lam0.1tau0.5W0.04	5.3497942386831275e-06	5.3497942386831275e-06	0.047601694938271605
lam0.1tau0.6W0.0	0.0	0.0	0.047601694938271605
lam0.1tau0.6W0.01	0.0	0.0	0.047601694938271605
lam0.1tau0.6W0.02	0.0	0.0	0.047601694938271605
lam0.1tau0.6W0.03	0.0	0.0	0.047601694938271605
lam0.1tau0.6W0.04	0.0	0.0	0.047601694938271605
lam0.1tau0.6W0.05	1.2551440329218106e-06	1.2551440329218106e-06	0.047601694938271605
lam0.15tau0.4W0.01	1.2551440329218106e-06	1.2551440329218106e-06	0.047601694938271605
lam0.15tau0.4W0.02	1.2551440329218106e-06	1.2551440329218106e-06	0.047601694938271605
lam0.15tau0.4W0.03	1.2551440329218106e-06	1.2551440329218106e-06	0.047601694938271605
lam0.15tau0.4W0.04	1.2551440329218106e-06	1.2551440329218106e-06	0.047601694938271605
lam0.15tau0.5W0.0	1.2757201646090535e-06	1.2757201646090535e-06	0.047601694938271605
lam0.15tau0.5W0.01	1.2757201646090535e-06	1.2757201646090535e-06	0.047601694938271605
lam0.15tau0.5W0.02	1.2757201646090535e-06	1.2757201646090535e-06	0.047601694938271605
lam0.15tau0.5W0.03	1.2757201646090535e-06	1.2757201646090535e-06	0.047601694938271605
lam0.15tau0.5W0.04	1.2757201646090535e-06	1.2757201646090535e-06	0.047601694938271605
lam0.15tau0.5W0.05	0.0	0.0	0.047601694938271605
lam0.15tau0.6W0.01	0.0	0.0	0.047601694938271605
lam0.15tau0.6W0.02	0.0	0.0	0.047601694938271605
lam0.15tau0.6W0.03	0.0	0.0	0.047601694938271605
lam0.15tau0.6W0.04	0.0	0.0	0.047601694938271605
lam0.2tau0.4W0.0	2.469135802469136e-07	2.469135802469136e-07	0.047601694938271605
lam0.2tau0.4W0.01	2.469135802469136e-07	2.469135802469136e-07	0.047601694938271605
lam0.2tau0.4W0.02	2.469135802469136e-07	2.469135802469136e-07	0.047601694938271605
lam0.2tau0.4W0.03	2.469135802469136e-07	2.469135802469136e-07	0.047601694938271605
lam0.2tau0.4W0.04	2.469135802469136e-07	2.469135802469136e-07	0.047601694938271605
lam0.2tau0.5W0.0	2.0576131687242797e-07	2.0576131687242797e-07	0.047601694938271605
lam0.2tau0.5W0.01	2.0576131687242797e-07	2.0576131687242797e-07	0.047601694938271605
lam0.2tau0.5W0.02	2.05761316872428e-07	2.05761316872428e-07	0.047601694938271605
lam0.2tau0.5W0.03	2.0576131687242797e-07	2.0576131687242797e-07	0.047601694938271605
lam0.2tau0.5W0.04	2.0576131687242797e-07	2.0576131687242797e-07	0.047601694938271605
lam0.2tau0.6W0.0	0.0	0.0	0.047601694938271605
lam0.2tau0.6W0.01	0.0	0.0	0.047601694938271605
lam0.2tau0.6W0.02	0.0	0.0	0.047601694938271605
lam0.2tau0.6W0.03	0.0	0.0	0.047601694938271605
lam0.2tau0.6W0.04	0.0	0.0	0.047601694938271605
lam0.2tau0.6W0.05	3.7860082304526754e-06	3.7860082304526754e-06	0.047601694938271605
lam0.25tau0.4W0.0	3.7860082304526754e-06	3.7860082304526754e-06	0.047601694938271605
lam0.25tau0.4W0.01	3.7860082304526754e-06	3.7860082304526754e-06	0.047601694938271605
lam0.25tau0.4W0.02	3.7860082304526754e-06	3.7860082304526754e-06	0.047601694938271605
lam0.25tau0.4W0.03	3.7860082304526746e-06	3.7860082304526746e-06	0.047601694938271605
lam0.25tau0.5W0.0	2.0576131687242797e-07	2.0576131687242797e-07	0.047601694938271605
lam0.25tau0.5W0.01	2.0576131687242797e-07	2.0576131687242797e-07	0.047601694938271605
lam0.25tau0.5W0.02	2.05761316872428e-07	2.05761316872428e-07	0.047601694938271605
lam0.25tau0.5W0.03	2.0576131687242797e-07	2.0576131687242797e-07	0.047601694938271605
lam0.25tau0.5W0.04	2.0576131687242797e-07	2.0576131687242797e-07	0.047601694938271605
lam0.25tau0.6W0.0	0.0	0.0	0.047601694938271605
lam0.25tau0.6W0.01	0.0	0.0	0.047601694938271605
lam0.25tau0.6W0.02	0.0	0.0	0.047601694938271605
lam0.25tau0.6W0.03	0.0	0.0	0.047601694938271605
lam0.25tau0.6W0.04	0.0	0.0	0.047601694938271605
lam0.3tau0.4W0.0	5.144032921810699e-07	5.144032921810699e-07	0.047601694938271605
lam0.3tau0.4W0.01	5.144032921810699e-07	5.144032921810699e-07	0.047601694938271605
lam0.3tau0.4W0.02	5.1440329218107e-07	5.1440329218107e-07	0.047601694938271605
lam0.3tau0.4W0.03	5.144032921810699e-07	5.144032921810699e-07	0.047601694938271605
lam0.3tau0.4W0.04	5.144032921810699e-07	5.144032921810699e-07	0.047601694938271605
lam0.3tau0.5W0.0	0.0	0.0	0.047601694938271605
lam0.3tau0.5W0.01	0.0	0.0	0.047601694938271605
lam0.3tau0.5W0.02	0.0	0.0	0.047601694938271605
lam0.3tau0.5W0.03	0.0	0.0	0.047601694938271605
lam0.3tau0.5W0.04	0.0	0.0	0.047601694938271605
lam0.3tau0.6W0.0	0.0	0.0	0.047601694938271605
lam0.3tau0.6W0.01	0.0	0.0	0.047601694938271605
lam0.3tau0.6W0.02	0.0	0.0	0.047601694938271605
lam0.3tau0.6W0.03	0.0	0.0	0.047601694938271605
lam0.3tau0.6W0.04	0.0	0.0	0.047601694938271605
lam0.35000000000000001tau0.4W0.0	0.0	0.0	0.047601694938271605
lam0.35000000000000001tau0.4W0.01	0.0	0.0	0.047601694938271605
lam0.35000000000000001tau0.4W0.02	0.0	0.0	0.047601694938271605
lam0.35000000000000001tau0.4W0.03	0.0	0.0	0.047601694938271605
lam0.35000000000000001tau0.4W0.04	0.0	0.0	0.047601694938271605
lam0.35000000000000001tau0.5W0.0	0.0	0.0	0.047601694938271605
lam0.35000000000000001tau0.5W0.01	0.0	0.0	0.047601694938271605
lam0.35000000000000001tau0.5W0.02	0.0	0.0	0.047601694938271605
lam0.35000000000000001tau0.5W0.03	0.0	0.0	0.047601694938271605
lam0.35000000000000001tau0.5W0.04	0.0	0.0	0.047601694938271605
lam0.35000000000000001tau0.6W0.0	0.0	0.0	0.047601694938271605
lam0.35000000000000001tau0.6W0.01	0.0	0.0	0.047601694938271605
lam0.35000000000000001tau0.6W0.02	0.0	0.0	0.047601694938271605
lam0.35000000000000001tau0.6W0.03	0.0	0.0	0.047601694938271605
lam0.4tau0.4W0.0	0.0	0.0	0.047601694938271605
lam0.4tau0.4W0.01	0.0	0.0	0.047601694938271605
lam0.4tau0.4W0.02	0.0	0.0	0.047601694938271605
lam0.4tau0.4W0.03	0.0	0.0	0.047601694938271605
lam0.4tau0.4W0.04	0.0	0.0	0.047601694938271605
lam0.4tau0.5W0.0	0.0	0.0	0.047601694938271605
lam0.4tau0.5W0.01	0.0	0.0	0.047601694938271605
lam0.4tau0.5W0.02	0.0	0.0	0.047601694938271605
lam0.4tau0.5W0.03	0.0	0.0	0.047601694938271605
lam0.4tau0.5W0.04	0.0	0.0	0.047601694938271605
lam0.4tau0.6W0.0	0.0	0.0	0.047601694938271605
lam0.4tau0.6W0.01	0.0	0.0	0.047601694938271605
lam0.4tau0.6W0.02	0.0	0.0	0.047601694938271605
lam0.4tau0.6W0.03	0.0	0.0	0.047601694938271605
lam0.4tau0.6W0.04	0.0	0.0	0.047601694938271605

Table 45: Trajectory change detection results (MLP)

Method	Mean	Std	Detection rate	Missing rate	Q25
BOCPDSUM	224.045	245.275	1.000	0.000	47.75
BOCPDbcpd2	160.742	163.829	1.000	0.000	75.50
BOCPDbcpd1	87.727	50.950	1.000	0.000	54.25
BOCPDbcpd3	1193.571	1258.164	0.530	0.470	427.50
EWMA3SUM	173.030	214.339	1.000	0.000	55.50
EWMA3	375.462	450.731	0.985	0.015	96.00
EWMA2	214.615	194.913	0.985	0.015	68.00
EWMA1	146.818	134.584	1.000	0.000	56.25
TSSUM	224.045	245.275	1.000	0.000	47.75
TS3			0.000	1.000	
TS2			0.000	1.000	
TS1			0.000	1.000	
ADWINSUM	44.879	53.753	1.000	0.000	7.25
ADWIN3			0.000	1.000	
ADWIN2			0.000	1.000	
ADWIN1	154.800	93.704	0.985	0.015	95.00
HDMMA3SUM	39.894	47.575	1.000	0.000	7.00
HDMMA3			0.000	1.000	
HDMMA2			0.000	1.000	
HDMMA1	181.169	112.182	0.985	0.015	96.00
ECDDWTSUM	1034.576	1378.736	1.000	0.000	111.50
ECDDWT3	184.938	103.829	0.985	0.015	100.00
ECDDWT2	169.754	107.305	0.985	0.015	89.00
HalfSpaceTreesSUM	3682.152	4562.858	1.000	0.000	450.00
HalfSpaceTrees1	201.000	0.000	1.000	0.000	201.00
HalfSpaceTrees2	202.000	0.000	1.000	0.000	202.00
HalfSpaceTrees3	203.000	0.000	1.000	0.000	203.00
martingaleSUM	2137.561	3475.156	1.000	0.000	16.25
martingale2SUM	579.561	1728.736	1.000	0.000	8.50
martingale1	178.827	220.730	0.788	0.212	37.00
martingale2	180.077	220.608	0.788	0.212	39.50
martingale3	181.154	220.575	0.788	0.212	40.50
martingale21	130.111	160.724	0.818	0.182	20.00
martingale22	133.981	161.036	0.803	0.197	23.00
martingale23	135.604	160.726	0.803	0.197	24.00
fixSUM	696.894	1108.513	1.000	0.000	73.00
fix1	36.015	22.080	1.000	0.000	22.25
fix2	83.591	70.994	1.000	0.000	30.00
fix3	165.200	244.328	0.985	0.015	51.00
addisSUM	696.894	1108.513	1.000	0.000	73.00
addis1	144.591	73.339	1.000	0.000	87.50
addis2	201.697	147.606	1.000	0.000	99.25
addis3	400.797	473.102	0.894	0.106	173.00
saffronSUM	696.894	1108.513	1.000	0.000	73.00
saffron1			0.000	1.000	
saffron2			0.000	1.000	
saffron3			0.000	1.000	
DDMSUM	4765.500	5750.305	1.000	0.000	555.50
DDM1	136.422	102.108	0.970	0.030	61.25
DDM2	137.422	102.108	0.970	0.030	62.25
DDM3	138.422	102.108	0.970	0.030	63.25
STEPDSUM	2705.045	3677.045	1.000	0.000	165.75
STEPD1	107.508	93.488	0.985	0.015	60.00
STEPD2	102.031	77.810	0.970	0.030	61.00
STEPD3	103.031	77.810	0.970	0.030	62.00
HDDMASUM	11.091	12.109	1.000	0.000	3.00
HDDMA1	267.538	155.718	0.985	0.015	129.00
HDDMA2			0.000	1.000	
HDDMA3			0.000	1.000	
ECDDWT1	140.121	97.220	1.000	0.000	73.00

Table 46: Trajectory change detection results (XGB)

Method	Mean	Std	Detection rate	Missing rate	Q25
BOCPDSUM	242.652	261.302	1.000	0.000	61.00
BOCPDbcpd2	122.652	70.945	1.000	0.000	68.25
BOCPDbcpd1	96.318	58.749	1.000	0.000	55.25
BOCPDbcpd3	1080.286	1233.426	0.636	0.364	339.25
EWMA3SUM	181.409	280.509	1.000	0.000	61.75
EWMA3	304.143	487.550	0.955	0.045	74.00
EWMA2	206.016	306.481	0.970	0.030	63.50
EWMA1	128.470	112.801	1.000	0.000	54.25
TSSUM	242.652	261.302	1.000	0.000	61.00
TS3			0.000	1.000	
TS2			0.000	1.000	
TS1			0.000	1.000	
ADWINSUM	35.348	46.788	1.000	0.000	3.00
ADWIN3			0.000	1.000	
ADWIN2			0.000	1.000	
ADWIN1	205.444	161.610	0.955	0.045	95.00
HDMMA3SUM	30.727	40.490	1.000	0.000	2.00
HDMMA3			0.000	1.000	
HDMMA2			0.000	1.000	
HDMMA1	206.164	131.359	0.924	0.076	128.00
ECDDWTSUM	626.500	941.293	1.000	0.000	36.00
ECDDWT3	185.312	158.175	0.970	0.030	85.75
ECDDWT2	166.455	153.241	1.000	0.000	73.50
HalfSpaceTreesSUM	4759.045	5684.773	1.000	0.000	450.00
HalfSpaceTrees1	201.000	0.000	1.000	0.000	201.00
HalfSpaceTrees2	202.000	0.000	1.000	0.000	202.00
HalfSpaceTrees3	203.000	0.000	1.000	0.000	203.00
martingaleSUM	165.939	539.346	1.000	0.000	0.00
martingale2SUM	70.803	229.730	1.000	0.000	0.00
martingale1	99.250	86.781	0.545	0.455	34.25
martingale2	101.000	87.882	0.530	0.470	34.50
martingale3	102.171	88.105	0.530	0.470	35.50
martingale21	66.000	78.007	0.545	0.455	19.00
martingale22	69.943	78.344	0.530	0.470	22.00
martingale23	69.353	78.702	0.515	0.485	25.00
fixSUM	274.727	323.754	1.000	0.000	39.25
fix1	37.091	21.840	1.000	0.000	23.00
fix2	90.985	110.313	1.000	0.000	30.25
fix3	157.636	165.429	1.000	0.000	46.50
addisSUM	274.727	323.754	1.000	0.000	39.25
addis1	131.470	65.159	1.000	0.000	80.00
addis2	213.409	154.745	1.000	0.000	108.00
addis3	292.357	217.322	0.848	0.152	141.25
saffronSUM	274.727	323.754	1.000	0.000	39.25
saffron1			0.000	1.000	
saffron2			0.000	1.000	
saffron3			0.000	1.000	
DDMSUM	4476.758	5718.749	1.000	0.000	522.50
DDM1	132.155	133.958	0.879	0.121	49.00
DDM2	133.190	133.940	0.879	0.121	50.00
DDM3	134.190	133.940	0.879	0.121	51.00
STEPDSUM	1228.121	2251.169	1.000	0.000	33.25
STEPD1	164.111	165.794	0.955	0.045	60.00
STEPD2	162.721	164.936	0.924	0.076	61.00
STEPD3	169.918	168.757	0.924	0.076	62.00
HDDMASUM	15.000	16.318	1.000	0.000	2.00
HDDMA1	240.458	204.909	0.894	0.106	88.00
HDDMA2			0.000	1.000	
HDDMA3			0.000	1.000	
ECDDWT1	116.864	81.780	1.000	0.000	53.75

Table 47: Trajectory change detection results (RF)

Method	Mean	Std	Detection rate	Missing rate	Q25
BOCPDSUM	227.742	253.916	1.000	0.000	56.50
BOCPDbcpd2	151.485	123.606	1.000	0.000	74.50
BOCPDbcpd1	108.424	75.786	1.000	0.000	56.25
BOCPDbcpd3	1381.750	2303.287	0.545	0.455	295.25
EWMA3SUM	109.818	90.945	1.000	0.000	44.00
EWMA3	266.922	312.759	0.970	0.030	100.75
EWMA2	194.846	152.852	0.985	0.015	83.00
EWMA1	137.030	107.097	1.000	0.000	64.25
TSSUM	227.742	253.916	1.000	0.000	56.50
TS3			0.000	1.000	
TS2			0.000	1.000	
TS1			0.000	1.000	
ADWINSUM	27.288	34.095	1.000	0.000	3.00
ADWIN3			0.000	1.000	
ADWIN2			0.000	1.000	
ADWIN1	223.169	155.443	0.985	0.015	103.00
HDMMASUM	25.076	32.027	1.000	0.000	2.00
HDMMA3			0.000	1.000	
HDMMA2			0.000	1.000	
HDMMA1	228.000	150.256	0.970	0.030	128.00
ECDDWTSUM	541.242	727.554	1.000	0.000	40.75
ECDDWT3	158.692	106.650	0.985	0.015	75.00
ECDDWT2	136.323	101.608	0.985	0.015	63.00
HalfSpaceTreesSUM	4783.333	5718.333	1.000	0.000	450.00
HalfSpaceTrees1	201.000	0.000	1.000	0.000	201.00
HalfSpaceTrees2	202.000	0.000	1.000	0.000	202.00
HalfSpaceTrees3	203.000	0.000	1.000	0.000	203.00
martingaleSUM	57.818	140.732	1.000	0.000	0.00
martingale2SUM	23.515	46.750	1.000	0.000	0.00
martingale1	98.607	111.082	0.424	0.576	37.00
martingale2	99.750	111.008	0.424	0.576	38.00
martingale3	109.393	115.093	0.424	0.576	39.00
martingale21	42.034	51.264	0.439	0.561	18.00
martingale22	45.148	52.763	0.409	0.591	20.50
martingale23	46.808	53.670	0.394	0.606	21.25
fixSUM	276.833	325.845	1.000	0.000	39.00
fix1	34.152	15.763	1.000	0.000	23.25
fix2	113.712	121.041	1.000	0.000	38.00
fix3	151.531	130.528	0.970	0.030	55.00
addisSUM	276.833	325.845	1.000	0.000	39.00
addis1	122.879	60.662	1.000	0.000	77.25
addis2	269.879	535.632	1.000	0.000	99.25
addis3	572.526	1482.219	0.864	0.136	112.00
saffronSUM	276.833	325.845	1.000	0.000	39.00
saffron1			0.000	1.000	
saffron2			0.000	1.000	
saffron3			0.000	1.000	
DDMSUM	4360.879	5565.834	1.000	0.000	571.00
DDM1	113.889	84.439	0.955	0.045	49.50
DDM2	133.730	139.944	0.955	0.045	50.50
DDM3	134.778	139.937	0.955	0.045	51.50
STEPDSUM	970.985	1429.224	1.000	0.000	38.75
STEPD1	141.730	142.161	0.955	0.045	60.00
STEPD2	147.730	144.826	0.955	0.045	61.00
STEPD3	149.079	145.110	0.955	0.045	62.00
HDDMASUM	15.394	17.674	1.000	0.000	2.25
HDDMA1	215.707	205.403	0.879	0.121	93.25
HDDMA2			0.000	1.000	
HDDMA3			0.000	1.000	
ECDDWT1	102.394	64.413	1.000	0.000	50.00